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Vericiguat for the treatment of coronary vasospasm (ViVA): a double-blind placebo-controlled randomized cross-over trial

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Abstract

Background

Coronary vasospasm is highly prevalent in patients with angina and no obstructive coronary arteries (ANOCA). A central mechanism of vasospasm in these patients arises from endothelial dysfunction and smooth muscle hyper-reactivity that disrupt the nitric oxide–soluble guanylate cyclase (sGC)–cyclic guanosine monophosphate pathway governing vasodilation. Vericiguat, an sGC stimulator that enhances sGC sensitivity to nitric oxide, may therefore be a therapeutic option for patients with coronary vasospasm.

Objectives

To evaluate the effects of vericiguat on angina symptoms, endothelial function, peripheral microvascular vasodilator responses, and safety in ANOCA patients with epicardial and/or microvascular vasospasm.

Methods

In this double-blind, placebo-controlled, randomized crossover trial, patients were randomized 1:1 to vericiguat followed by placebo, or placebo followed by vericiguat. Each treatment period lasted 10 weeks. The primary functional outcome was the difference in peripheral vasodilator function during acetylcholine iontophoresis using laser speckle contrast imaging; the primary symptom outcome was the difference in daily angina episodes, recorded via the ORBITA-app, between treatments.

Results

Among the 57 patients enrolled in the trial, the median age was 55 years [50, 61], and 44 (77%) were female. Vericiguat did not improve peripheral vasodilator function, assessed by LASCA during acetylcholine iontophoresis, compared with placebo on AUC (intervention effect estimate $-1221 \text{ APU}\cdot\text{s}$, 95% CI -4237 to 1796 , $p=0.42$). For the primary symptom endpoint, the final-day odds ratio for reduction in angina episodes recorded via the ORBITA-app, the posterior was distributed more towards the direction favoring placebo (OR 0.84, 95% CrI 0.65 to 1.04); however, it did not meet the prespecified efficacy or harm criterion (posterior probability of benefit with vericiguat versus placebo 0.07).; episode counts and angina-free days over 70 days were similar between arms. Vericiguat was generally tolerated, with modest reductions in blood pressure, no clinically relevant laboratory abnormalities, and no treatment-related serious adverse events.

Conclusions

The Vericiguat in Vasospastic Angina trial did not demonstrate evidence of improved peripheral vasodilator function or anginal symptoms in patients with coronary vasospasm. Vericiguat was nonetheless generally tolerated, with no treatment-related serious adverse events. Larger studies with longer treatment exposure could further evaluate the potential clinical effects of vericiguat in patients with coronary vasospasm.

ClinicalTrials.gov NCT06415227

Condensed abstract

Coronary vasospasm is a common cause of ANOCA and is associated with dysfunction of the nitric oxide–sGC–cGMP pathway. We evaluated the efficacy and safety of the sGC stimulator vericiguat in a randomized, placebo-controlled crossover trial in 57 patients with coronary vasospasm. Vericiguat did not demonstrate evidence of improved peripheral microvascular function or anginal symptoms compared with placebo. Vericiguat was generally tolerated, with no treatment-related serious adverse events. Larger studies with longer treatment exposure could further evaluate the potential clinical effects of vericiguat in patients with coronary vasospasm.

Key words

Angina with non-obstructive coronary arteries, coronary vasospasm, pharmacotherapy, vasospastic angina, vericiguat

Abbreviations

ACh	acetylcholine
ANOCA	angina with non-obstructive coronary arteries
APU	arbitrary perfusion units
AUC	area under the curve
cGMP	cyclic guanosine monophosphate
CVC	cutaneous vascular conductance
NO	nitric oxide
RHI	reactive hyperemia index
sGC	soluble guanylate cyclase
SNP	sodium nitroprusside

Introduction

Nearly half of patients undergoing invasive coronary angiography for suspected coronary artery disease have angina with non-obstructive coronary arteries (ANOCA).¹ In many of these patients, the underlying problem is coronary vasomotor dysfunction, most often related to epicardial and/or microvascular vasospasm.²⁻⁴ Despite its prevalence, coronary vasospasm remains difficult to treat. Current recommended therapy relies largely on calcium channel blockers and organic nitrates, but treatment responses are variable, leaving many patients with persistent symptoms.¹

Coronary vasospasm reflects an imbalance between vasodilator and vasoconstrictor influences. Among several potential mechanisms underlying coronary vasospasm, impaired nitric oxide (NO) signaling may play a central role, alongside enhanced vasoconstrictor pathways such as endothelin-1 signaling and Rho-kinase activation, which contribute to vascular smooth muscle hypercontractility and impaired vasomotor regulation.^{5,6} Under normal conditions, endothelium-derived NO activates soluble guanylate cyclase (sGC) to generate cyclic guanosine monophosphate (cGMP), promoting smooth muscle relaxation. In vasospasm-prone vessels, endothelial dysfunction and oxidative stress reduce NO bioavailability and blunt sGC responsiveness, impairing vasodilation and predisposing to inappropriate vasoconstriction. This disrupted NO–sGC–cGMP pathway may also help explain the heterogeneous response to organic nitrate therapy in patients with vasospastic angina.⁷⁻¹⁷ Restoring sGC signaling therefore represents a rational and mechanistically targeted therapeutic strategy in coronary vasospasm.

Two drug classes act on this pathway: sGC activators, which directly activate oxidized or heme-free sGC independently of NO, and sGC stimulators, which enhance the sensitivity of sGC to endogenous NO.^{12,17} Whereas the clinical use of sGC activators has been limited by side effects, the sGC stimulator vericiguat has established safety and efficacy in patients with

heart failure and pulmonary hypertension.^{18–20} Importantly, the VISOR and VENICE trials demonstrated that vericiguat is well tolerated in patients with coronary artery disease, even when combined with nitrates.^{21,22} Together with its favorable pharmacokinetic profile, these characteristics make vericiguat a compelling candidate for the treatment of coronary vasospasm.

The Vericiguat in Vasospastic Angina (ViVA) trial was designed to evaluate the effects of vericiguat on angina symptoms, endothelial function, and peripheral vasodilator responses in patients with ANOCA due to coronary vasospasm, as well as its safety and tolerability.

Methods

Study design

The ViVA trial is a prospective, double-blind, placebo-controlled, randomized cross-over trial evaluating vericiguat versus placebo as add-on therapy in patients with epicardial and/or microvascular coronary vasospasm and persistent angina despite guideline-directed medical therapy. The trial design has been detailed previously.²³

The trial was embedded within the Netherlands Registry of Invasive Coronary Vasomotor Function Testing (NL-CFT; NCT06083155), a national collaboration that collects data from clinically indicated coronary function tests (CFT) performed under a standardized protocol across 16 Dutch hospitals.²⁴ All NL-CFT centers could refer eligible patients, with trial-specific procedures conducted at Amsterdam UMC (Amsterdam, the Netherlands).

The protocol was approved by a central medical ethics committee in the Netherlands (Amsterdam UMC, location AMC, EU CT number 2022-502998-42-00). The trial was conducted in accordance with Good Clinical Practice and is registered with ClinicalTrials.gov

(NCT06415227). Data that support the findings of this study are available from the corresponding author upon reasonable request.

Participants

Eligible patients had epicardial and/or microvascular vasospasm according to COVADIS criteria as diagnosed by comprehensive invasive CFT following the NL-CFT protocol, and had persistent angina despite guideline-directed medical therapy.²⁵ Epicardial coronary vasospasm was defined as the presence of all three criteria during acetylcholine (ACh) provocation testing: significant epicardial vasoconstriction ($\geq 90\%$ luminal narrowing), reproduction of recognizable chest pain, and ischemic electrocardiographic changes. Microvascular spasm was defined as the reproduction of recognizable chest pain and ischemic electrocardiographic changes in the absence of significant epicardial vasoconstriction ($< 90\%$ luminal narrowing). Combined epicardial and microvascular spasm was diagnosed by ACh rechallenge, in which the previously identified epicardial spasm-provoking ACh dose was readministered after intracoronary nitrate administration. As nitrate administration may relieve the epicardial vasoconstrictive component, this approach allows assessment of a concomitant microvascular spasm component.²⁶ Endothelial dysfunction was not assessed as a separate phenotype, and there was no prespecified time interval between CFT diagnosis and trial enrollment. CFT-based diagnoses were established during the diagnostic work-up before trial entry, with patients enrolled after symptoms persisted despite guideline-directed medical therapy. Both epicardial and microvascular spasm were included to explore the effects of vericiguat across different coronary vasospasm endotypes. Consistent with the 2024 ESC Guidelines for chronic coronary syndromes, persistent angina was defined as at least one weekly episode of chest pain while receiving at least two anti-anginal agents directed at coronary vasospasm, or the maximally tolerated regimen when fewer than two agents were used.¹ Other inclusion criteria included age over 18 years, and absence of flow-limiting

coronary artery disease (no stenosis >50%, fractional flow reserve (FFR) ≤ 0.80 , or non-hyperemic pressure ratio (NHPR) ≤ 0.89). Women were eligible if postmenopausal or surgically sterile (prior hysterectomy, bilateral salpingectomy, or bilateral oophorectomy). Those of childbearing potential required a negative pregnancy test within 24 hours before first dosing and use of highly effective contraception or sexual abstinence throughout the treatment period and for at least 1 month thereafter.

Key exclusion criteria were impaired left ventricular function (LVEF <50%) or significant valvular disease, contraindications to background sublingual nitrate therapy as judged by the treating cardiologist, and contraindications to vericiguat, including systolic blood pressure <100 mmHg, severe renal impairment (eGFR <15 mL/min/1.73 m²), or severe hepatic dysfunction. Patients were also excluded for known hypersensitivity to study drug components, concomitant use of other sGC stimulators (e.g., riociguat) or phosphodiesterase-5 inhibitors (e.g., sildenafil), hereditary galactose intolerance, lactase deficiency, or glucose-galactose malabsorption, life expectancy <1 year, or inability to provide informed consent.

Randomization and masking

After providing written informed consent, patients were randomized 1:1 to receive either placebo followed by vericiguat or vericiguat followed by placebo. Randomization was performed using a secure web-based system (Castor Electronic Data Capture, Amsterdam, The Netherlands) with variable block sizes of 2, 4, and 6, stratified by sex and vasospasm type (epicardial, microvascular, or mixed). Allocation was communicated automatically and exclusively to the trial pharmacy at Amsterdam UMC, which dispensed study medication while maintaining concealment from investigators and treating physicians. Vericiguat and placebo tablets were identical in appearance, and participants, treating clinicians, outcome assessors, and data analysts all remained masked to treatment assignment throughout the trial.

Procedures

Each treatment period lasted approximately 10 weeks. Patients initiated treatment with oral vericiguat 2.5 mg or placebo once daily and were uptitrated to 5 mg at 14 ± 4 days, and to the target dose of 10 mg at 28 ± 4 days. The target dose was maintained for six weeks, with endpoint assessment at 70 ± 4 days. A two-week washout period then preceded crossover to the alternate treatment phase according to the concealed allocation sequence. Placebo was uptitrated in the same manner to maintain blinding (Supplementary Figure S2).

Peripheral microvascular function testing

Given the cross-over design of the ViVA trial, a non-invasive assessment of vascular function was preferred to allow repeated measurements within the same participant. Laser speckle contrast imaging (LASCA) was selected because this technique was established and routinely available at our center, with local expertise in its application and interpretation. Systemic endothelial function and vasodilator capacity were assessed at baseline before the first study drug dose and again at the end of each 10-week treatment period, by blinded investigators. Microvascular reactivity was measured by LASCA (PeriCam PSI, Perimed, Järfälla, Sweden) combined with iontophoresis of vasoactive agents. This non-invasive optical technique derives a relative index of skin microvascular perfusion from fluctuations in laser speckle patterns generated by red blood cell movement. Reduced blood cell movement produces higher speckle contrast, corresponding to lower perfusion, whereas increased blood cell movement decreases speckle contrast, corresponding to higher perfusion. Measurements were performed under standardized conditions to minimize artifacts. A dedicated iontophoresis chamber (PeriIont, Perimed) placed on the forearm, with a dispersive electrode completing the circuit, enabled controlled transdermal delivery of acetylcholine and insulin (endothelium-dependent vasodilation) and sodium nitroprusside (endothelium-independent vasodilation). Data were analyzed with PIMSoft (Perimed). Outcomes were expressed as peak responses and baseline-corrected area under the perfusion–time curve (AUC; arbitrary perfusion

units·seconds, APU•s). The AUC was calculated automatically using PIMSoft and corrected for baseline perfusion by subtracting the baseline perfusion AUC from the total AUC (Supplementary Figure S3). Skin blood flow was normalized to mean arterial pressure to derive cutaneous vascular conductance (CVC). Prior studies demonstrated that LASCA allows for the detection of microvascular dysfunction in ANOCA and heart failure with preserved ejection fraction, and that LASCA-derived measures reflect peripheral microvascular abnormalities associated with coronary vascular dysfunction.^{27–30}

Systemic endothelial function was additionally assessed by peripheral arterial tonometry (EndoPAT; Itamar Medical, Israel) during post-occlusive reactive hyperemia. Finger probes recorded pulse amplitude signals before and after 5 minutes of brachial artery occlusion, and the reactive hyperemia index (RHI) was calculated as the ratio of post- to pre-occlusion signals, adjusted for the contralateral control arm. An RHI <1.67 indicated endothelial dysfunction.^{27,31,32}

Symptom assessment

Angina burden was recorded daily with the ORBITA app (ORBITA Systems Ltd., London, UK) from two weeks before randomization until study end.³³ Patients also completed the Seattle Angina Questionnaire (SAQ), Rose Dyspnea Scale, EQ-5D-5L, and iPCQ/iMCQ at baseline and at 10-, 12-, and 22-week follow-up.

Endpoints

The primary functional endpoint was the within-patient difference in the AUC for cutaneous vascular perfusion during acetylcholine iontophoresis (arbitrary perfusion units x seconds, APU•s), measured at the end of each 10-week treatment period. The primary symptom endpoint was the difference in daily angina episodes recorded with the ORBITA-app between the vericiguat and placebo periods.

Secondary endpoints, each comparing the vericiguat and placebo periods, were microvascular function by laser speckle imaging during ACh, sodium nitroprusside (SNP), and insulin iontophoresis, endothelial function by EndoPAT, and quality of life measured by the SAQ Summary Score (SAQSS), Rose Dyspnea Scale, EQ-5D-5L, and EQ-VAS. Additionally, analyses were performed to assess whether coronary spasm type or sex modified the treatment effects on laser speckle imaging outcomes, ORBITA-app outcomes, and SAQSS outcomes.

Adverse events were assessed at every study visit. Major adverse cardiac events (MACE), defined as a composite of hospitalization for angina, spontaneous myocardial infarction, unplanned coronary revascularization, and all-cause death, were systematically recorded throughout the trial.

Statistical analysis

Because ViVA was a pilot study, no formal sample size calculation was performed. The intended enrollment was informed by effect sizes from prior studies using laser speckle imaging. These estimates were derived from parallel-group designs and were not adjusted for the within-subject correlation expected in a cross-over design to inform a conservative sample size. Assuming a clinically relevant AUC difference of $3536 \text{ APU} \cdot \text{s} \pm 4493$, 34 patients per group would provide 90% power to detect a between-group difference at $\alpha=0.05$.³⁴ Assuming a clinically relevant difference in peak CVC during ACh iontophoresis of $0.20 \pm 0.21 \text{ APU/mmHg}$, 25 patients per group would provide 90% power at $\alpha=0.05$.³⁵

We aimed for 50 patients to complete the protocol, anticipating that 20% would not reach the target vericiguat dose of 10 mg, and allowing for uncertainty in the treatment effect. The protocol permitted replacement of patients unable to tolerate the 5 mg dose, which was considered the minimal effective dose; this was expected to occur in $\leq 10\%$ of participants, based on previous studies. Patients who were unable to complete the study because of

intolerance and those who withdrew consent after randomization were replaced, and the protocol allowed up to 55 patients to complete the study protocol.

Treatment effects on peripheral vasodilator function were estimated using linear mixed-effects models with fixed effects for treatment, period, sequence, and baseline values, and a random participant intercept. Carry-over effects were not modeled because the prespecified 2-week washout period was considered adequate given the pharmacokinetic profile of vericiguat. With an elimination half-life of approximately 30 hours, the washout period represents 11–12 half-lives, during which residual drug concentrations are expected to decline to negligible levels. Furthermore, both functional and SAQ assessments were performed at the end of each 10-week treatment period, making clinically relevant residual treatment effects from the preceding period unlikely. Therefore, clinically relevant carry-over effects were considered unlikely. Treatment effects are reported as estimated mean differences with 95% confidence intervals (CI) and p-value. Differences in Seattle Angina (SAQSS) were analyzed using a linear mixed-effects model adjusted for baseline SAQSS.

The primary symptom endpoint was the daily odds ratio (OR) of shifting to fewer ORBITA-app recorded angina episodes with vericiguat versus placebo ($OR > 1$ indicating fewer episodes with vericiguat, $OR < 1$ fewer episodes with placebo). Treatment effects were estimated using a Bayesian Markov ordinal state-transition model that incorporated the full daily angina record and adjusted for previous-day angina symptom count, pre-randomization mean angina frequency, period, sequence, treatment allocation, study day, and a patient-level random intercept.³⁶ Treatment-by-time interaction terms and restricted cubic splines modeled time-varying and non-linear effects. The model also allowed derivation of clinically interpretable estimates to supplement the OR, including expected daily angina episodes and time spent without angina for a representative patient profile. A skeptical prior was placed on the treatment effect, centered at 0 on the log-OR scale, treating no treatment effect as the most

plausible outcome. The prior standard deviation was constructed such that the transition OR was constrained within 1/4 to 4 with 90% probability mass, chosen to be skeptical against large treatment effects considered implausible for this therapy based on current understanding, and to maintain appropriate skepticism given the small trial size. The ORBITA score is a composite combining angina episode frequency and intensity on a single 0–30 scale. A score of 0 represents no angina. Scores of 1–10 represent mild-intensity angina, with the value within this range reflecting the episode frequency (1 to 10 episodes). Scores of 11–20 represent moderate-intensity angina, again with the value reflecting episode frequency (1 to 10 episodes) within that band. Scores of 21–30 represent severe-intensity angina, with the value reflecting episode frequency (1 to 10 episodes) within that band. Statistical significance was defined as a posterior probability of benefit ($\text{Pr}(\text{benefit})$) or harm ≥ 0.975 , with results reported as posterior medians and 95% credible intervals (CrI).

All efficacy analyses were performed according to the intention-to-treat principle using a superiority framework. The analyses were conducted using linear mixed-effects models fitted by maximum likelihood, including all available observations from randomized participants. Missing outcome data were handled implicitly under the assumption that data were missing at random. To evaluate the robustness of the findings, a sensitivity analysis was performed using a per-protocol approach including only participants who completed the study. Continuous variables are presented as mean $\pm$ SD or median [Q1, Q3], as appropriate, and categorical variables as n (%). Statistical significance in frequentist analyses was defined as two-sided $p < 0.05$. Analyses were performed in R (version 4.2.1). An independent Data and Safety Monitoring Board oversaw the trial.

Role of the funding source

The ViVA study is an investigator-initiated trial funded by Amsterdam UMC and Bayer AG (Leverkusen, Germany). The authors are solely responsible for the study design and conduct, all analyses, the drafting and editing of the paper, and its final contents.

Results

Study visits took place between April 2025 and July 2026. Of 109 patients screened, 57 provided written informed consent and were randomized: 28 (49%) to vericiguat followed by placebo, and 29 (51%) to placebo followed by vericiguat (Figure 1). ORBITA-app data were available for 95.0% of the expected 7980 patient-days across these 57 participants. 6 patients discontinued owing to side effects (n=3) or for unspecified reasons (n=3). Among the 51 patients whom completed the study protocol, 25 (49%) were randomized to vericiguat followed by placebo, and 26 (51%) to placebo followed by vericiguat. The corresponding per-protocol ORBITA-app analysis, with 99.5% data availability across 7140 expected patient-days, is provided in the Supplementary Material. All participants who completed the study reached the target dose of 10 mg. Therefore, dose-stratified analyses were considered not informative and were not performed. Compliance was assessed at every study visit by asking patients whether they had forgotten any doses and by counting the number of tablets returned. Compliance was defined as the proportion of prescribed doses actually taken by each patient. Compliance was excellent, with a median compliance of 100.0% [100.0–100.0]. Three of the 57 participants (5%) missed a single dose during the study, corresponding to an individual compliance of 98.6%; all remaining participants had 100% compliance. Five patient visits (1%) occurred outside the protocol-specified visit window of ± 4 days. However, as patients had sufficient remaining tablets to ensure continuous treatment without interruption until endpoint assessment, these were considered minor protocol deviations and were not expected to affect treatment exposure. No other protocol deviations occurred.

Baseline patient characteristics

The median age was 55 years [50, 61], and most participants were female (n=44, 77%) (Table 1). The vasospasm endotype was epicardial spasm in 24 patients (42%), microvascular spasm in 25 (44%), and combined epicardial and microvascular spasm in 6 (11%). No patients had concomitant coronary microvascular dysfunction (CMD), confirmed by a normal coronary flow reserve (median 3.5 [2.9, 4.9]) and normal microvascular resistance (median hyperemic microvascular resistance 1.3 [1.2, 1.5] mmHg/cm/s; or median index of microvascular resistance 14.0 [8.0, 20.0]). The most common baseline anti-anginal therapies were short-acting nitrates (n=47, 82%), long-acting nitrates (n=39, 68%) and non-dihydropyridine calcium channel blockers (n=36, 63%). Median systolic blood pressure was 123 mmHg [110, 135], median diastolic blood pressure was 76 mmHg [70, 80], and mean arterial pressure (MAP) was 92 mmHg [86, 99]. Laboratory values (hemoglobin, CRP, creatinine, troponinT, and NT-proBNP) were all within the normal range.

At baseline, the median AUC during ACh iontophoresis was 32172 APU•s [25193, 43040], with a median CVC of 0.8 APU/mmHg [0.6, 0.9] and a median perfusion response (peak perfusion – baseline perfusion) of 70.1 APU [50.5, 84.1]. During SNP iontophoresis, the median AUC was 41548 APU•s [34056, 50478], CVC was 1.0 APU/mmHg [0.9, 1.2], and perfusion response was 92.5 APU [76.7, 107.4]. The median RHI measured by EndoPAT was 1.7 [1.4, 2.1], with 27 patients (47%) meeting the criterion for endothelial dysfunction.

The median number of daily angina episodes recorded by the ORBITA-app during the run-in period was 3 [2, 6]. The median SAQSS at baseline was 38 [29, 47], indicating poor angina-related quality of life.

Functional endpoints

Vericiguat had no significant effect on peripheral microvascular function. During ACh iontophoresis, neither AUC (treatment effect -1221 APU•s, 95% CI -4237 to 1796; p=0.42) nor CVC (treatment effect 0.0 APU/mmHg, 95% CI -0.1 to 0.0; p=0.23) differed between

treatments (Table 2), although, conversely, the perfusion response to ACh was slightly lower after vericiguat (treatment effect -5.2 APU, 95% CI -9.7 to -0.8; $p=0.02$). SNP and insulin iontophoresis likewise showed no significant differences in AUC, CVC, or perfusion response (Figure 2). EndoPAT-assessed RHI also did not show an effect of vericiguat treatment (treatment effect 0.1, 95% CI -0.1 to 0.3; $p=0.17$), and the number of patients meeting the endothelial dysfunction criterion did not change after vericiguat treatment (OR 0.5, 95% CI 0.1 to 1.5; $p=0.19$).

Incorporating type of coronary spasm (epicardial, microvascular, or combined epicardial and microvascular) into the model revealed no significant interaction between spasm type and change in AUC, CVC, or perfusion response during ACh iontophoresis, nor for the corresponding measures during SNP or insulin iontophoresis (Supplementary Table S5). In the post-hoc exploratory subgroup analysis, patients with epicardial spasm showed a significantly greater reduction in perfusion response with vericiguat compared with placebo (estimate -7.74, 95% CI -14.31 to -1.17, $p=0.02$).

In a prespecified exploratory subgroup analysis, sex did not modify the treatment effect on any outcome regarding LASCA ACh iontophoresis in exploratory subgroup analyses (interaction $p>0.05$ for all, see Supplementary Table S9).

Symptom endpoints

For the primary symptom endpoint, the final-day odds ratio for reduction in angina episodes recorded via the ORBITA-app, the posterior was distributed more towards the direction favoring placebo (OR 0.84, 95% CrI 0.65 to 1.04); however, it did not meet the prespecified efficacy or harm criterion (posterior probability of benefit with vericiguat versus placebo 0.07). This indicates a 7% posterior probability that vericiguat reduced angina episodes compared with placebo. For a representative patient (at the median), the expected number of angina episodes on the final day of each treatment period was similar with vericiguat

compared with placebo (vericiguat 2.8, 95% CrI 2.1 to 3.6, vs placebo 2.6, 95% CrI 1.9 to 3.4; placebo-controlled benefit -0.2 , 95% CrI -0.5 to 0.1 ; $\text{Pr}(\text{benefit})=0.07$). Over the full 70-day follow-up period, expected angina episode counts were likewise similar with vericiguat compared with placebo (vericiguat 184.8, 95% CrI 136.4 to 237.3, vs placebo 185.3, 95% CrI 133.9 to 234.8; placebo-controlled benefit 0.5 , 95% CrI -7.2 to 8.0 ; $\text{Pr}(\text{benefit})=0.55$), as were angina-free days (vericiguat 5.6, 95% CrI 2.0 to 10.8, vs placebo 5.5, 95% CrI 1.9 to 10.6; placebo-controlled benefit 0.1 , 95% CrI -0.7 to 0.8 ; $\text{Pr}(\text{benefit})=0.58$) (Table 3, Figure 3).

Additional analyses of the composite ORBITA angina symptom score, combining episode frequency and symptom intensity, and of symptom intensity alone are available Table 3 and Supplementary Results. They were consistent with the primary symptom analysis and showed no treatment effect (composite final-day OR 0.86 , 95% CrI 0.69 to 1.08 , posterior probability of benefit 0.10 ; intensity final-day OR 1.11 , 95% CrI 0.85 to 1.42 , posterior probability of benefit 0.78 ; Supplemental Figures S14-S15-S19-S20).

Endotype-stratified analyses for ORBITA angina frequency are available in the Supplementary Material. These post-hoc, exploratory analyses showed divergent treatment effects in patients with epicardial versus microvascular spasm, with a final day transition OR for symptom reduction of 0.49 (95% CI 0.38 to 0.64) in patients with epicardial spasm and of 1.14 (95% CI 0.87 to 1.45) in patients with microvascular spasm. The effect modification of sex on the ORBITA-app symptom frequency outcome was assessed in the intention-to-treat population, as a secondary, exploratory analysis. The final day transition OR for reduction in daily angina episodes with vericiguat vs placebo was smaller in female vs male participants (interaction OR 0.74 , 95% CrI 0.61 to 0.92 ; posterior probability of smaller treatment effect in female vs male participants = 1.00 , Supplementary Table S13).

For quality of life assessed by the SAQ, neither the change in SAQSS nor any individual domain score differed between vericiguat and placebo treatment (treatment effect for SAQSS 0.3U, 95% CI -4.0 to 4.5; $p=0.91$) (Figure 4). Likewise, the EQ-5D-5L and Rose Dyspnea Scale showed no significant improvements with vericiguat compared with placebo (Table 2). Type of coronary spasm did not significantly influence the intervention effect on SAQSS ($\chi^2(4) = 7.10$, $p=0.13$) (Supplementary Table S5), nor did sex ($\chi^2(1) = 0.62$, $p=0.43$) (Supplementary Table S9). Outcomes from the iPCQ and iMCQ questionnaires, assessing productivity, daily functioning, and healthcare resource use, are presented in Supplementary Table S4.

Because responder status was defined as the within-patient difference in SAQSS change between the vericiguat and placebo periods, this analysis was restricted to the per-protocol population including 51 of 57 participants with complete SAQSS data in both treatment periods. When comparing the change from baseline between treatments, 20 patients (39%) achieved a ≥ 5 U clinically relevant improvement in SAQSS with vericiguat vs placebo, compared with 13 patients (25%) who achieved a ≥ 5 U improvement with placebo vs vericiguat; however, this difference was not statistically significant ($p=0.30$). Using a more stringent threshold, eight patients (16%) achieved a ≥ 10 U clinically relevant improvement with vericiguat vs placebo, compared with seven patients (14%) who achieved a ≥ 10 U improvement with placebo vs vericiguat ($p=1.00$).^{2,37}

Univariable linear regression analyses identified no significant predictors of a clinically relevant treatment effect on SAQSS, defined as a difference in change from baseline of ≥ 5 U between vericiguat and placebo. Similarly, no significant predictors were identified for a difference in change from baseline of ≥ 10 U between vericiguat and placebo. Neither type of coronary vasospasm, nor sex was associated with the likelihood of achieving clinically relevant improvement in SAQSS (Table 4).

Safety

Laboratory parameters were unchanged after the vericiguat and placebo periods. Vericiguat resulted in a statistically significant, but clinically modest reduction in blood pressure, lowering systolic blood pressure by 7 mmHg (95% CI -19 to 0; $p<0.01$), diastolic blood pressure by 5 mmHg (95% CI -7 to 4; $p=0.06$), and MAP by 5 mmHg (95% CI -12 to 0; $p<0.01$) (Table 2). All patients in the per-protocol population were successfully uptitrated to the maximum vericiguat dose of 10 mg.

Seventeen patients (30%) experienced a MACE, all of which were hospitalizations for angina. In every case, acute coronary syndrome was excluded without the need for additional non-invasive ischemia testing or invasive coronary angiography, and the symptoms were attributed to the patients' underlying coronary vasospasm (Table 5). Of these 17 hospitalizations, seven occurred during the placebo treatment period compared with four during the vericiguat treatment period (7/17 (41%) vs 4/17 (24%), McNemar's test $p=0.55$). The remaining six (35%) hospitalizations occurred during the pre-treatment or washout periods.

Treatment-related side effects during vericiguat treatment were most commonly vertigo, occurring in 17 patients (31%), and heartburn/acid reflux, occurring in 14 patients (25%), followed by edema and headache, each occurring in 9 patients (16%).

Discussion

The ViVA trial is the first randomized, placebo-controlled study to evaluate the effects of vericiguat on systemic vasodilator function and angina symptoms in patients with coronary vasospasm. In patients with refractory angina due to epicardial and/or microvascular spasm, vericiguat neither reduced daily angina episodes nor improved peripheral vasodilator function compared with placebo.

The CorMicA and ILIAS ANOCA trials demonstrated that endotype-guided treatment improves quality of life in patients with ANOCA,^{2,37} yet evidence-based pharmacological therapies targeting specific ANOCA endotypes remain scarce. Most trials predate contemporary comprehensive ANOCA endotyping or lacked a placebo arm. This is particularly relevant in a population as symptomatic as the one enrolled in ViVA, where uncontrolled study designs are prone to overestimating the treatment effect due to expectation effects and regression to the mean. Consequently, treatment recommendations continue to rely largely on expert consensus rather than robust evidence.

Coronary vasospasm is the most prevalent ANOCA endotype, and reflects an imbalance between endothelial vasodilator and vasoconstrictor mechanisms, in which dysfunction of the NO–sGC–cGMP pathway is thought to play a key role.^{2,38} Reduced NO bioavailability and impaired sGC responsiveness from oxidative stress limit smooth muscle relaxation, increase susceptibility to vasoconstriction, and may contribute to variable nitrate responses in the treatment of coronary vasospasm in clinical practice. Restoring sGC signaling therefore offered a mechanistically rational novel treatment strategy for coronary vasospasm, and vericiguat was the logical agent.^{11–13,17} The ongoing V-COM trial (NCT06239974) will evaluate whether vericiguat improves myocardial blood flow, myocardial perfusion reserve, and anginal symptoms in patients with coronary microvascular dysfunction. In contrast, ViVA exclusively enrolled patients with coronary vasospasm without concomitant coronary microvascular dysfunction, thereby addressing a distinct pathophysiological phenotype. Consequently, the two trials investigate complementary patient populations and will provide insight into the potential role of vericiguat across different endotypes of ANOCA.

Consistent with its established safety profile in heart failure and coronary artery disease, vericiguat was generally well-tolerated in ViVA, although treatment was associated

with modest reductions in blood pressure and adverse effects including vertigo, heartburn/acid reflux, edema, and headache, which contributed to treatment discontinuation in some participants. As heart rate was not recorded, we cannot determine whether the observed blood pressure reductions were accompanied by a reflex chronotropic response. No treatment-related serious adverse events or excess cardiovascular events were observed compared with placebo.^{18,20–22,39} Despite the observed pharmacodynamic effects and safety profile, forearm LASCA measurements assessing endothelium-dependent and endothelium-independent vasodilator responses did not provide evidence of a treatment-related functional benefit. However, the confidence intervals around the estimated treatment effects were relatively wide, and modest beneficial or adverse effects cannot be excluded. Therefore, these findings should be interpreted as an inability to demonstrate efficacy within the precision of this pilot study rather than definitive evidence of no treatment effect. In addition, the neutral findings may reflect the heterogeneous pathophysiology of coronary vasospasm. Although dysregulation of the NO–sGC–cGMP pathway is implicated in the pathophysiology of coronary vasospasm, alternative mechanisms, such as Rho-kinase–mediated vascular smooth muscle hypercontractility, may play a dominant role in some patients with ANOCA.^{5,6} Consequently, targeting the NO–sGC–cGMP pathway alone may not be effective across the entire patient population.

The nominally significant reductions in ACh-mediated peak perfusion and perfusion response with vericiguat warrant caution. We did not identify a plausible mechanism for an isolated impairment, particularly in the absence of consistent changes across the other functional measures. These isolated findings are therefore most likely a reflection of measurement variability. Furthermore, the exploratory subgroup analyses examining the impact of coronary spasm type on LASCA outcomes should be interpreted cautiously given the small subgroup size and lack of adjustment for multiple testing. These findings should

therefore be considered hypothesis-generating rather than confirmatory. LASCA was nonetheless the most appropriate non-invasive functional endpoint in the context of the chosen cross-over study design. A cross-over design was preferred over a parallel-group design because it allows each participant to serve as their own control, thereby reducing between-participant variability and providing a more informative assessment of treatment effects in a disease as heterogeneous as coronary vasospasm. However, this design requires repeated assessments within the same participant, and repeated invasive coronary function testing was considered unethical in the context of a placebo-controlled cross-over trial. Therefore, LASCA provided a feasible approach to assess microvascular function while maintaining the advantages of a cross-over design.

Beyond these functional measures, the data on the placebo-controlled impact of vericiguat on angina symptoms support the functional results. Angina burden was captured through daily ORBITA-app entries in addition to periodic assessment of quality-of-life questionnaires. Despite the dense, patient-reported record of day-to-day changes in angina burden, vericiguat treatment did not meaningfully reduce daily angina episodes compared with placebo treatment (OR 0.84, 95% CrI 0.65 to 1.04; Pr(benefit)=0.07). In addition, the expected angina count for a representative patient was near-identical on both treatments, whether assessed on the final treatment day (2.8 vs 2.6 episodes for vericiguat and placebo, respectively) or across the full 70-day period (184.8 vs 185.3 episodes, respectively). There was also no difference in angina-free days (5.6 vs. 5.5, respectively). Notably, previous studies employing the ORBITA-app for angina symptom assessment have demonstrated significant changes in daily angina episodes with vastly smaller sample sizes, suggesting that our sample size would have been sufficient to detect a clinically meaningful treatment effect.^{40,41} Moreover, quality of life questionnaire data were consistent with the findings from

ORBITA-app. This concordance between the functional and symptom endpoints strengthens confidence in the neutral result.

Several recent randomized trials evaluating pharmacological therapies for coronary vasospasm have yielded heterogeneous results, likely impacted by differences in study populations and the extent of patient phenotyping. In patients with vasospastic angina, amlodipine significantly reduced angina frequency compared with placebo, while nebivolol and diltiazem reduced vasospasm on repeat acetylcholine provocation testing and improved SAQ scores after 12 weeks.^{42,43} In contrast, the EDIT-CMD trial, which included patients with coronary microvascular dysfunction and/or vasospasm, found no significant effect of diltiazem on vasospasm, coronary flow reserve, angina symptoms, or quality of life, although a reduction in epicardial vasospasm was observed in a subgroup of patients.⁴⁴ Similarly, a proof-of-concept trial investigating endothelin receptor antagonism with macitentan in patients with vasospastic angina did not demonstrate a reduction in anginal burden compared with placebo.⁴⁵ The PRIZE and WARRIOR trials evaluated broader ANOCA populations without comprehensive endotype-based selection and also failed to demonstrate significant improvements in angina symptoms or clinical outcomes.^{46,47} Together, these findings highlight the limited and heterogeneous evidence base for pharmacological treatment of coronary vasospasm as a pathophysiological substrate of ANOCA, emphasizing the need for adequately powered randomized, placebo-controlled trials evaluating endotype-specific therapies. Moreover, the underlying pathophysiological mechanisms driving coronary vasospasm remain incompletely understood and are potentially heterogeneous across patients, meaning that current trials are unable to selectively target specific mechanistic pathways, a limitation that may partly explain the absence of evidence-based medical therapeutics to date. The neutral results of the ViVA trial further reinforce the importance of rigorous placebo-

controlled designs to establish whether pharmacological treatments provide true clinical benefit.

Some limitations should be taken into account. First, LASCA measurements showed greater variability than anticipated from the variance reported in a previous study in a different population. Consequently, confidence intervals around the estimated treatment effects were wider, and the absence of a statistically significant difference should not be interpreted as evidence of no treatment effect. Second, peripheral measures of vascular function represent surrogates of coronary vascular function. Therefore, treatment effects on coronary vasomotor function not captured by peripheral vascular assessments cannot be excluded. Third, a 10-week treatment period may be insufficient to discern treatment effects from symptom variations influenced by factors such as weather conditions, physical activity, and psychological stress, which notably affect disease burden in coronary vasospasm. Fourth, the inclusion of both epicardial and microvascular spasm may have introduced heterogeneity in treatment response. Although randomization was stratified by spasm type and prespecified subgroup analyses were performed, the sample size limits definitive conclusions regarding differential effects between types of coronary vasospasm. Fifth, patients enrolled in the ViVA trial were highly symptomatic despite a high background use of vasodilator therapies, representing a refractory patient population on optimal medical therapy, as also reflected by the substantial burden of angina-related hospitalizations. The advanced disease burden and high background use of available medical therapy may have limited the potential incremental effect of vericiguat. Sixth, the intensive nature of the study protocol, including frequent study visits, daily app assessments, and travel requirements, may have introduced selection bias where participants may represent a subgroup of patients with greater ability and willingness to engage in intensive follow-up, potentially limiting generalizability. Finally, blinding was not formally assessed, and the possibility of functional unblinding cannot be excluded. However,

functional unblinding might bias symptom reporting in favor of the vericiguat arm of the study, but we did not find evidence of this across any of the endpoints evaluated.

Conclusion

The ViVA trial did not demonstrate evidence of improved peripheral vasodilator function or anginal symptoms with vericiguat compared with placebo in patients with coronary vasospasm. Vericiguat was nonetheless generally tolerated, with no treatment-related serious adverse events. Larger studies with longer treatment durations and more reproducible functional outcome measures could further evaluate the potential role of vericiguat in patients with coronary vasospasm.

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Tables and figures

Table 1: Baseline characteristics

	Overall (n=57)
Randomization allocation	
- Vericiguat first, placebo second	28 (49%)
- Placebo first, vericiguat second	29 (51%)
Age, years	55 [50, 61]
Sex	
- Female	44 (77%)
- Male	13 (23%)
BMI, kg/m²	27 [25, 31]
Medical history	
- Hypertension	18 (32%)
- Hypercholesterolemia	21 (37%)
- Diabetes	
Insulin dependent	2 (4%)
Insulin independent	2 (4%)
- Smoking status	
Current	2 (4%)
Former	20 (35%)
- Family history of CVD	41 (72%)
- Previous myocardial infarction	13 (23%)
- Previous percutaneous coronary intervention	12 (21%)
Female specific risk factors	
- Pregnancy	39 (89%)
Pregnancy-related hypertension	9 (23%)
Pregnancy-related diabetes	1 (3%)
Pre-eclampsia/HELLP	4 (10%)
History of miscarriages	10 (26%)
- Hormonal therapy >60 years	1 (2%)
- Post-menopausal	24 (55%)
Early menopause <40 years	0 (0%)
- History of hysterectomy	8 (18%)
- History of ovariectomy	1 (2%)
- History of polycystic ovary syndrome	0 (0%)
Acetylcholine provocation test result	
- Epicardial spasm	24 (42%)
- Microvascular spasm	25 (44%)
- Combined epicardial + microvascular spasm	6 (11%)
- Inconclusive*	2 (4%)
Acetylcholine rechallenge performed	26 (46%)
- <i>Result:</i>	
No spasm	6 (23%)
Microvascular spasm	14 (54%)

Epicardial spasm	3 (12%)
Inconclusive*	3 (12%)
- <i>Conclusion:</i>	
Nitrate unresponsive	7 (27%)
Nitrate responsive	5 (19%)
Partially nitrate responsive†	14 (54%)
Vasodilator testing result	
- CFR	3.5 [2.9, 4.9]
- HMR, mmHg/cm/s	1.3 [1.2, 1.5]
- IMR	14.0 [8.0, 20.0]
- Concomitant CMD ^Δ	0 (0%)
Baseline medication use	
- Non-DHP CCB	36 (63%)
- DHP CCB	26 (46%)
- Betablocker	11 (19%)
- Long-acting nitrate	39 (68%)
- Short-acting nitrate	47 (82%)
- Nicorandil	28 (49%)
- ACEi/ARB	15 (26%)
- Lipid lowering drug	37 (65%)
- Antiplatelets	30 (53%)
- Proton pump inhibitors	30 (53%)
- Molsidomine	2 (4%)
- Ivabradine	4 (7%)

Intention-to-treat population with maximum likelihood estimation, assuming missings at random. *Inconclusive acetylcholine provocation test/rechallenge: recognizable symptoms, but no ischemic electrocardiographic changes or vasoconstriction. †Partially nitrate responsive: improvement of vasoconstriction and/or ischemic electrocardiographic changes, but no relief of anginal symptoms. ^Δ Based on CFR <2.5. ACEi, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; BMI, body mass index; CCB, calcium channel blocker; CFR, coronary flow reserve; CMD, coronary microvascular dysfunction; CVD, cardiovascular disease; DHP, dihydropyridine; HELLP, hemolysis, elevated liver enzymes, and low platelet count; HMR, hyperemic microvascular resistance; IMR, index of microvascular resistance.

Table 2: Outcomes after vericiguat versus placebo treatment

	Baseline	After 10-week vericiguat treatment	Change from baseline	After 10-week placebo treatment	Change from baseline	Intervention effect*		
						Estimate	95% CI	P- value
Laboratory results								
- Hemoglobin, mmol/L	8.8 [8.3, 9.3]	8.8 [8.5, 9.2]	-0.1 [-0.4, 0.1]	8.7 [8.4, 9.3]	-0.1 [-0.4, 0.2]	0.0	-0.1 to	0.58
- CRP, mg/L	1.5 [0.7, 2.8]	1.5 [0.7, 2.7]	0.0 [-0.5, 0.3]	1.3 [0.7, 3.0]	0.0 [-0.3, 0.5]	0.2	0.2	0.74
- Creatinine, μ mol/L	71 [65, 80]	72 [66, 82]	2 [-3, 5]	71 [65, 78]	-1 [-6, 3]	1.3	-1.1 to	0.23
- TroponinT hs, ng/L	6 [4, 8]	6 [4, 7]	0 [-1, 0]	6 [4, 8]	0 [-1, 0]	0.2	1.5	0.52
- NT-proBNP, ng/L	70 [38, 124]	79 [49, 121]	6 [-24, 30]	70 [39, 120]	1 [-16, 22]	-5.8	-0.8 to 3.4 -0.3 to 0.7 -31.2 to 19.6	0.65
Blood pressure								
- Systolic, mmHg	123 [110, 135]	110 [105, 122]	-7 [-19, 0]	120 [110, 125]	-4 [-10, 1]	-6.8	-10.9 to	<0.01
- Diastolic, mmHg	76 [70, 80]	70 [70, 79]	-5 [-7, 4]	75 [70, 80]	0 [-5, 4]	-2.3	-2.7	0.06
- MAP, mmHg	92 [86, 99]	85 [80, 92]	-5 [-12, 0]	90 [83, 95]	-2 [-5, 3]	-3.8	-4.6 to 0.1 -6.3 to - 1.2	<0.01
LASCA assessment								
- <i>ACh</i>								
Baseline perfusion, APU	36.5 [29.8,	34.2 [30.0, 40.9]	-0.6 [-4.7, 4.6]	36.7 [32.5, 43.0]	0.8 [-3.9, 5.8]	-1.1	-3.6 to	0.37
Peak perfusion, APU	42.3]	100.0 [87.7, 113.6]	-5.3 [-15.1,	104.1 [91.9, 127.9]	5.2 [-10.0,	-6.3	1.4	0.02
Perfusion response, APU	103.3 [88.3,	64.1 [51.9, 78.3]	9.4]	68.5 [53.9, 86.0]	14.9]	-5.2	-11.5 to	0.02
CVC, APU/mmHg	122.8]	0.8 [0.6, 0.9]	-2.6 [-13.0,	0.7 [0.6, 1.0]	0.5 [-10.1,	0.0	-1.0	0.23
AUC, APU*s	70.1 [50.5,	33811 [25324, 40769]	7.0]	34422 [25335,	13.0]	-1221	-9.7 to -	0.42
- <i>SNP</i>	84.1]		0.0 [-0.1, 0.1]	41919]	0.0 [-0.1, 0.2]		0.8	
Baseline perfusion, APU	0.8 [0.6, 0.9]	29.7 [26.0, 34.7]				-2.5		0.21

	Peak perfusion, APU	32172 [25193,	119.9 [105.7, 142.1]	-3026 [-9928,	32.0 [27.2, 38.6]	-915 [-6091,	-2.9	-0.1 to	0.44
	Perfusion response, APU	43040]	87.1 [73.3, 105.7]	6932]	118.0 [107.1, 136.6]	6084]	-0.4	0.0	0.89
	CVC, APU/mmHg		1.0 [0.8, 1.2]		89.0 [72.3, 103.9]		0.0	-4237 to	0.31
	AUC, APU•s	31.2 [27.9,	37529 [30888, 46297]	-0.9 [-5.6, 4.6]	1.0 [0.8, 1.2]	-0.3 [-5.2, 5.7]	-618	1796	0.66
-	<i>Insulin</i>	37.9]		-4.0 [-18.0,	37426 [32900,	-1.3 [-20.0,			
	Baseline perfusion, APU	126.4 [107.3,	28.3 [25.2, 34.4]	10.4]	43497]	14.8]	-1.5	-6.5 to	0.12
	Peak perfusion, APU	143.6]	46.6 [39.3, 60.0]	-4.5 [-17.1,		-1.3 [-16.2,	-0.1	1.4	0.98
	Perfusion response, APU	92.5 [76.7,	15.3 [8.7, 33.2]	10.2]	30.5 [26.5, 36.4]	13.1]	1.4	-10.4 to	0.55
	CVC, APU/mmHg	107.4]	0.2 [0.1, 0.4]	0.0 [-0.1, 0.2]	50.4 [41.6, 60.5]	0.0 [-0.2, 0.2]	0.0	4.6	0.37
	AUC, APU•s	1.0 [0.9, 1.2]	7343 [3402, 17141]	-3695 [-6781,	19.4 [8.8, 31.4]	-2578 [-8578,	-46	-6.1 to	0.97
-	<i>NaCl</i>	41548 [34056,		2586]	0.2 [0.1, 0.4]	1953]		5.3	
	Baseline perfusion, APU	50478]	31.6 [27.3, 40.8]		10432 [3929, 18017]		-0.5	0.0 to	0.63
	Peak perfusion, APU		48.5 [39.8, 62.3]	-1.7 [-5.3, 2.3]		0.2 [-4.9, 4.9]	-3.7	0.1	0.24
	Perfusion response, APU	30.8 [24.3,	9.8 [2.4, 32.3]	0.0 [-14.9,	32.4 [28.7, 38.9]	-0.1 [-11.3,	-3.2	-3377 to	0.30
	CVC, APU/mmHg	36.1]	0.1 [0.0, 0.4]	11.1]	45.1 [38.4, 77.1]	6.9]	0.0	2142	0.48
	AUC, APU•s	48.7 [40.4,	3242 [-48, 14296]	-0.7 [-10.0,	7.1 [2.1, 40.8]	-1.5 [-7.9, 3.1]	-1398		0.35
		64.8]		5.2]	0.1 [0.0, 0.5]	0.0 [-0.1, 0.1]		-3.4 to	
		19.1 [10.2,		0.0 [-0.1, 0.1]	1828 [-964, 18382]	156 [-4266,		0.4	
		30.6]		-899 [-6294,		4861]		-4.9 to	
		0.2 [0.1, 0.3]		4236]				4.8	
		8627 [3804,				-2.1 [-6.9, 0.8]		-3.4 to	
		15294]		-3.2 [-7.9, 2.5]		-3.1 [-11.8,		6.2	
				1.8 [-17.1, 8.9]		7.5]		0.0 to	
		37.4 [28.8,		1.6 [-5.8, 11.3]		0.6 [-5.3, 7.1]		0.1	
		42.9]		0.0 [-0.1, 0.1]		0.0 [-0.1, 0.1]		-2876 to	
		50.8 [38.4,		1390 [-3634,		-48 [-4271,		2783	
		66.2]		7393]		5925]			
		7.7 [2.4, 30.1]						-2.8 to	
		0.1 [0.0, 0.4]						1.7	
		2671 [-1583,						-10.0 to	
		13153]						2.5	

							-9.3 to 3.0 -0.1 to 0.0 -4388 to 1592	
EndoPAT assessment - RHI - Endothelial dysfunction based on RHI <1.67	1.7 [1.4, 2.1] 27 (47%)	1.9 [1.6, 2.3] 15 (28%)	0.2 [-0.1, 0.5] -12 patients	1.8 [1.5, 2.3] 20 (38%)	0.0 [-0.3, 0.4] -7 patients	0.1 OR 0.5	-0.1 to 0.3 0.1 to 1.5	0.17 0.19
Seattle Angina Questionnaire - Summary score - Angina limitation - Angina stability - Angina frequency - Treatment satisfaction - Quality of life	38 [29, 47] 46 [33, 58] 50 [25, 50] 30 [10, 50] 75 [63, 88] 33 [25, 42]	38 [28, 51] 44 [31, 61] 50 [50, 75] 40 [10, 60] 69 [63, 88] 33 [25, 50]	3 [-4, 8] 0 [-10, 6] 0 [-25, 50] 0 [0, 20] 0 [-19, 6] 0 [-8, 8]	41 [27, 49] 43 [33, 59] 50 [25, 50] 30 [20, 50] 75 [63, 88] 33 [25, 50]	1 [-5, 8] -1 [-9, 8] 0 [-25, 25] 0 [0, 10] -6 [-13, 0] 0 [-8, 8]	0.3 0.1 7.7 2.2 0.0 -2.1	-4.0 to 4.5 -5.0 to 5.3 -2.7 to 18.1 -4.6 to 9.0 -4.0 to 4.1 -6.8 to 2.8	0.91 0.96 0.14 0.52 0.98 0.40
EQ-5D-5L questionnaire - VAS - Index score	55 [40, 65] 0.7 [0.5, 0.8]	50 [38, 66] 0.6 [0.4, 0.8]	-3 [-18, 5] 0.0 [-0.2, 0.1]	50 [37, 60] 0.7 [0.5, 0.8]	-3 [-15, 4] 0.0 [-0.1, 0.1]	0.7 0.0	-4.7 to 6.1 -0.1 to 0.0	0.79 0.47
Rose Dyspnea Scale - 0 - 1	3 [2, 4] 2 (4%) 6 (11%)	3 [2, 3] 3 (6%) 8 (15%)	0 [-1, 0]	3 [2, 4] 5 (9%) 6 (11%)	0 [-1, 0]	-0.1	-0.4 to 0.2	0.69

- 2	12 (21%)	14 (26%)		12 (23%)				
- 3	18 (32%)	15 (28%)		14 (26%)				
- 4	19 (33%)	13 (25%)		16 (30%)				

Intention-to-treat analysis with maximum likelihood estimation, assuming missings at random. *Values are derived from a mixed linear regression model and depicted as mean (95% CI) for continuous outcomes, and from a logistic mixed-effects model, depicted as odds ratio (95% CI) for binary outcomes, adjusting for period, randomization sequence, baseline value, and participant. AUC, area under the curve; ACh, acetylcholine; APU, arbitrary perfusion units; CI, confidence interval; CRP, C-reactive protein; CVC, cutaneous vascular conductance; LASCA, laser speckle contrast analysis; MAP, mean arterial pressure; NT-proBNP, N-terminal pro-B-type natriuretic peptide; OR, odds ratio; RHI, reactive hyperemia index; SAQ, Seattle Angina Questionnaire, higher scores represent less angina symptoms; SNP, sodium nitroprusside; VAS, visual analogue scale, of EQ-5D validated quality of life with higher scores indicating better quality of life. Rose Dyspnea Scale, scores range from 0 to 4, with higher scores indicating dyspnea with milder activities.

Table 3: Bayesian ORBITA-app angina symptom outcomes

Outcome		Effect size (95% CrI)	Probability of greater benefit with vericiguat versus placebo
Primary outcome			
Final day transition OR for symptom reduction with vericiguat versus placebo		0.84 (0.65 to 1.04)	0.07
Supplementary derived outcomes from primary symptom model			
Expected number of angina episodes on final day under each treatment	Vericiguat	2.8 (2.1 to 3.6)	0.07
	Placebo	2.6 (1.9 to 3.4)	
	Placebo-controlled benefit	−0.2 (−0.5 to 0.1)	
Expected number of angina episodes over the follow-up period under each treatment	Vericiguat	184.8 (136.4 to 237.3)	0.55
	Placebo	185.3 (133.9 to 234.8)	
	Placebo-controlled benefit	0.5 (−7.2 to 8.0)	
Expected number of days free from angina over the follow-up period under each treatment	Vericiguat	5.6 (2.0 to 10.8)	0.58
	Placebo	5.5 (1.9 to 10.6)	
	Placebo-controlled benefit	0.1 (−0.7 to 0.8)	
Secondary outcomes			
Final day transition OR for ORBITA score improvement (reduction) with vericiguat versus placebo		0.86 (0.69 to 1.08)	0.10
Final day transition OR for symptom intensity reduction with vericiguat versus placebo		1.11 (0.85 to 1.42)	0.78

Values are derived from a Bayesian ordinal longitudinal regression model fit to daily angina episode counts recorded via the ORBITA-app and are presented as OR for the primary outcome and expected count for supplementary derived outcomes. Posterior median effect sizes are summarized with 95% CrI. The final-day transition odds ratio is oriented so that OR >1 favors vericiguat. Supplementary derived outcomes are evaluated for a representative patient profile, defined at the median baseline value of 3 angina episodes per day. Placebo-controlled benefit is the vericiguat-minus-placebo difference, oriented so that a value >0 favors vericiguat. The pre-specified posterior probability threshold of ≥ 0.975 used to declare a beneficial treatment effect was not reached for any outcome. CrI, credible interval; OR, odds ratio.

Table 4: Univariate logistic regression for predictors of SAQSS improvement

Variable	Net SAQSS responder ≥ 5 U		Net SAQSS responder ≥ 10 U	
	OR (95% CI)	P-value	OR (95% CI)	P-value
Age	0.98 (0.92 to 1.03)	0.45	0.95 (0.88 to 1.02)	0.13
Sex, male	0.37 (0.07 to 1.44)	0.18	1.98 (0.36 to 9.62)	0.40
BMI	1.07 (0.93 to 1.24)	0.36	1.06 (0.88 to 1.28)	0.50
Diabetes	1.72 (0.75 to 5.22)	0.23	1.13 (0.28 to 2.76)	0.81
Hypertension	1.63 (0.49 to 5.40)	0.42	0.62 (0.08 to 3.10)	0.59
Hypercholesterolemia	0.31 (0.08 to 1.03)	0.06	0.16 (0.01 to 1.04)	0.10
Smoking, former	0.36 (0.10 to 1.19)	0.11	0.94 (0.17 to 4.37)	0.94
Family history of CVD	0.95 (0.28 to 3.39)	0.94	0.65 (0.14 to 3.53)	0.59
Previous MI	0.37 (0.07 to 1.44)	0.18	*	
Previous PCI	1.14 (0.29 to 4.25)	0.84	1.10 (0.15 to 5.71)	0.92
Type of coronary vasospasm [†]				
- vs microvascular spasm	2.67 (0.78 to 9.86)	0.13	2.22 (0.39 to 17.44)	0.39
- vs both epicardial and microvascular spasm	0.67 (0.03 to 5.77)	0.74	2.50 (0.10 to 33.37)	0.49
Nitrate responder based on ACh rechallenge				
- partial responder	0.40 (0.05 to 2.96)	0.36	1.09 (0.09 to 26.38)	0.95
- responder	0.44 (0.02 to 5.92)	0.56	*	
CFR	0.92 (0.61 to 1.01)	0.66	0.57 (0.21 to 0.94)	0.18
HMR	*		*	
IMR	1.00 (0.98 to 1.04)	0.90	1.11 (1.00 to 1.26)	0.09

Analysis restricted to participants with complete paired placebo and vericiguat data (n = 51/57); participants missing an outcome measurement in either treatment period could not be included in this paired-difference outcome. Net SAQSS responder = (Δ SAQSS, after vericiguat treatment minus baseline) – (Δ SAQSS, after placebo treatment minus baseline), difference ≥ 5 or ≥ 10 U. *Not estimable due to complete separation in the logistic regression model. [†] Reference category used is epicardial vasospasm. ACh, acetylcholine; CFR, coronary flow reserve; CI, confidence interval; CVD, cardiovascular disease; HMR, hyperemic microvascular resistance; IMR, index of microvascular resistance; MI, myocardial infarction; OR, odds ratio; PCI, percutaneous coronary intervention; SAQSS, Seattle Angina Questionnaire summary score.

Table 5: Adverse events throughout the trial

	Pre-treatment (n=57)	During vericiguat treatment (n=55)	During washout after vericiguat (n=53)	During placebo treatment (n=56)	During washout after placebo (n=53)	P-value ^π
MACE						
- Hospitalization for angina*	3 (5%)	4 (7%)	2 (4%)	7 (13%)	1 (2%)	0.55
- Myocardial infarction	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	
- Unplanned coronary revascularization	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	
- All-cause death	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	
Other						

- Hospitalization for abdominal pain [†]	0 (0%)	0 (0%)	0 (0%)	1 (2%)	0 (0%)	
- Hospitalization for knee surgery	0 (0%)	0 (0%)	0 (0%)	1 (2%)	1 (2%)	
- Hospitalization for gastric bypass surgery	0 (0%)	1 (2%)	0 (0%)	0 (0%)	0 (0%)	
- Hospitalization for neurologic disorder ^Δ	0 (0%)	1 (2%)	0 (0%)	0 (0%)	1 (2%)	
Potential side effects						
- Hypotension (systolic blood pressure <100 mmHg)		0 (0%)		0 (0%)		
- Vertigo or other symptoms of hypotension		17 (31%)		14 (25%)		
- Headache		9 (16%)		9 (16%)		
- Oedema		9 (16%)		2 (4%)		
- Abdominal pain		5 (9%)		0 (0%)		
- Heartburn/acid reflux		14 (25%)		2 (4%)		
- Nausea		4 (7%)		4 (7%)		
- Urticaria		2 (4%)		0 (0%)		

Intention-to-treat population with maximum likelihood estimation, assuming missings at random. Data are number of events (%). ^ΔDerived from McNemar's test comparing proportion of patients experiencing hospitalization for angina during vericiguat and placebo treatment periods. *No acute coronary syndrome; symptoms attributed to underlying condition (coronary vasospasm). [†]No underlying pathology identified; spontaneous recovery. ^ΔNo neurologic pathology identified; spontaneous recovery. MACE: major adverse cardiac events.

Central Illustration: Design and main findings of the Vericiguat in Vasospastic Angina trial

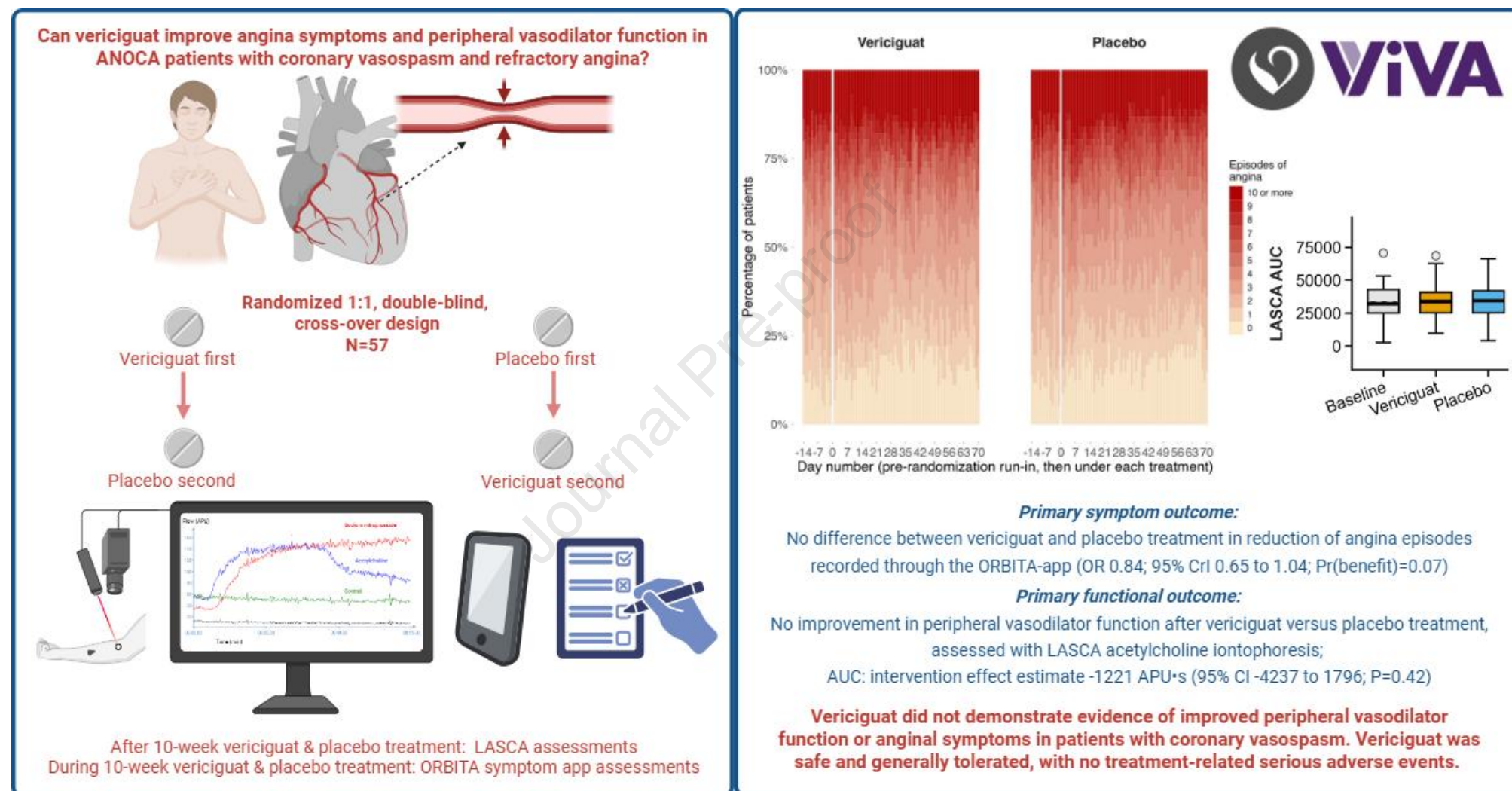


Figure 1: CONSORT 2025 Flow Diagram

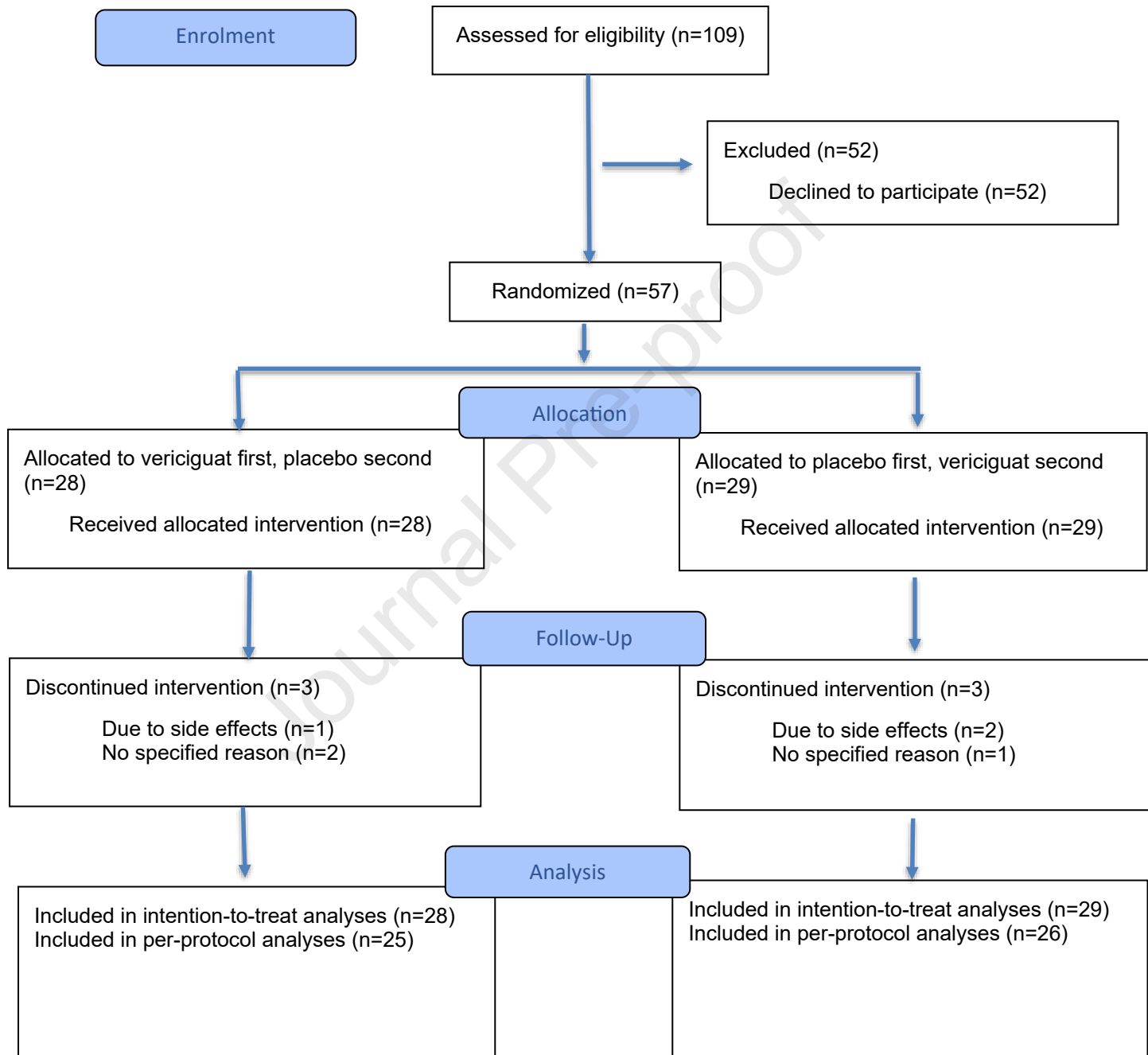
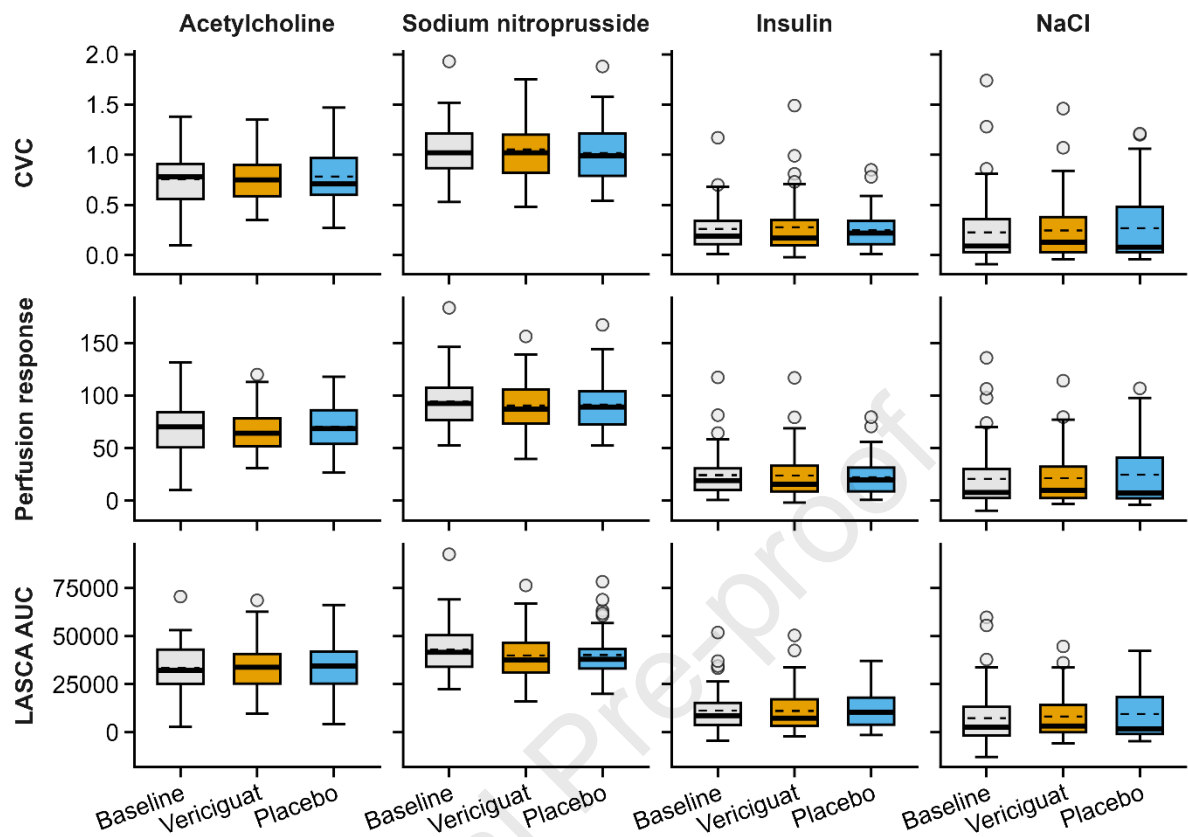


Figure 2: Boxplots of LASCA results, at baseline and after treatment



Intention-to-treat population with maximum likelihood estimation, assuming missings at random. Boxplots show median (thick solid line) and interquartile range (box edges, Q1–Q3), with whiskers extending to $1.5 \times \text{IQR}$; the dashed line indicates the mean; open circles represent outliers. Perfusion response = peak perfusion – baseline perfusion, in APU. APU, arbitrary perfusion unit; AUC, area under the curve in APU•s; CVC, cutaneous vascular conductance, in APU/mmHg; LASCA, laser speckle contrast analysis; NaCl, sodium chloride.

Figure 3: Distribution of daily angina episodes recorded via the ORBITA-app

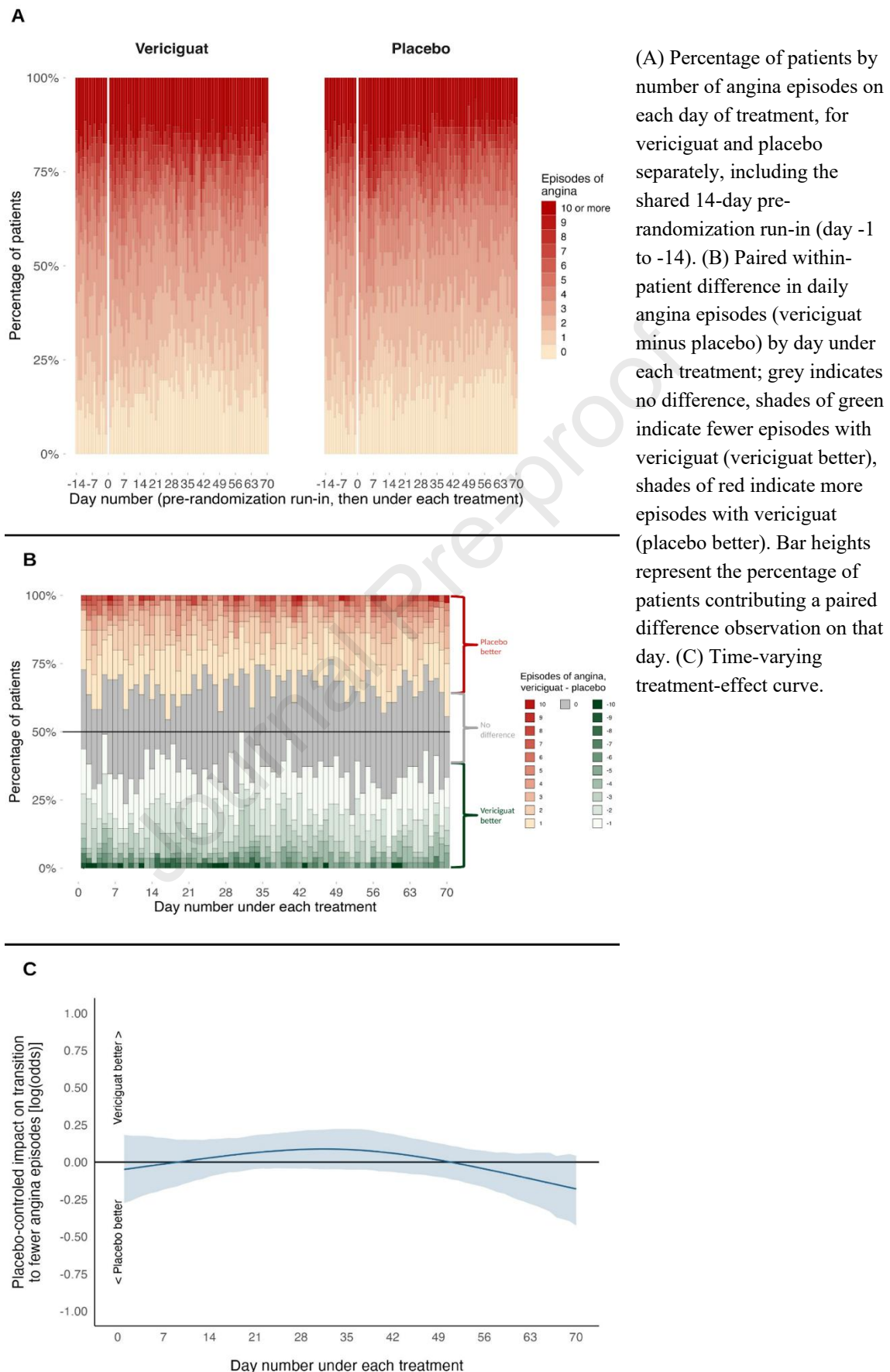
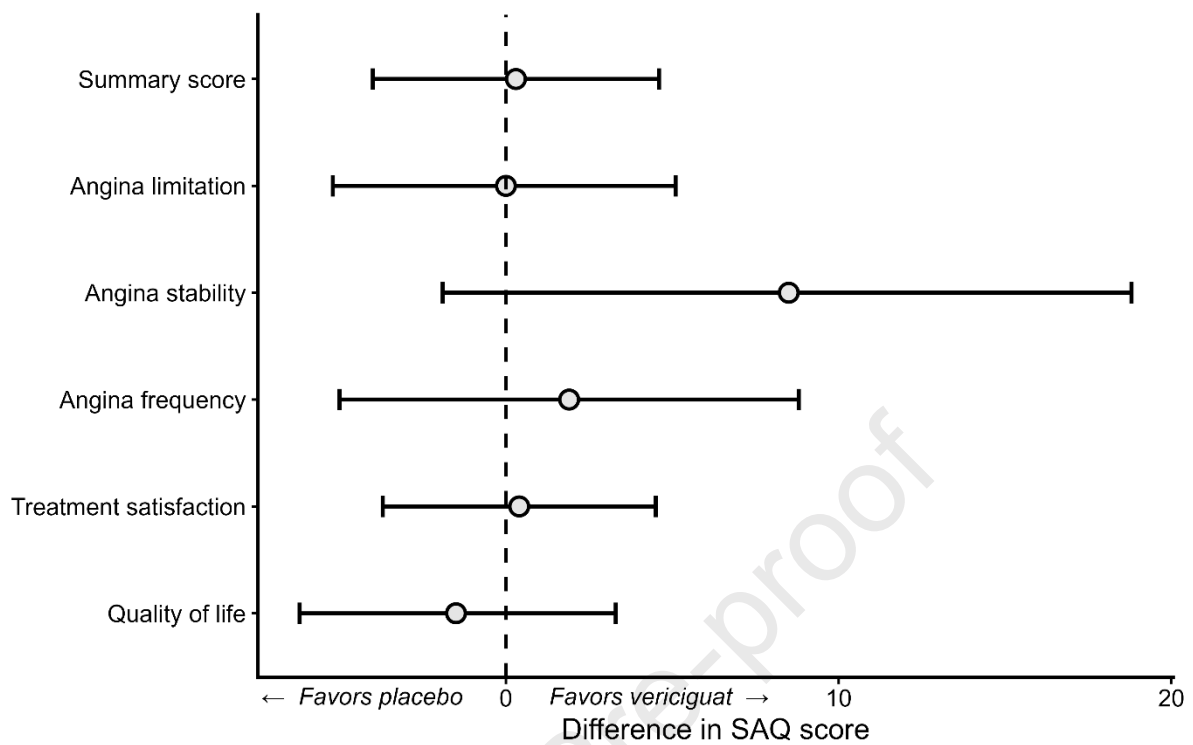
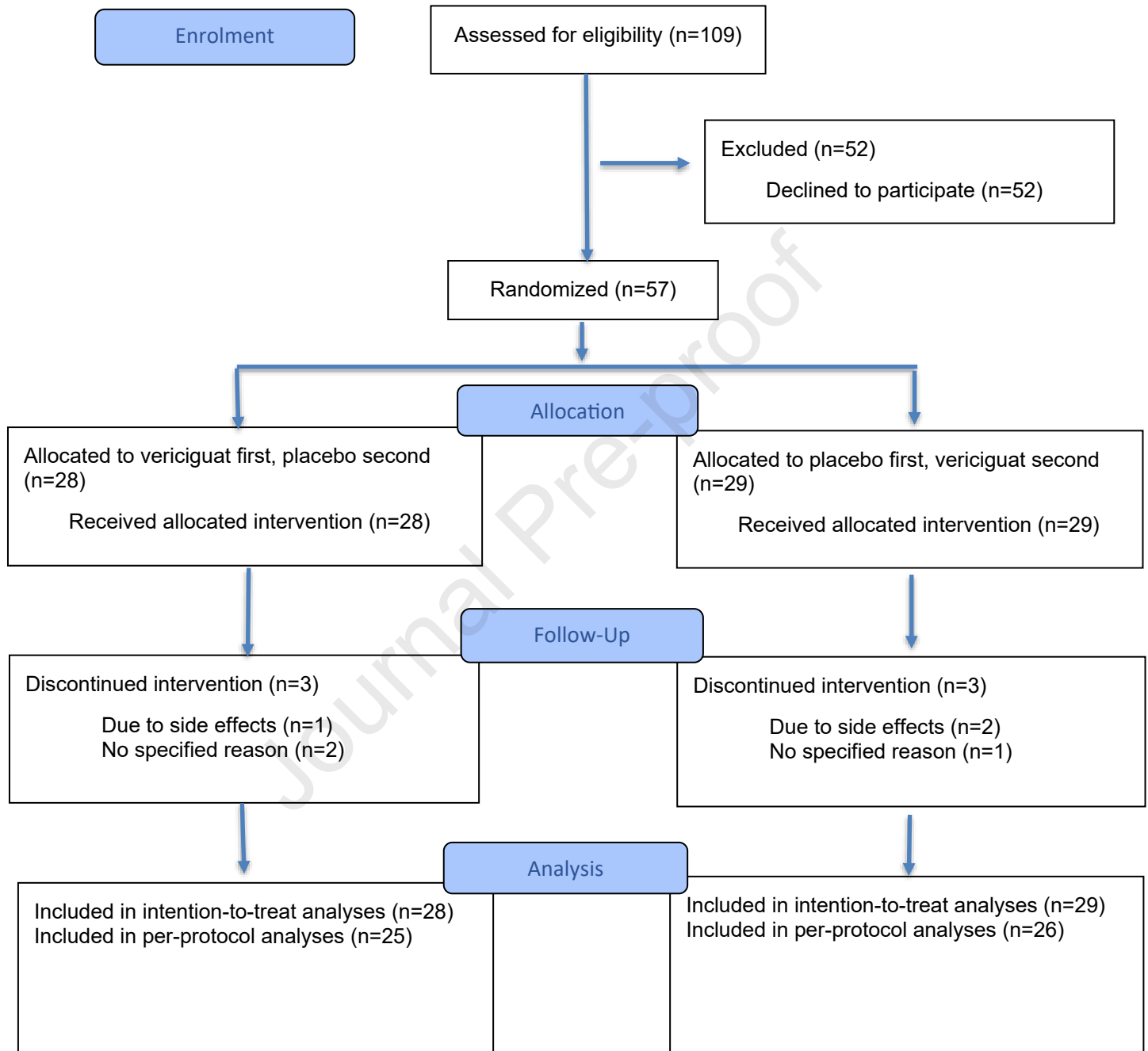
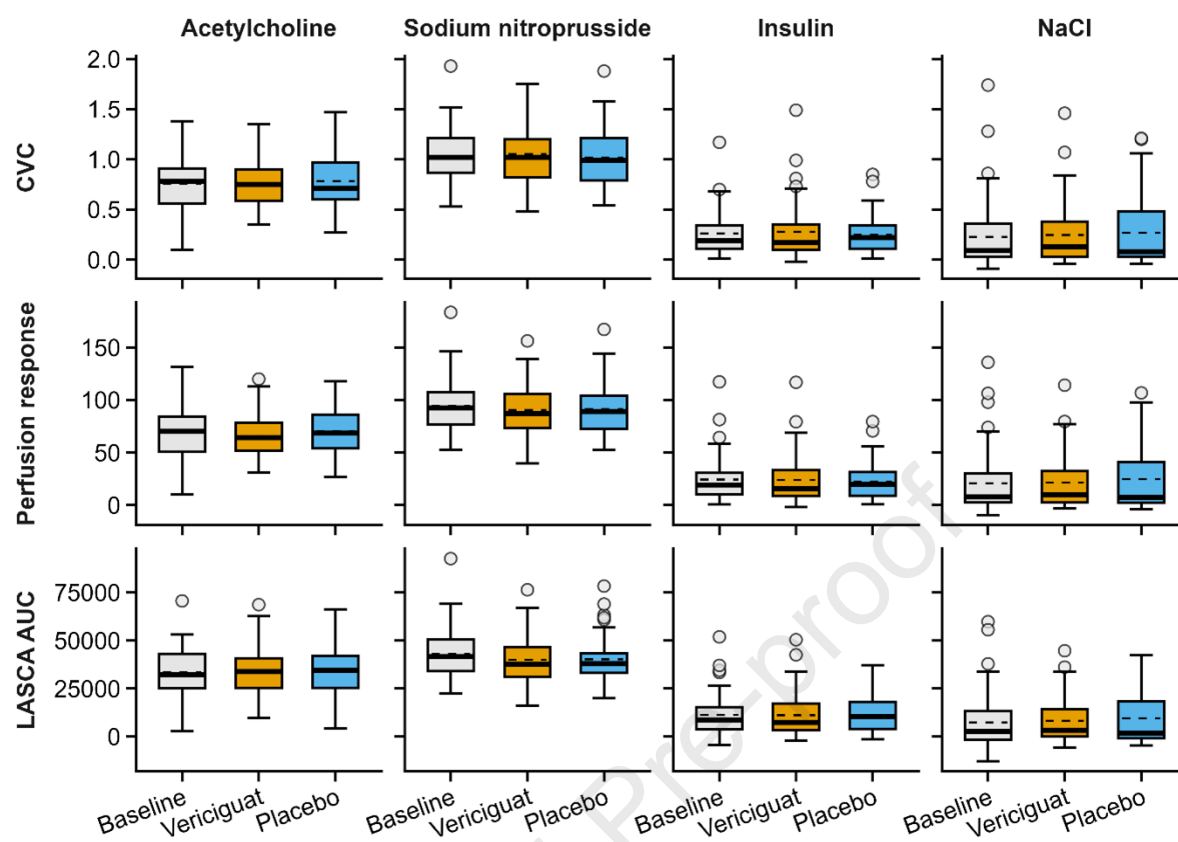


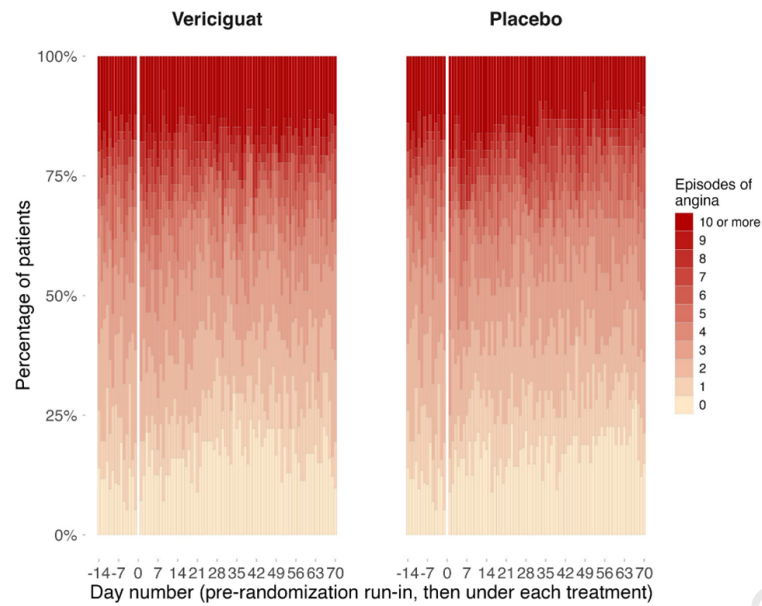
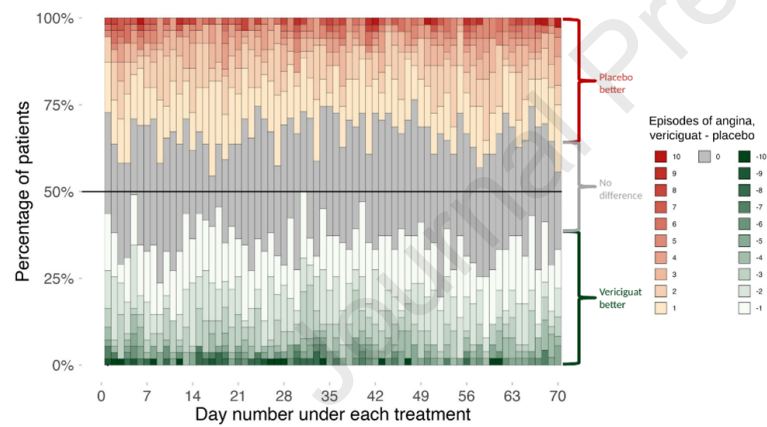
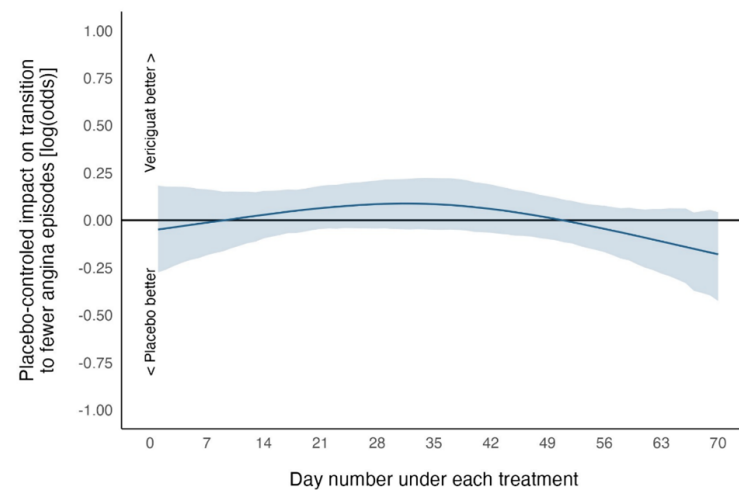
Figure 4: Forest plot of Seattle Angina Questionnaire domains

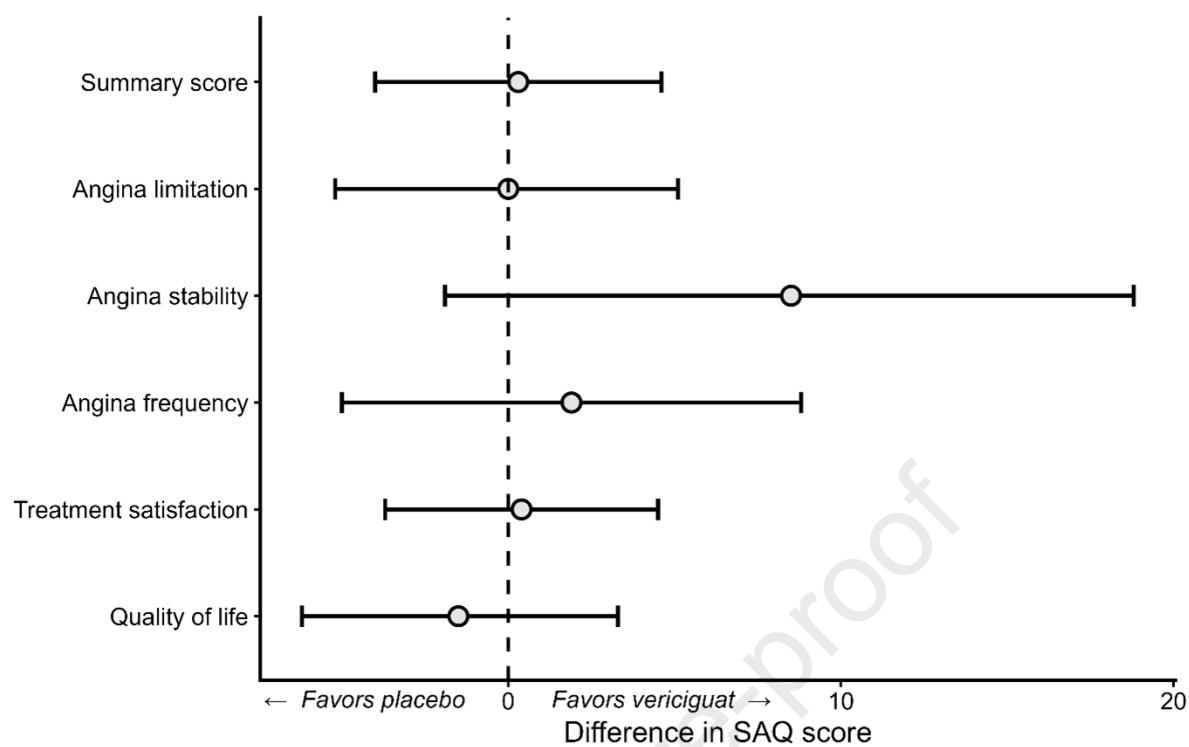


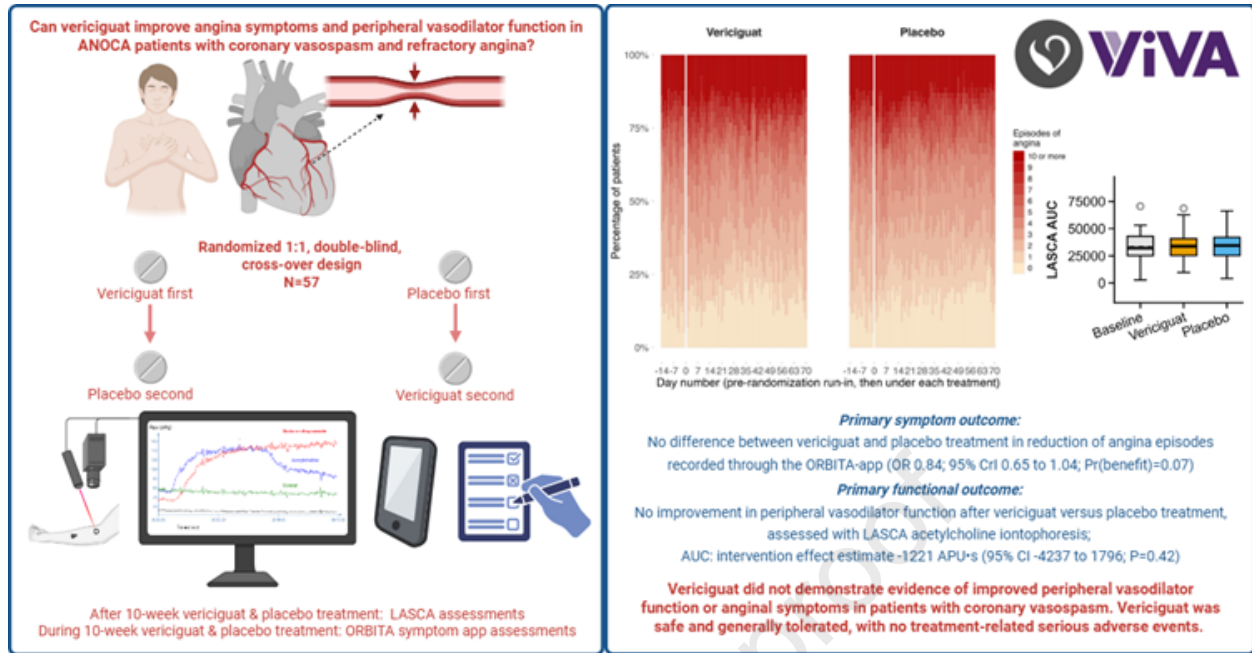
Intention-to-treat population with maximum likelihood estimation, assuming missings at random. Data are estimated intervention effects (open circles) with 95% confidence intervals (horizontal lines) from linear mixed-effects (LME) models, comparing vericiguat versus placebo on each Seattle Angina Questionnaire (SAQ) domain. The dashed vertical line indicates no difference between treatments; values to the right favor vericiguat, values to the left favor placebo.





A**B****C**





Supplementary Data

Supplementary Methods

Symptom application concept and design

Supplementary Figure S1: Screenshots of the ORBITA-app

Supplementary Figure S2: Study flowchart

Laser speckle contrast imaging iontophoresis protocol

Supplementary Figure S3: Laser speckle contrast imaging

Supplementary Results

Supplementary Table S1: Baseline characteristics based on 51-patient per-protocol analysis

Supplementary Table S2: Outcomes after vericiguat versus placebo treatment, based on 51-patient per-protocol analysis

Supplementary Table S3: Adverse events throughout the trial, based on 51-patient per-protocol analysis

Supplementary Table S4: iPCQ and iMCQ outcomes

Supplementary Table S5: Treatment effect on functional outcomes and SAQSS by coronary spasm type, based on 57-patient intention-to-treat analysis

Supplementary Table S6: Bayesian ORBITA-app angina symptom outcomes based on 51-patient per-protocol analysis

Supplementary Table S7: Bayesian ORBITA-app angina symptom outcomes based on 57-patient intention-to-treat analysis

Supplementary Table S8: Missing outcome data per treatment sequence

Supplementary Table S9: Treatment effect on functional outcomes and SAQSS by sex, based on 57-patient intention-to-treat analysis

Supplementary Table S10: Sensitivity analysis of treatment effects based on 55-patient intention-to-treat analysis, excluding those with inconclusive spasm provocation test results

Supplementary Table S11: Sensitivity analysis of treatment effects using period-1 data only, analyzed as a parallel-group comparison

Supplementary Table S12: Sensitivity analysis of treatment effects based on 48-patient intention-to-treat analysis, excluding those with changes in antianginal medication during enrolment

Supplementary Table S13: Sensitivity analyses of treatment effect in terms of ORBITA-app daily angina episodes

Supplementary Figure S4: Regression model and coefficients for symptom app model based on 57-patient intention-to-treat analysis

Supplementary Figure S5: Coefficient density plots: symptom app model based on 57-patient intention-to-treat analysis

Supplementary Figure S6: Chain plot of MCMC draws for symptom app model based on 57-patient intention-to-treat analysis

Supplementary Figure S7: Regression model and coefficients for symptom app model based on 51-patient per-protocol analysis

Supplementary Figure S8: Coefficient density plots: symptom app model based on 51-patient per-protocol analysis

Supplementary Figure S9: Chain plot of MCMC draws for symptom app model based on 51-patient per-protocol analysis

Supplementary Figure S10: Log-odds ratio for transitioning to fewer angina episodes with vericiguat versus placebo, by day number under each treatment based on 51-patient per-protocol analysis

Supplementary Figure S11: Regression model and coefficients for ORBITA score app model based on 57-patient intention-to-treat analysis

Supplementary Figure S12: Coefficient density plots for ORBITA score app model based on 57-patient intention-to-treat analysis

Supplementary Figure S13: Chain plot of MCMC draws for ORBITA score app model based on 57-patient intention-to-treat analysis

Supplementary Figure S14: Log-odds ratio for transitioning to better ORBITA score with vericiguat versus placebo, by day number under each treatment based on 57-patient intention-to-treat analysis

Supplementary Figure S15: Percentage of patients by ORBITA score on each day of treatment, for vericiguat and placebo separately, including the shared 14-day pre-randomization run-in (day -1 to -14)

Supplementary Figure S16: Regression model and coefficients for ORBITA-app symptom intensity model based on 57-patient intention-to-treat analysis

Supplementary Figure S17: Coefficient density plots for ORBITA-app symptom intensity model based on 57-patient intention-to-treat analysis

Supplementary Figure S18: Chain plot of MCMC draws for ORBITA-app symptom intensity model based on 57-patient intention-to-treat analysis

Supplementary Figure S19: Log-odds ratio for transitioning to better ORBITA-app symptom intensity with vericiguat versus placebo, by day number under each treatment based on 57-patient intention-to-treat analysis

Supplementary Figure S20: Percentage of patients by ORBITA score on each day of treatment, for vericiguat and placebo separately, including the shared 14-day pre-randomization run-in (day -1 to -14)

Supplementary Figure S21: Regression model and coefficients for ORBITA-app symptom frequency model stratified by vasospasm type based on 57-patient intention-to-treat analysis

Supplementary Figure S22: Coefficient density plots for ORBITA-app symptom frequency model stratified by vasospasm type based on 57-patient intention-to-treat analysis

Supplementary Figure S23: Chain plot of MCMC draws for ORBITA-app symptom frequency model stratified by vasospasm type based on 57-patient intention-to-treat analysis

Supplementary Figure S24: Odds ratio for transitioning to better ORBITA-app symptom frequency with vericiguat versus placebo, by day number under each treatment and by vasospasm type, based on 57-patient intention-to-treat analysis

Supplementary Figure S25: Percentage of patients by ORBITA-app symptom frequency on each day of treatment, for vericiguat and placebo separately and stratified by vasospasm type, including the shared 14-day pre-randomization run-in (day -1 to -14)

Supplementary Methods

Symptom application concept and design

The smartphone application used in this trial is similar in concept and design to the one deployed in the ORBITA-2 trial (ORBITA-app).

At initiation of the application, participants were asked to define their angina symptom in their own words (i.e. “pressure in the chest”, “tightness in the chest and throat”, “heaviness in the chest and left arm”). They additionally selected two activities which currently reliably bring on their angina symptoms from a pre-specified list (i.e. walking up a flight of stairs without stopping, walking briskly for 10 minutes). Participants were encouraged to undertake these activities at least once a week.

Throughout the trial, from enrolment to trial exit, participants were asked to record daily:

- Did you have angina? (Yes or no)
- How many episodes did you have? (1-10)
- How severe was the worst episode of angina? (Visual analogue scale of mild, moderate, or severe, generating a score of 0-600)

On a weekly basis, participants were asked if they experienced angina with the two pre-specified angina triggering activities.

At trial enrolment, participants completed a short training module with the ORBITA- app, to ensure they were comfortable with its use. Throughout the trial they had 24/7 access to a blinded member of the trial team for technical support as required. If data was not entered for more than 3 days, a blinded member of the team sent a text reminder. If necessary, a follow up telephone call was also performed.

Supplementary Figure S1: Screenshots of the ORBITA-app

James

Yesterday, Fri 12 Jan:

Did you have tightness in my chest?

Yes No

James

Yesterday, Fri 12 Jan:

Had tightness in my chest [Edit](#)

How many times did you have tightness in my chest?

1 2 3

4 5 6

7 8 9

10 or more

James

Yesterday, Fri 12 Jan:

2 episodes of tightness in my chest [Edit](#)

How bad was the worst episode?

Mild Moderate Severe

OK

James

Yesterday, Fri 12 Jan:

Did you avoid any activity, to avoid getting tightness in my chest?

Yes No

James

During the 7 days from Mon 1 Jan to Sun 7 Jan:

When you **walk briskly for 10 minutes**, do you get tightness in my chest?

Yes No

James

During the 7 days from Mon 1 Jan to Sun 7 Jan:

When you **walk up a hill without stopping**, do you get tightness in my chest?

Yes No

James

Thank you!

You are up to date. Your next entry will be tomorrow.

Once a week you will also be asked whether you get symptoms when you walk briskly for 10 minutes, and when you walk up a hill without stopping.

All data are saved. App can be closed.

James

Help

Use this App every day throughout the trial to report the symptoms you had the day before. Once a week, the App will additionally ask whether you get angina when you walk briskly for 10 minutes and when you walk up a hill without stopping. The App will say when you are up to date with data entry.

What happens if I miss a day?

If you miss a day, for example from lack of internet access, you can enter the data later. Our team will phone you if there is too much delay.

Do I count episodes of angina I get when I walk briskly for 10 minutes or walk up a hill without stopping?

Yes, include such episodes within your daily angina counts. Then at the weekly question about angina during those specific activities, please answer yes. When we process the data, we know not to double-count the answers to the weekly questions as extra episodes.

James

How do I use the mild-moderate-severe slider?

Use the slider to record your personal opinion on the severity of the angina you had on the day being asked about. This is a personal judgement for you to make at the time you are asked. We expect the way you make these judgements may naturally change over the course of the study. We are asking your independent view on how you feel each day.

What happens if I make a mistake?

Make a note of what you intended to enter and then call our helpline.

James

Why can't I see my previous entries?

During the trial we need your independent answers from the separate days and therefore the app does not show previous entries. After you finish the trial, we can show you all your entries.

What if the app isn't working properly?

Please call our helpline.

What should I do if I don't feel well?

If you want to discuss this with us, please call our helpline at any time.

Helpline number

0757551868

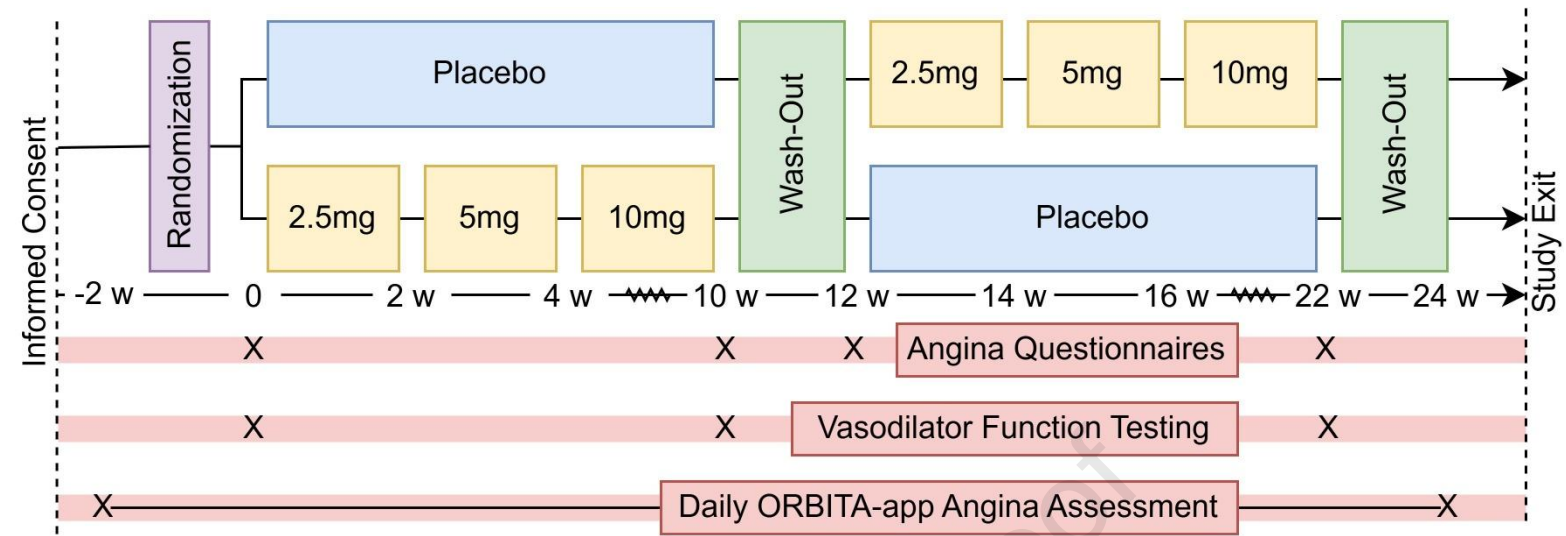
Advanced Tools

The researchers may ask you to click this button

[Advanced Tools](#)

Close Help

Supplementary Figure S2: Study flowchart



*Laser speckle contrast imaging iontophoresis protocol***Acetylcholine Iontophoresis Protocol**

Iontophoresis settings: **20 s at 100 μ A (0.10 mA)**, followed by **40 s rest**, repeated **7 times**.

Time	Procedure
0:00–1:00 min	Baseline recording
1:00–1:20 min	First acetylcholine dose
2:00–2:20 min	Second acetylcholine dose
3:00–3:20 min	Third acetylcholine dose
4:00–4:20 min	Fourth acetylcholine dose
5:00–5:20 min	Fifth acetylcholine dose
6:00–6:20 min	Sixth acetylcholine dose
7:00–7:20 min	Seventh acetylcholine dose
14:30 min	Stop recording; save and back up the data

Sodium Nitroprusside (SNP) Iontophoresis Protocol

Iontophoresis settings: **20 s at 200 μ A (0.20 mA)**, followed by **70 s rest**, repeated **9 times**.

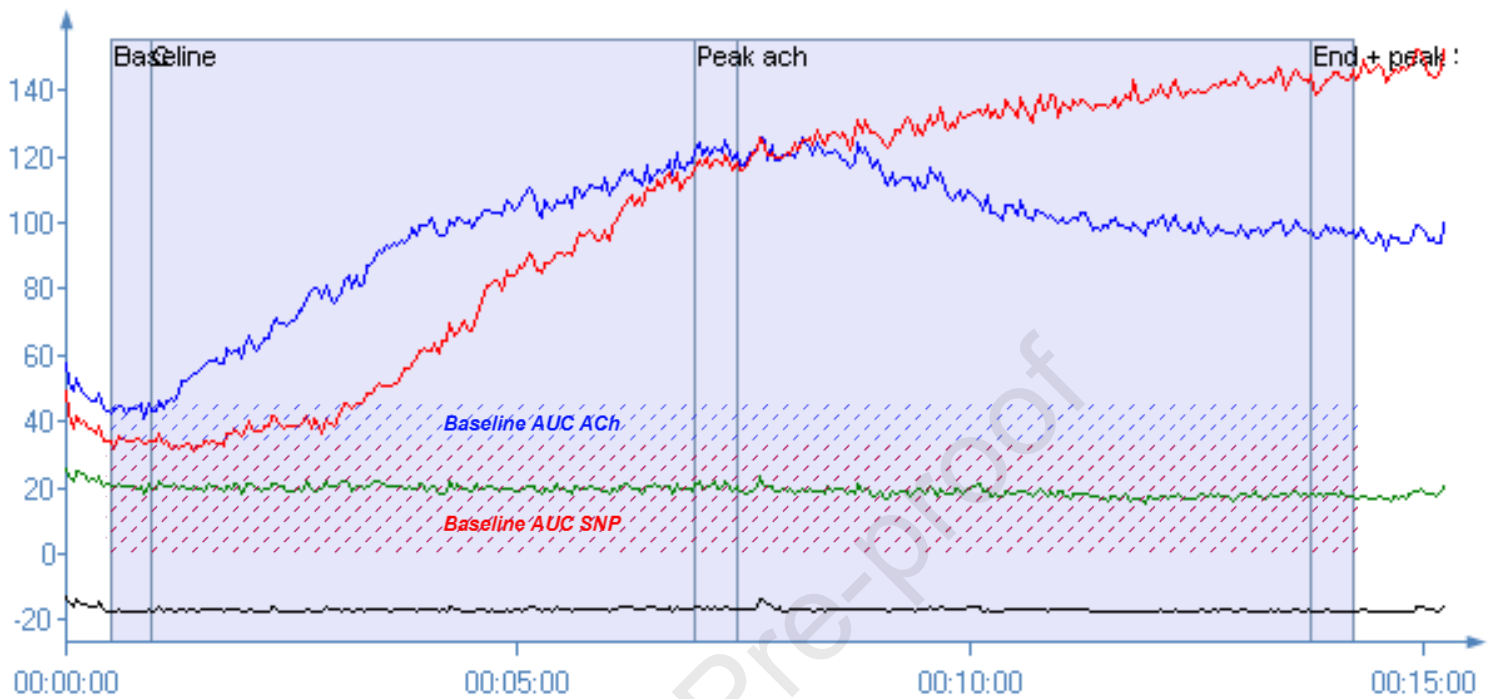
Time	Procedure
0:00–1:00 min	Baseline recording
1:00–1:20 min	First SNP dose
2:30–2:50 min	Second SNP dose
4:00–4:20 min	Third SNP dose
5:30–5:50 min	Fourth SNP dose
7:00–7:20 min	Fifth SNP dose
8:30–8:50 min	Sixth SNP dose
10:00–10:20 min	Seventh SNP dose
11:30–11:50 min	Eighth SNP dose
13:00–13:20 min	Ninth SNP dose
14:30 min	Stop recording; save and back up the data

Insulin/Saline (NaCl) Iontophoresis Protocol

Iontophoresis settings: **continuous delivery for 18 min at 20 μ A (0.02 mA)**.

Time	Procedure
0:00–1:00 min	Baseline recording
1:00–19:00 min	Continuous insulin/NaCl iontophoresis (20 μ A)
19:00 min	Stop recording; save and back up the data

Supplementary Figure S3: Laser speckle contrast imaging



Representative PIMSoft output from iontophoresis measurements following administration of acetylcholine (ACh, blue) and sodium nitroprusside (SNP, red). The analysis identifies the baseline, peak response, and end of the recording. Total area under the curve (AUC) was calculated automatically for each vasodilatory response and corrected for baseline perfusion by subtracting the baseline AUC (striped areas), yielding the baseline-corrected AUC used for subsequent analyses, in APU•s.

Supplementary Results

Supplementary Table S1: Baseline characteristics based on 51-patient per-protocol analysis

	Overall (n=51)
Randomization allocation	
Vericiguat first, placebo second	25 (49%)
Placebo first, vericiguat second	26 (51%)
Age, years	55 [51, 63]
Sex	
Female	38 (75%)
Male	13 (25%)
BMI, kg/m ²	27 [25, 31]
Medical history	
Hypertension	17 (33%)
Hypercholesterolemia	21 (41%)
Diabetes	
Insulin dependent	2 (4%)
Insulin independent	2 (4%)
Smoking status	
Current	2 (4%)
Former	19 (37%)
Family history of CVD	36 (71%)
Previous myocardial infarction	13 (25%)
Previous percutaneous coronary intervention	12 (24%)
Female specific risk factors	
Pregnancy	34 (89%)
Pregnancy-related hypertension	8 (24%)
Pregnancy-related diabetes	1 (3%)
Pre-eclampsia/HELLP	4 (12%)
History of miscarriages	8 (24%)
Hormonal therapy >60 years	0 (0%)
Post-menopausal	21 (55%)
Early menopause <40 years	0 (0%)
History of hysterectomy	6 (16%)
History of ovariectomy	1 (3%)
History of polycystic ovary syndrome	0 (0%)
Acetylcholine provocation test result	

Epicardial spasm	22 (43%)
Microvascular spasm	22 (43%)
Combined epicardial + microvascular spasm	5 (10%)
Inconclusive*	2 (4%)
Acetylcholine rechallenge performed	24 (47%)
Result:	
No spasm	5 (21%)
Microvascular spasm	13 (54%)
Epicardial spasm	3 (13%)
Inconclusive*	3 (13%)
Conclusion:	
Nitrate unresponsive	7 (29%)
Nitrate responsive	4 (17%)
Partially nitrate responsive [†]	13 (54%)
Vasodilator testing result	
CFR	3.5 [2.9, 4.9]
HMR, mmHg/cm/s	1.3 [1.2, 1.5]
IMR	14.0 [8.0, 19.5]
Concomitant CMD ^Δ	0 (0%)
Baseline medication use	
Non-DHP CCB	33 (65%)
DHP CCB	25 (49%)
Betablocker	10 (20%)
Long-acting nitrate	37 (73%)
Short-acting nitrate	43 (84%)
Nicorandil	25 (49%)
ACEi/ARB	15 (29%)
Lipid lowering drug	35 (69%)
Antiplatelets	28 (55%)
Proton pump inhibitors	28 (55%)

*Inconclusive acetylcholine provocation test/rechallenge: recognizable symptoms, but no ischemic electrocardiographic changes or vasoconstriction. [†]Partially nitrate responsive: improvement of vasoconstriction and/or ischemic electrocardiographic changes, but no relief of anginal symptoms. ^Δ Based on CFR <2.5. ACEi, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; BMI, body mass index; CCB, calcium channel blocker; CFR, coronary flow reserve; CMD, coronary microvascular dysfunction; CVD, cardiovascular disease; DHP, dihydropyridine; HELLP, hemolysis, elevated liver enzymes, and low platelet count; HMR, hyperemic microvascular resistance; IMR, index of microvascular resistance.

Supplementary Table S2: Outcomes after vericiguat versus placebo treatment, based on 51-patient per-protocol analysis

	Baseline	After 10-week vericiguat treatment	Change from baseline	After 10-week placebo treatment	Change from baseline	Intervention effect*		
						Estimate	95% CI	P-value
Laboratory results								
- Hemoglobin, mmol/L	8.9 [8.4, 9.4]	8.8 [8.6, 9.2]	-0.1 [-0.4, 0.1]	8.8 [8.4, 9.3]	-0.1 [-0.4, 0.2]	0.0	-0.1 to 0.2	0.61
- CRP, mg/L	1.5 [0.7, 2.7]	1.6 [0.7, 2.7]	0.0 [-0.6, 0.3]	1.2 [0.7, 2.8]	0.0 [-0.3, 0.5]	0.2	-1.1 to 1.5	0.75
- Creatinine, μ mol/L	71 [66, 82]	72 [66, 83]	2 [-3, 5]	72 [65, 79]	-1 [-6, 4]	1.3	-0.9 to 3.4	0.23
- TroponinT hs, ng/L	6 [4, 9]	6 [4, 8]	0 [-1, 1]	6 [4, 8]	0 [-1, 0]	0.2	-0.4 to 0.7	0.53
- NT-proBNP, ng/L	69 [37, 121]	76 [44, 120]	2 [-25, 30]	70 [42, 115]	0 [-17, 18]	-8.5	-34.4 to 17.3	0.51
Blood pressure								
- Systolic, mmHg	124 [112, 135]	110 [105, 120]	-11 [-18, -3]	120 [110, 130]	-4 [-10, 0]	-5.6	-9.4 to -1.9	<0.01
- Diastolic, mmHg	76 [71, 80]	70 [70, 80]	-3 [-8, 5]	75 [70, 80]	0 [-5, 5]	-2.2	-3.9 to -0.4	0.02
- MAP, mmHg	92 [86, 100]	85 [80, 92]	-5 [-10, 1]	90 [83, 96]	-2 [-6, 2]	-3.3	-5.5 to -1.2	<0.01
LASCA assessment								
- <i>ACH</i>								
Baseline perfusion, APU	37.1 [30.7, 42.6]	34.2 [30.2, 40.9]	-0.6 [-4.9, 4.5]	36.7 [32.4, 42.9]	0.8 [-3.9, 5.9]	-1.2	-3.7 to 1.3	0.35
Peak perfusion, APU	102.7 [88.0, 124.1]	100.0 [87.0, 113.3]	-5.9 [-16.0, 9.7]	102.5 [91.7, 127.0]	5.1 [-8.8, 13.3]	-6.3	-11.6 to -1.0	0.02
Perfusion response, APU	69.2 [47.6, 82.6]	63.8 [51.5, 78.0]	-2.6 [-13.0, 7.0]	67.6 [53.6, 86.0]	0.5 [-10.8, 13.0]	-5.2	-9.7 to -0.7	0.03
CVC, APU/mmHg [†]	0.8 [0.6, 0.9]	0.7 [0.6, 0.9]	0.0 [-0.1, 0.1]	0.7 [0.6, 1.0]	0.0 [-0.1, 0.2]	0.0	-0.1 to 0.0	0.22
AUC, APU•s ^Δ	31129 [24378, 43370]	33435 [24976, 41204]	-3298 [-10125, 6593]	33896 [25080, 41445]	-915 [-5796, 5239]	-1201	-4250 to 1848	0.43
- <i>SNP</i>								
Baseline perfusion, APU ^Δ	31.3 [29.6, 38.2]	30.1 [26.3, 35.2]	-0.8 [-5.6, 4.7]	32.0 [27.7, 38.5]	-0.3 [-4.9, 5.4]	-2.6	-6.6 to 1.4	0.20
Peak perfusion, APU ^Δ	122.5 [106.8, 144.0]	118.0 [106.9, 136.3]	-4.1 [-18.1, 10.4]	118.0 [106.9, 136.3]	-1.3 [-20.3, 14.8]	-2.9	-10.6 to 4.7	0.44
Perfusion response, APU ^Δ	93.1 [73.1, 107.3]	87.1 [72.0, 105.6]	-4.7 [-17.3, 10.4]	88.8 [72.1, 103.3]	-1.3 [-16.2, 12.2]	-0.3	-6.1 to 5.5	0.91
CVC, APU/mmHg ^Δ	1.0 [0.8, 1.2]	1.0 [0.8, 1.2]	0.0 [-0.1, 0.2]	1.0 [0.8, 1.2]	0.0 [-0.2, 0.2]	0.0	0.0 to 0.1	0.32
AUC, APU•s ^Δ	41186 [32965, 50409]	37529 [30888, 46297]	-3695 [-6781, 2586]	37426 [32900, 43497]	-2578 [-8578, 1953]	-661	-3459 to 2137	0.64
- <i>Insulin</i>								
Baseline perfusion, APU	30.9 [26.3, 36.4]	28.3 [25.2, 34.2]	-1.7 [-5.3, 2.2]	30.5 [26.5, 35.6]	0.2 [-4.6, 5.0]	-1.5	-3.4 to 0.4	0.12
Peak perfusion, APU	50.2 [40.7, 66.2]	46.6 [39.0, 61.0]	-0.9 [-15.1, 10.4]	50.8 [40.7, 61.6]	0.0 [-10.3, 7.3]	-0.2	-5.1 to 4.8	0.95
Perfusion response, APU	19.1 [10.2, 33.7]	14.6 [8.5, 34.0]	-1.4 [-10.1, 5.8]	19.4 [9.5, 31.4]	-1.5 [-7.9, 4.3]	1.3	-3.5 to 6.2	0.59
CVC, APU/mmHg	0.2 [0.1, 0.4]	0.2 [0.1, 0.4]	0.0 [-0.1, 0.1]	0.2 [0.1, 0.4]	0.0 [-0.1, 0.1]	0.0	0.0 to 0.1	0.41
AUC, APU•s	8986 [3804, 19264]	7317 [3376, 17192]	-931 [-6307, 4113]	10432 [4244, 18299]	156 [-4645, 4597]	-224	-3113 to 2664	0.88
- <i>NaCl</i>								
Baseline perfusion, APU	37.9 [31.0, 42.9]	32.5 [27.4, 40.8]	-3.2 [-7.9, 2.3]	32.4 [28.7, 38.9]	-2.1 [-6.9, 0.4]	-0.4	-2.7 to 1.9	0.70
Peak perfusion, APU	50.8 [37.5, 61.8]	48.5 [40.4, 62.2]	1.7 [-17.3, 8.3]	43.8 [38.2, 75.7]	-3.1 [-12.1, 6.6]	-4.2	-10.4 to 2.1	0.18
Perfusion response, APU	7.7 [2.0, 33.8]	8.9 [2.3, 31.8]	0.5 [-6.8, 9.9]	7.0 [1.9, 40.5]	-0.6 [-6.7, 7.1]	-3.8	-9.9 to 2.3	0.22
CVC, APU/mmHg	0.1 [0.0, 0.4]	0.1 [0.0, 0.4]	0.0 [-0.1, 0.1]	0.1 [0.0, 0.5]	0.0 [-0.1, 0.1]	0.0	-0.1 to 0.0	0.32
AUC, APU•s	2512 [-2049, 15400]	3223 [-158, 13589]	1036 [-3858, 7135]	1828 [-1050, 17522]	-181 [-4334, 5617]	-1658	-4639 to 1324	0.27
EndoPAT assessment								
- RHI	1.7 [1.4, 2.1]	1.9 [1.7, 2.3]	0.2 [-0.1, 0.5]	1.8 [1.6, 2.3]	0.0 [-0.3, 0.4]	0.1	-0.1 to 0.3	0.17

- Endothelial dysfunction based on RHI <1.67	23 (45%)	14 (28%)	-9 patients (-17%)	19 (37%)	-4 patients (-8%)	OR 0.5	0.1 to 1.5	0.19
Seattle Angina Questionnaire								
- Summary score	39 [30, 47]	40 [30, 52]	4 [-4, 8]	41 [28, 49]	1 [-4, 8]	0.3	-4.0 to 4.6	0.90
- Angina limitation	46 [34, 61]	44 [32, 62]	0 [-9, 7]	43 [33, 60]	0 [-8, 8]	0.0	-5.2 to 5.1	1.00
- Angina stability	50 [25, 50]	50 [50, 75]	0 [0, 50]	50 [25, 50]	0 [-13, 25]	8.5	-1.9 to 18.8	0.11
- Angina frequency	30 [20, 50]	40 [15, 60]	0 [0, 20]	30 [20, 50]	0 [0, 10]	1.9	-5.0 to 8.8	0.58
- Treatment satisfaction	81 [63, 91]	75 [63, 88]	0 [-19, 6]	75 [59, 88]	-6 [-16, 0]	0.4	-3.7 to 4.5	0.85
- Quality of life	33 [25, 42]	33 [25, 50]	0 [-8, 8]	33 [25, 54]	0 [-8, 8]	-1.5	-6.2 to 3.3	0.54
EQ-5D-5L questionnaire								
- VAS	55 [40, 70]	50 [39, 66]	-3 [-18, 5]	50 [38, 60]	-2 [-14, 4]	0.6	-4.9 to 6.1	0.83
- Index score	0.7 [0.4, 0.8]	0.6 [0.4, 0.8]	0.0 [-0.2, 0.1]	0.7 [0.5, 0.8]	0.0 [-0.1, 0.1]	0.0	-0.1 to 0.0	0.33
Rose Dyspnea Scale								
- 0	3 [2, 4]	3 [2, 4]	0 [-1, 0]	3 [2, 4]	0 [-1, 0]	-0.1	-0.4 to 0.2	0.70
- 1	2 (4%)	3 (6%)		5 (10%)				
- 2	6 (12%)	8 (16%)		6 (12%)				
- 3	10 (20%)	14 (28%)		11 (22%)				
- 4	17 (33%)	13 (26%)		14 (28%)				
- 4	16 (31%)	13 (26%)		15 (29%)				

*Values are derived from a mixed linear regression model and depicted as mean (95% CI) for continuous outcomes, and from a logistic mixed-effects model, depicted as odds ratio (95% CI) for binary outcomes, adjusting for period, randomization sequence, baseline value, and participant. ^Δ Value is missing for n=5 patients. [†] Value is missing for n=1 patient. AUC, area under the curve; ACh, acetylcholine; APU, arbitrary perfusion units; CI, confidence interval; CRP, C-reactive protein; CVC, cutaneous vascular conductance; LASCA, laser speckle contrast analysis; MAP, mean arterial pressure; NT-proBNP, N-terminal pro-B-type natriuretic peptide; OR, odds ratio; RHI, reactive hyperemia index; SAQ, Seattle Angina Questionnaire, higher scores represent less angina symptoms; SNP, sodium nitroprusside; VAS, visual analogue scale, of EQ-5D validated quality of life with higher scores indicating better quality of life. Rose Dyspnea Scale, scores range from 0 to 4, with higher scores indicating dyspnea with milder activities.

Supplementary Table S3: Adverse events throughout the trial, based on 51-patient per-protocol analysis

	Pre-treatment	During vericiguat treatment	During washout after vericiguat	During placebo treatment	During washout after placebo
MACE					
- Hospitalization for angina*	2 (4%)	3 (6%)	2 (4%)	7 (14%)	1 (2%)
- Myocardial infarction	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
- Unplanned coronary revascularization	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
- All-cause death	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Other					
- Hospitalization for abdominal pain [†]	0 (0%)	0 (0%)	0 (0%)	1 (2%)	0 (0%)
- Hospitalization for knee surgery	0 (0%)	0 (0%)	0 (0%)	1 (2%)	1 (2%)
- Hospitalization for gastric bypass surgery	0 (0%)	1 (2%)	0 (0%)	0 (0%)	0 (0%)
- Hospitalization for neurologic disorder ^Δ	0 (0%)	1 (2%)	0 (0%)	0 (0%)	1 (2%)
Potential side effects					
- Hypotension (systolic blood pressure <100 mmHg)		0 (0%) 14 (27%) 8 (16%)		0 (0%) 12 (24%) 8 (16%)	
- Vertigo or other symptoms of hypotension		9 (18%)		2 (4%)	
- Headache		4 (8%)		0 (0%)	
- Oedema		14 (27%)		1 (2%)	
- Abdominal pain		3 (6%)		3 (6%)	
- Heartburn/acid reflux		2 (4%)		0 (0%)	
- Nausea					
- Urticaria					

Data are number of events (%). *No acute coronary syndrome; symptoms attributed to underlying condition (coronary vasospasm). [†] No underlying pathology identified; spontaneous recovery. ^ΔNo neurologic pathology identified; spontaneous recovery. MACE: major adverse cardiac events.

Supplementary Table S4: iPCQ and iMCQ outcomes

Socioeconomic and functional characteristics	At baseline (n=57)	After vericiguat treatment (n=53)	After placebo treatment (n=53)
Educational level			
- None	0 (0%)		
- Primary/lower vocational education	18 (32%)		
- Lower secondary education	14 (25%)		
- Intermediate vocational education	38 (67%)		
- Upper secondary education	8 (14%)		
- Higher professional education	30 (53%)		
- University education	6 (11%)		
Daily functioning			
- School/student	0 (0%)		
- Paid employment	19 (33%)		
- Self-employed	6 (11%)		
- Homemaker	2 (4%)		
- Unemployed	1 (2%)		
- Disabled/unable to work	24 (42%)		
- Retired/early retired	5 (9%)		
Absence from work (paid employed or self-employed) because of illness in the past 4 weeks	22 (88%)	25 (100%)	24 (96%)
Days worked per week	4 [3, 5]	3 [3, 5]	4 [3, 5]
Hours worked per week	29 [20, 36]	28 [24, 36]	29 [24, 36]
Healthcare resource	At baseline (n=57)	After vericiguat treatment (n=53)	After placebo treatment (n=53)
General practitioner	32 (56%)	30 (57%)	29 (55%)
Social worker	8 (14%)	1 (2%)	3 (6%)
Physiotherapy	22 (39%)	21 (40%)	16 (30%)
Occupational therapy	11 (19%)	5 (9%)	5 (9%)
Speech therapy	1 (2%)	1 (2%)	0 (0%)
Dietitian	9 (16%)	6 (11%)	8 (15%)
Homeopath	2 (4%)	3 (6%)	1 (2%)
Psychologist/psychiatrist	13 (23%)	9 (17%)	7 (13%)
Occupational physician	15 (26%)	11 (21%)	14 (26%)
Home care	5 (9%)	7 (13%)	5 (9%)
Emergency department	12 (21%)	12 (23%)	18 (34%)

Ambulance	10 (18%)	9 (17%)	15 (28%)
Outpatient clinic	43 (75%)	33 (62%)	34 (64%)

Values are median [Q1, Q3] or n (%). iPCQ and iMCQ questionnaire outcomes at baseline and after vericiguat or placebo treatment, reflecting healthcare consumption, productivity, and daily functioning. iPCQ questionnaire reflects functional characteristics during the preceding 4 weeks; iMCQ questionnaire reflects healthcare consumption during the preceding 3 months. Educational level and daily functioning did not change over time and were therefore not considered informative to report after the treatment periods.

Supplementary Table S5: Treatment effect on functional outcomes and SAQSS by coronary spasm type, based on 57-patient intention-to-treat analysis

Outcome	Interaction χ^2 , df=4	Interaction p-value	Type of coronary spasm	Estimate	95% CI	P-value
LASCA ACh						
- Perfusion response	3.51	0.48	Epicardial	-7.74	-14.31 to -1.17	0.02
			Microvascular	-1.71	-8.39 to 4.97	0.61
			Combined	-11.15	-24.76 to 2.46	0.11
- CVC	2.54	0.64	Epicardial	-0.06	-0.14 to 0.02	0.16
			Microvascular	0.00	-0.09 to 0.08	0.91
			Combined	-0.08	-0.25 to 0.09	0.33
- AUC	2.20	0.70	Epicardial	-1692.68	-6293.40 to 2908.03	0.46
			Microvascular	293.21	-4344.89 to 4931.32	0.90
			Combined	-4857.72	-14056.19 to 4340.75	0.29
LASCA SNP						
- Perfusion response	9.28	0.05	Epicardial	-2.94	-11.60 to 5.73	0.50
			Microvascular	4.94	-3.61 to 13.49	0.25
			Combined	-7.20	-24.55 to 10.16	0.41
- CVC	4.60	0.33	Epicardial	0.02	-0.08 to 0.13	0.66
			Microvascular	0.07	-0.04 to 0.18	0.20
			Combined	-0.02	-0.24 to 0.20	0.85
- AUC	7.23	0.12	Epicardial	161.80	-4160.36 to 4483.95	0.94
			Microvascular	-16.70	-4281.76 to 4248.37	0.99
			Combined	-6725.76	-15379.81 to 1928.29	0.12
LASCA Insulin						
- Perfusion response	2.04	0.73	Epicardial	2.42	-4.83 to 9.66	0.51
			Microvascular	0.56	-6.64 to 7.77	0.88
			Combined	-4.85	-19.79 to 10.09	0.52
- CVC	1.45	0.83	Epicardial	0.04	-0.05 to 0.13	0.38
			Microvascular	0.01	-0.08 to 0.10	0.78
			Combined	-0.04	-0.23 to 0.14	0.64
- AUC	2.56	0.63	Epicardial	799.39	-3522.36 to 5121.14	0.71
			Microvascular	-769.18	-5060.77 to 3522.41	0.72
			Combined	-3376.77	-12239.27 to 5485.72	0.45
LASCA NaCl						
- Perfusion response	4.22	0.38	Epicardial	-9.78	-19.24 to -0.33	0.04
			Microvascular	1.77	-7.64 to 11.18	0.71
			Combined	3.80	-15.75 to 23.34	0.70

- CVC	4.97	0.29	Epicardial	-0.10	-0.21 to 0.00	0.05
			Microvascular	0.03	-0.08 to 0.13	0.58
			Combined	0.07	-0.15 to 0.29	0.53
- AUC	6.61	0.16	Epicardial	-4695.84	-9264.07 to -127.61	0.04
			Microvascular	1278.93	-3266.13 to 5824.00	0.58
			Combined	1998.28	-7436.84 to 11433.39	0.67
RHI	1.71	0.79	Epicardial	0.10	-0.16 to 0.35	0.45
			Microvascular	0.01	-0.24 to 0.27	0.91
			Combined	0.17	-0.36 to 0.71	0.52
SAQSS	7.10	0.13	Epicardial	-5.59	-12.00 to 0.81	0.09
			Microvascular	3.87	-2.50 to 10.23	0.23
			Combined	2.84	-10.32 to 16.00	0.67

Linear mixed-effects models (LME) were fit for each outcome, including fixed effects for treatment, study period, randomization sequence group, baseline value of the outcome, and the treatment $\times$ type of coronary spasm interaction, with a random intercept for participant to account for the crossover design and within-patient correlation across periods. Models were estimated using maximum likelihood. All participants with available data at each assessment were included under a missing-at-random assumption, consistent with the intention-to-treat analysis population; no imputation was performed. For each outcome, the treatment $\times$ spasm-type interaction was tested via likelihood ratio test comparing the model with and without the interaction term and spasm type as a main effect (χ^2 , 4 degrees of freedom, reflecting the addition of the 3-level spasm-type factor [epicardial, microvascular, combined] and its interaction with the 2-level treatment factor). Estimated marginal means and treatment contrasts (vericiguat – placebo) within each spasm-type subgroup were derived from the interaction model, with Kenward-Roger degrees-of-freedom approximation; subgroup contrasts are presented for descriptive purposes and should be interpreted as exploratory, particularly for outcomes with a non-significant omnibus interaction test. AUC, area under the curve; ACh, acetylcholine; APU, arbitrary perfusion units; CI, confidence interval; CVC, cutaneous vascular conductance; LASCA, laser speckle contrast analysis; RHI, reactive hyperemia index; SAQSS, Seattle Angina Questionnaire summary score, higher scores represent less angina symptoms; SNP, sodium nitroprusside.

Supplementary Table S6: Bayesian ORBITA-app angina symptom outcomes based on 51-patient per-protocol analysis

Outcome		Effect size (95% CrI)	Posterior probability of greater benefit with vericiguat versus placebo
Primary outcome			
Final day transition OR for symptom reduction with vericiguat versus placebo		0.85 (0.67 to 1.08)	0.09
Supplementary derived outcomes from primary symptom model			
Expected number of angina episodes on final day under each treatment	Vericiguat	2.4 (1.8 to 3.0)	
	Placebo	2.2 (1.7 to 2.8)	
	Placebo-controlled benefit	-0.2 (-0.4 to 0.1)	0.10
Expected number of angina episodes over the follow-up period under each treatment	Vericiguat	159.4 (121.2 to 199.9)	
	Placebo	160.1 (121.7 to 200.9)	
	Placebo-controlled benefit	0.7 (-6.2 to 7.7)	0.58
Expected number of days free from angina over the follow-up period under each treatment	Vericiguat	8.3 (3.8 to 14.1)	
	Placebo	8.2 (3.5 to 13.8)	
	Placebo-controlled benefit	0.1 (-0.8 to 1.1)	0.63

Supplementary Table S7: Bayesian ORBITA-app angina symptom outcomes based on 57-patient intention-to-treat analysis

Outcome	Coronary spasm type	Effect size (95% CrI)
Final day transition OR for symptom reduction with vericiguat versus placebo	Epicardial	0.49 (0.38 to 0.64)
	Microvascular	1.14 (0.87 to 1.45)
	Combined	1.01 (0.71 to 1.44)
	Combined vs epicardial	2.08 (1.53 to 2.82)
	Combined vs microvascular	0.89 (0.66 to 1.21)
	Microvascular vs epicardial	2.35 (1.94 to 2.84)
Placebo-controlled benefit in expected number of angina episodes over the follow-up period	Epicardial	-41.5 (-55.2 to -28.4)
	Microvascular	+27.4 (16.1 to 38.8)
	Combined	+21.6 (-6.9 to 53.9)
	Combined vs epicardial	+63.3 (31.6 to 98.8)
	Combined vs microvascular	-6.3 (-34.7 to 29.7)
	Microvascular vs epicardial	+68.9 (51.9 to 89.0)
Placebo-controlled benefit in expected number of days free from angina over the follow-up period	Epicardial	-4.7 (-8.7 to -1.5)
	Microvascular	+2.5 (0.8 to 5.0)
	Combined	+0.6 (-0.5 to 3.0)
	Combined vs epicardial	+5.5 (1.9 to 10.2)
	Combined vs microvascular	-1.8 (-4.6 to 0.8)
	Microvascular vs epicardial	+7.4 (3.0 to 13.1)

Supplementary Table S8: Missing outcome data per treatment sequence

Outcome variable	Vericiguat 1st period, missing n (%)	Vericiguat 2nd period, missing n (%)	Placebo 1st period, missing n (%)	Placebo 2nd period, missing n (%)
LASCA ACh				
- Perfusion response	1 (2%)	4 (7%)	1 (2%)	3 (5%)
- CVC	1 (2%)	4 (7%)	1 (2%)	3 (5%)
- AUC	1 (2%)	4 (7%)	1 (2%)	3 (5%)
LASCA SNP				
- Perfusion response	1 (2%)	3 (5%)	1 (2%)	3 (5%)
- CVC	1 (2%)	3 (5%)	1 (2%)	3 (5%)
- AUC	1 (2%)	3 (5%)	1 (2%)	3 (5%)
LASCA Insulin				
- Perfusion response	1 (2%)	3 (5%)	1 (2%)	3 (5%)
- CVC	1 (2%)	3 (5%)	1 (2%)	3 (5%)
- AUC	1 (2%)	3 (5%)	1 (2%)	3 (5%)
LASCA NaCl				
- Perfusion response	1 (2%)	3 (5%)	1 (2%)	3 (5%)
- CVC	1 (2%)	3 (5%)	1 (2%)	3 (5%)
- AUC	1 (2%)	3 (5%)	1 (2%)	3 (5%)
SAQSS	1 (2%)	3 (5%)	1 (2%)	3 (5%)

Data are n (%), of total n=57. AUC, area under the curve; ACh, acetylcholine; CVC, cutaneous vascular conductance; LASCA, laser speckle contrast analysis; SAQSS, Seattle Angina Questionnaire summary score; SNP, sodium nitroprusside.

Supplementary Table S9: Treatment effect on functional outcomes and SAQSS by sex, based on 57-patient intention-to-treat analysis

Outcome	Interaction χ^2 , df=1	Interaction p-value	Sex	Estimate	95% CI	P-value
LASCA ACh						
- Perfusion response	0.07	0.78	Female	-5.6	-10.8 to -0.37	0.04
			Male	-4.2	-13.1 to 4.7	0.35
- CVC	0.03	0.87	Female	-0.04	-0.10 to 0.03	0.27
			Male	-0.02	-0.13 to 0.08	0.65
- AUC	0.19	0.66	Female	-854	-4319 to 2611	0.62
			Male	-2379	-8689 to 3931	0.45
SAQSS	0.62	0.43	Female	-0.7	-5.6 to 4.2	0.78
			Male	3.1	-5.5 to 11.7	0.47

Linear mixed-effects models (LME) were fit for each outcome, including fixed effects for treatment, study period, randomization sequence group, baseline value of the outcome, and the treatment $\times$ sex interaction, with a random intercept for participant to account for the crossover design and within-patient correlation across periods. Models were estimated using maximum likelihood. All participants with available data at each assessment were included under a missing-at-random assumption, consistent with the intention-to-treat analysis population; no imputation was performed. For each outcome, the treatment $\times$ sex interaction was tested via likelihood ratio test comparing the model with and without the interaction term and sex as a main effect. Estimated marginal means and treatment contrasts (vericiguat – placebo) within each sex subgroup were derived from the interaction model, with Kenward-Roger degrees-of-freedom approximation; subgroup contrasts are presented for descriptive purposes and should be interpreted as exploratory, particularly for outcomes with a non-significant omnibus interaction test. AUC, area under the curve; ACh, acetylcholine; APU, arbitrary perfusion units; CI, confidence interval; CVC, cutaneous vascular conductance; LASCA, laser speckle contrast analysis; SAQSS, Seattle Angina Questionnaire summary score, higher scores represent less angina symptoms.

Supplementary Table S10: Sensitivity analysis of treatment effects based on 55-patient intention-to-treat analysis, excluding those with inconclusive spasm provocation test results

	Intervention effect		
	Estimate	95% CI	P-value
LASCA assessment ACh			
- Perfusion response, APU	-5.4	-9.8 to -0.9	0.02
- CVC, APU/mmHg	-0.04	-0.09 to 0.02	0.19
- AUC, APU•s	-1120	-4175 to 1934	0.46
Seattle Angina Questionnaire summary score	-0.3	-4.7 to 4.0	0.88

Values are derived from a mixed linear regression model and depicted as mean (95% CI) for continuous outcomes, adjusting for period, randomization sequence, baseline value, and participant. AUC, area under the curve; ACh, acetylcholine; APU, arbitrary perfusion units; CI, confidence interval; CVC, cutaneous vascular conductance; LASCA, laser speckle contrast analysis; SAQ, Seattle Angina Questionnaire, higher scores represent less angina symptoms.

Supplementary Table S11: Sensitivity analysis of treatment effects using period-1 data only, analyzed as a parallel-group comparison

	Intervention effect		
	Estimate	95% CI	P-value
LASCA assessment ACh			
- Perfusion response, APU	0.7	-8.0 to 9.5	0.87
- CVC, APU/mmHg	0.01	-0.10 to 0.11	0.91
- AUC, APU•s	3505	-2260 to 9269	0.23
Seattle Angina Questionnaire summary score	-0.4	-7.0 to 6.3	0.91

Sensitivity analysis restricting the crossover trial data to Period 1 only, analyzed as a parallel-group (non-crossover) comparison between vericiguat and placebo, to assess for potential residual carryover effects from an insufficient washout period between treatment periods. Because treatment assignment and Randomization Group are collinear within Period 1 (each participant's Period 1 treatment is fully determined by their randomized treatment sequence), Randomization Group was not included as a covariate in these models. Models were adjusted for the respective baseline (week 0) value of each outcome. Estimates represent the treatment difference (vericiguat – placebo) with 95% confidence intervals, derived from linear models fit to Period 1 data only. Concordance between these estimates and those from the primary crossover analysis (all periods, both treatment sequences) supports adequacy of the washout period; discordance would suggest possible carryover effects. AUC, area under the curve; ACh, acetylcholine; APU, arbitrary perfusion units; CI, confidence interval; CVC, cutaneous vascular conductance; LASCA, laser speckle contrast analysis; SAQ, Seattle Angina Questionnaire, higher scores represent less angina symptoms.

Supplementary Table S12: Sensitivity analysis of treatment effects based on 48-patient intention-to-treat analysis, excluding those with changes in antianginal medication during enrolment

	Intervention effect		
	Estimate	95% CI	P-value
LASCA assessment ACh			
- Perfusion response, APU	-5.4	-10.4 to -0.5	0.03
- CVC, APU/mmHg	-0.03	-0.10 to 0.03	0.26
- AUC, APU•s	-1702	-5053 to 1649	0.31
Seattle Angina Questionnaire summary score	0.4	-4.4 to 5.2	0.87

Changes in antianginal medication during the enrolment period were defined as dose adjustments or initiation or discontinuation of nitrates, calcium channel blockers, nicorandil, angiotensin-converting enzyme inhibitors, angiotensin receptor blockers, or beta-blockers. Values are derived from a mixed linear regression model and depicted as mean (95% CI) for continuous outcomes, adjusting for period, randomization sequence, baseline value, and participant. AUC, area under the curve; ACh, acetylcholine; APU, arbitrary perfusion units; CI, confidence interval; CVC, cutaneous vascular conductance; LASCA, laser speckle contrast analysis; SAQ, Seattle Angina Questionnaire, higher scores represent less angina symptoms.

Supplementary Table S13: Sensitivity analyses of treatment effect in terms of ORBITA-app daily angina episodes

Final day transition odds ratio for reduction in ORBITA-app daily angina episodes with vericiguat versus placebo	Odds ratio	95% credible interval	Posterior probability of benefit with vericiguat vs placebo
48-patient intention-to-treat analysis, excluding those with changes in antianginal medication during enrolment	0.89	0.67 to 1.14	0.18
49-patient per-protocol analysis, excluding those with inconclusive spasm provocation test results	0.82	0.64 to 1.05	0.06
57-patient intention-to-treat analysis, female versus male patients*	Interaction odds ratio 0.74	0.61 to 0.92	1.00
57-patient intention-to-treat analysis, including treatment by sequence interaction term ^Δ	Interaction odds ratio 1.03	0.04 to 2.42	0.51
Period-1 only parallel-group comparison	0.78	0.36 to 1.61	0.25

*The final day transition OR for reduction in daily angina episodes with vericiguat vs placebo was smaller in female vs male participants; posterior probability of smaller treatment effect in female vs male participants = 0.996. This sex-treatment interaction was a secondary, hypothesis-generating analysis and should be interpreted with caution given the marked imbalance in sex distribution and small number of male participants available to model. ^ΔThe formal crossover assessment was done by adding a treatment by sequence interaction term to the primary symptom model. This was included as a fixed term applying uniformly across all days. There was no evidence of a period-by-treatment carryover effect (interaction OR 1.03, 95% CrI 0.04 to 24.2). The posterior probability of a positive interaction was 0.51, consistent with negligible directional evidence either way. As an additional sensitivity analysis, period 1 was analyzed in isolation as a parallel-group comparison; the final day transition OR for reduction in daily angina episodes with vericiguat vs placebo was 0.78 (95% CrI 0.36 to 1.61; posterior probability of benefit with vericiguat vs placebo = 0.25).

Supplementary Figure S4: Regression model and coefficients for symptom app model based on 57-patient intention-to-treat analysis

Bayesian Constrained Partial Proportional Odds Ordinal Logistic Model

Dirichlet Priors With Concentration Parameter 0.215 for Intercepts

```
blrm(formula = symptom_frequency_num ~ rcs(symptom_frequency_num_lag1,
c(2, 5, 8)) + rcs(day_num_period, 3) * Treatment + rcs(symptom_frequency_num_pre_mean,
3) + period + sequence + cluster(orbita_id), ppo = ~day_num_period,
cppo = function(y) y, data = model_d, pcontrast = pcon, backend = "cmdstan",
iter = 8000, chains = 4, refresh = 100, progress = file.path(out_dir,
"model.progress.txt"), loo = FALSE, ppairs = NULL, method = "sampling",
file = file.path(out_dir, "res1.blrm.rds"))
```

Frequencies of Responses

```
0 1 2 3 4 5 6 7 8 9 10
1315 801 1011 1018 728 437 387 275 340 179 1089
```

Frequencies of Missing Values Due to Each Variable

```
symptom_frequency_num 367 symptom_frequency_num_lag1 361 day_num_period 0 Treatment symptom_frequency_num_pre_mean 0 period 0 sequence 0 cluster(orbita_id) 0
```

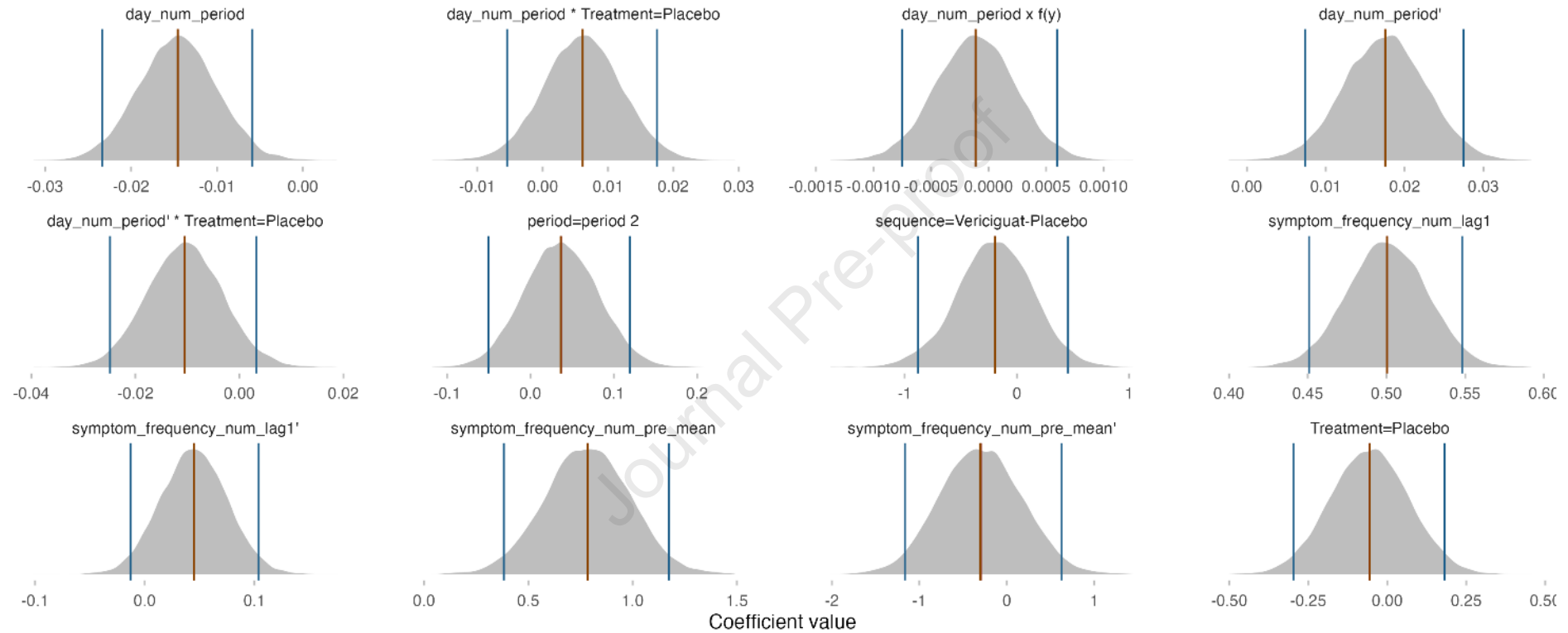
	Mixed Calibration/ Discrimination Indexes	Discrimination Indexes	Rank Discrim. Indexes
Obs 7580	B 0.109 [0.105, 0.113]	g 3.875 [3.554, 4.21]	C 0.866 [0.862, 0.87]
Draws 16000		g _p 0.409 [0.388, 0.429]	D _{xy} 0.732 [0.724, 0.74]
Chains 4		EV 0.547 [0.493, 0.591]	
Time 405.1s		v 11.933 [10.036, 14.119]	
p 11		vp 0.132 [0.118, 0.145]	
Cluster on orbita_id			
Clusters 57			
α _y 1.2052 [0.9598, 1.4713]			

	Mean β	Median β	S.E.	Lower	Upper	Pr(β>0)	Symmetry
symptom_frequency_num_lag1	0.5004	0.5003	0.0250	0.4508	0.5481	1.0000	0.98
symptom_frequency_num_lag1'	0.0449	0.0448	0.0300	-0.0130	0.1039	0.9324	1.01
day_num_period	-0.0146	-0.0146	0.0045	-0.0234	-0.0059	0.0008	0.99
day_num_period'	0.0175	0.0175	0.0052	0.0074	0.0275	0.9995	1.00
Treatment=Placebo	-0.0556	-0.0548	0.1228	-0.2968	0.1814	0.3269	1.00
symptom_frequency_num_pre_mean	0.7843	0.7842	0.2025	0.3826	1.1743	1.0000	0.99
symptom_frequency_num_pre_mean'	-0.2969	-0.3023	0.4643	-1.1655	0.6283	0.2609	1.02
period=period 2	0.0367	0.0364	0.0436	-0.0503	0.1193	0.7988	1.02
sequence=Vericiguat-Placebo	-0.1972	-0.1943	0.3367	-0.8813	0.4554	0.2826	0.95
day_num_period × Treatment=Placebo	0.0061	0.0061	0.0059	-0.0054	0.0175	0.8525	1.00
day_num_period' × Treatment=Placebo	-0.0105	-0.0105	0.0072	-0.0249	0.0033	0.0715	1.03
day_num_period × f(y)	-0.0001	-0.0001	0.0003	-0.0008	0.0006	0.3726	1.00

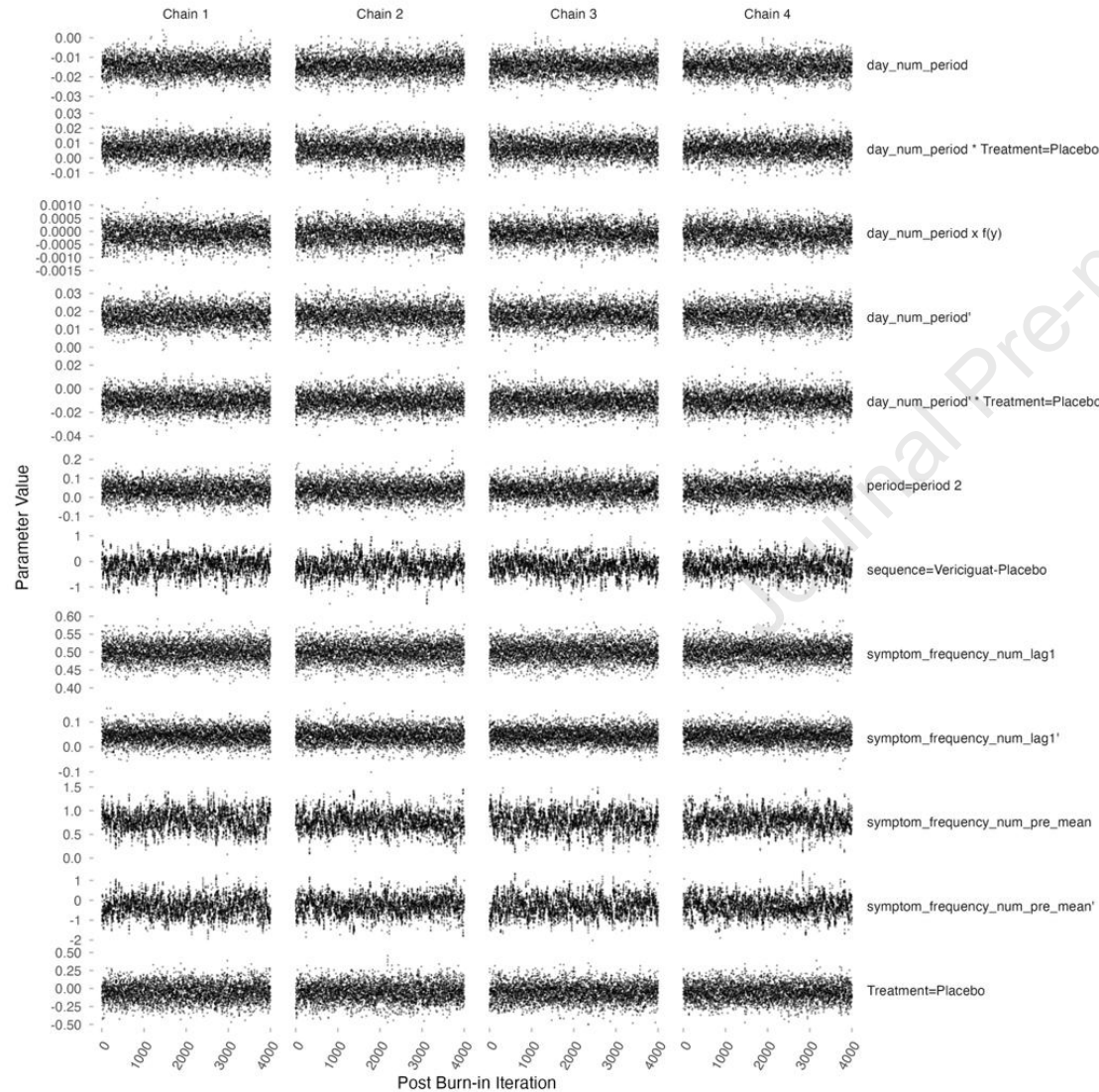
Contrasts Given Priors

```
[1] "list(c1 = list(Treatment = \"Vericiguat\"), c2 = list(Treatment = \"Placebo\"), \" \" contrast = expression(c1 - c2), sd = 0.842807127883599)"
```

Supplementary Figure S5: Coefficient density plots: symptom app model based on 57-patient intention-to-treat analysis



Supplementary Figure S6: Chain plot of MCMC draws for symptom app model based on 57-patient intention-to-treat analysis



Supplementary Figure S7: Regression model and coefficients for symptom app model based on 51-patient per-protocol analysis

Bayesian Constrained Partial Proportional Odds Ordinal Logistic Model

Dirichlet Priors With Concentration Parameter 0.215 for Intercepts

```
blrm(formula = symptom_frequency_num ~ rcs(symptom_frequency_num_lag1,
      c(2, 5, 8)) + rcs(day_num_period, 3) * Treatment + rcs(symptom_frequency_num_pre_mean,
      3) + period + sequence + cluster(orbita_id), ppo = ~day_num_period,
      cppo = function(y) y, data = model_d, pcontrast = pcon, backend = "cmdstan",
      iter = 8000, chains = 4, refresh = 100, progress = file.path(out_dir,
      "model.progress.txt"), too = FALSE, ppairs = NULL, method = "sampling",
      file = file.path(out_dir, "res1.blrm.rds"))
```

Frequencies of Responses

```
0 1 2 3 4 5 6 7 8 9 10
1313 798 974 972 695 408 355 247 325 173 845
```

Frequencies of Missing Values Due to Each Variable

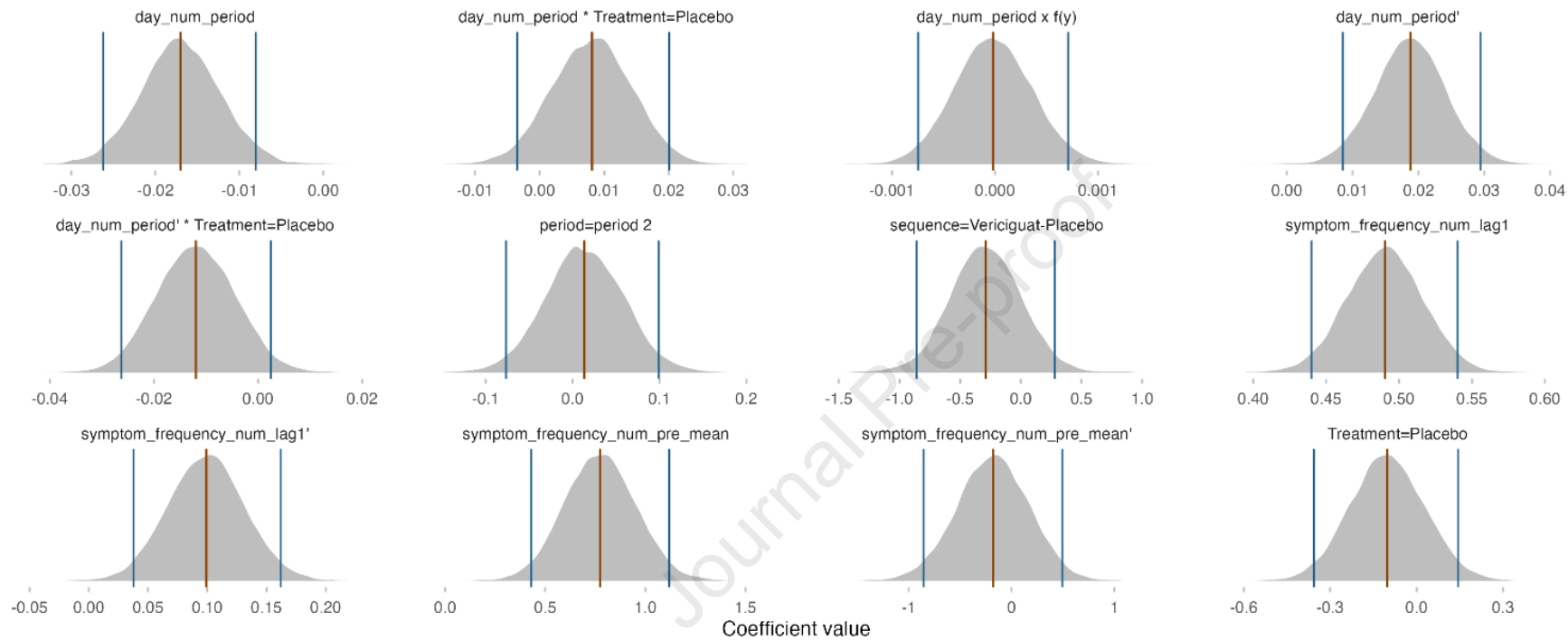
	symptom_frequency_num	symptom_frequency_num_lag1	day_num_period	Treatment	symptom_frequency_num_pre_mean	period	sequence	cluster(orbita_id)
	2	2	0	0	0	0	0	0
Obs	7105							
Draws	16000							
Chains	4							
Time	332.6s							
p	11							
Cluster on	orbita_id							
Clusters	51							
σ_y	1.008 [0.7994, 1.2409]							

	Mean β	Median β	S.E.	Lower	Upper	Pr($\beta > 0$)	Symmetry
symptom_frequency_num_lag1	0.4902	0.4904	0.0257	0.4399	0.5402	1.0000	0.99
symptom_frequency_num_lag1'	0.0993	0.0994	0.0320	0.0377	0.1621	0.9987	1.02
day_num_period	-0.0170	-0.0170	0.0046	-0.0262	-0.0081	0.0003	1.02
day_num_period'	0.0187	0.0188	0.0053	0.0085	0.0294	0.9996	1.00
Treatment=Placebo	-0.1013	-0.1023	0.1281	-0.3567	0.1438	0.2158	1.00
symptom_frequency_num_pre_mean	0.7745	0.7735	0.1777	0.4296	1.1196	0.9999	1.02
symptom_frequency_num_pre_mean'	-0.1809	-0.1774	0.3478	-0.8549	0.4962	0.2999	0.97
period=period 2	0.0138	0.0133	0.0447	-0.0766	0.0993	0.6181	1.01
sequence=Vericiguat-Placebo	-0.2875	-0.2900	0.2927	-0.8595	0.2793	0.1573	1.00
day_num_period x Treatment=Placebo	0.0081	0.0082	0.0061	-0.0035	0.0201	0.9074	0.99
day_num_period' x Treatment=Placebo	-0.0119	-0.0119	0.0074	-0.0262	0.0025	0.0540	1.01
day_num_period x f(y)	0.0000	0.0000	0.0004	-0.0007	0.0007	0.4763	1.02

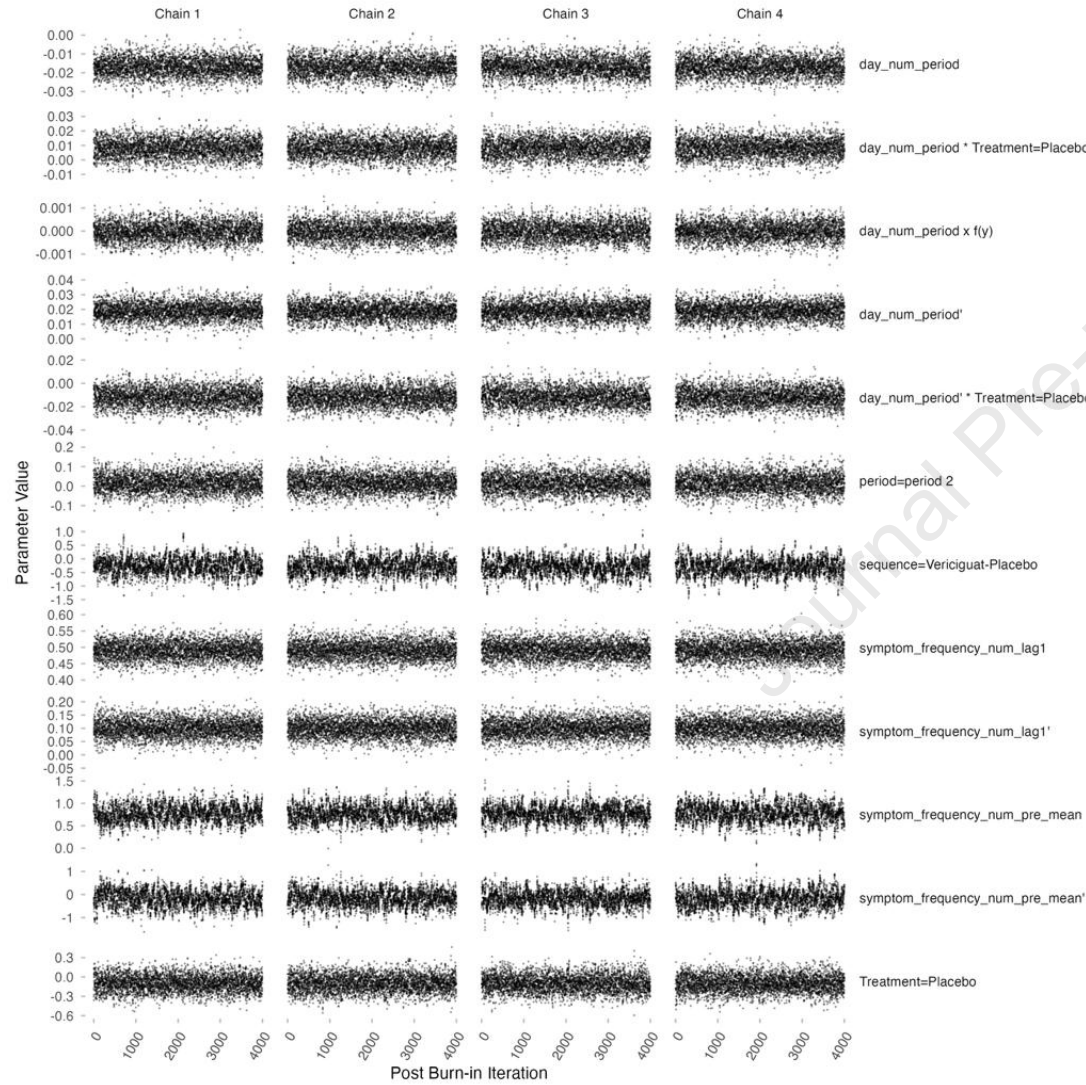
Contrasts Given Priors

```
[1] "list(c1 = list(Treatment = \"Vericiguat\"), c2 = list(Treatment = \"Placebo\"), \" \" contrast = expression(c1 - c2), sd = 0.842807127883599)\""
```

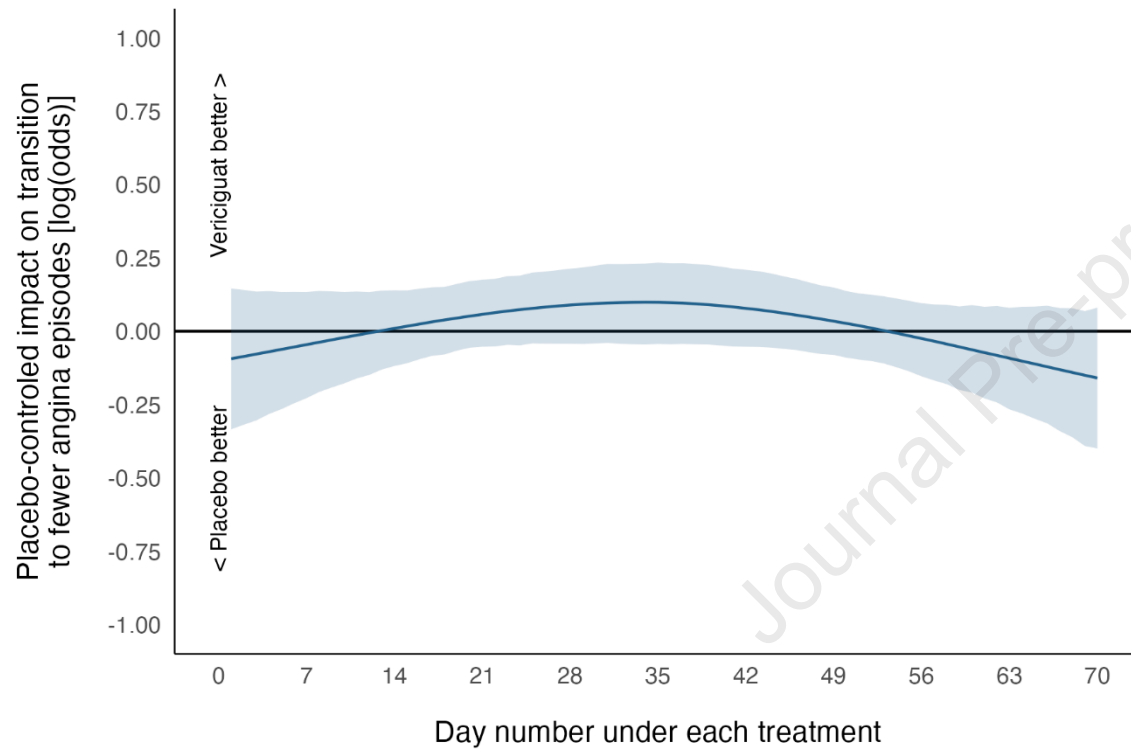
Supplementary Figure S8: Coefficient density plots: symptom app model based on 51-patient per-protocol analysis



Supplementary Figure S9: Chain plot of MCMC draws for symptom app model based on 51-patient per-protocol analysis



Supplementary Figure S10: Log-odds ratio for transitioning to fewer angina episodes with vericiguat versus placebo, by day number under each treatment based on 51-patient per-protocol analysis



Supplementary Figure S11: Regression model and coefficients for ORBITA score app model based on 57-patient intention-to-treat analysis

Model Summary

Bayesian Constrained Partial Proportional Odds Ordinal Logistic Model

Dirichlet Priors With Concentration Parameter 0.086 for Intercepts

```
blrm(formula = orbita_score ~ rcs(orbita_score_lag1, c(5, 15, 25)) + rcs(day_num_period, 3) * Treatment + rcs(orbita_score_pre_mean, 3) + period + sequence + cluster(orbita_id), ppo = ~day_num_period, cppo = function(y) y, data = model_d, pcontrast = pcon, backend = "cmdstan", iter = 8000, chains = 4, refresh = 100, progress = file.path(out_dir, "model.progress.txt"), loo = FALSE, ppairs = NULL, method = "sampling", file = file.path(out_dir, "res1.blrm.rds"))
```

Frequencies of Responses

```
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
1315 480 248 73 392 461 158 208 501 309 119 353 256 97 234 106 73 247 67 28 198 49 36 170 134 17 69 93 90 415 584
```

Frequencies of Missing Values Due to Each Variable

```
orbita_score 367 orbita_score_lag1 361 day_num_period 0 Treatment orbita_score_pre_mean 0 period 0 sequence 0 cluster(orbita_id) 0
```

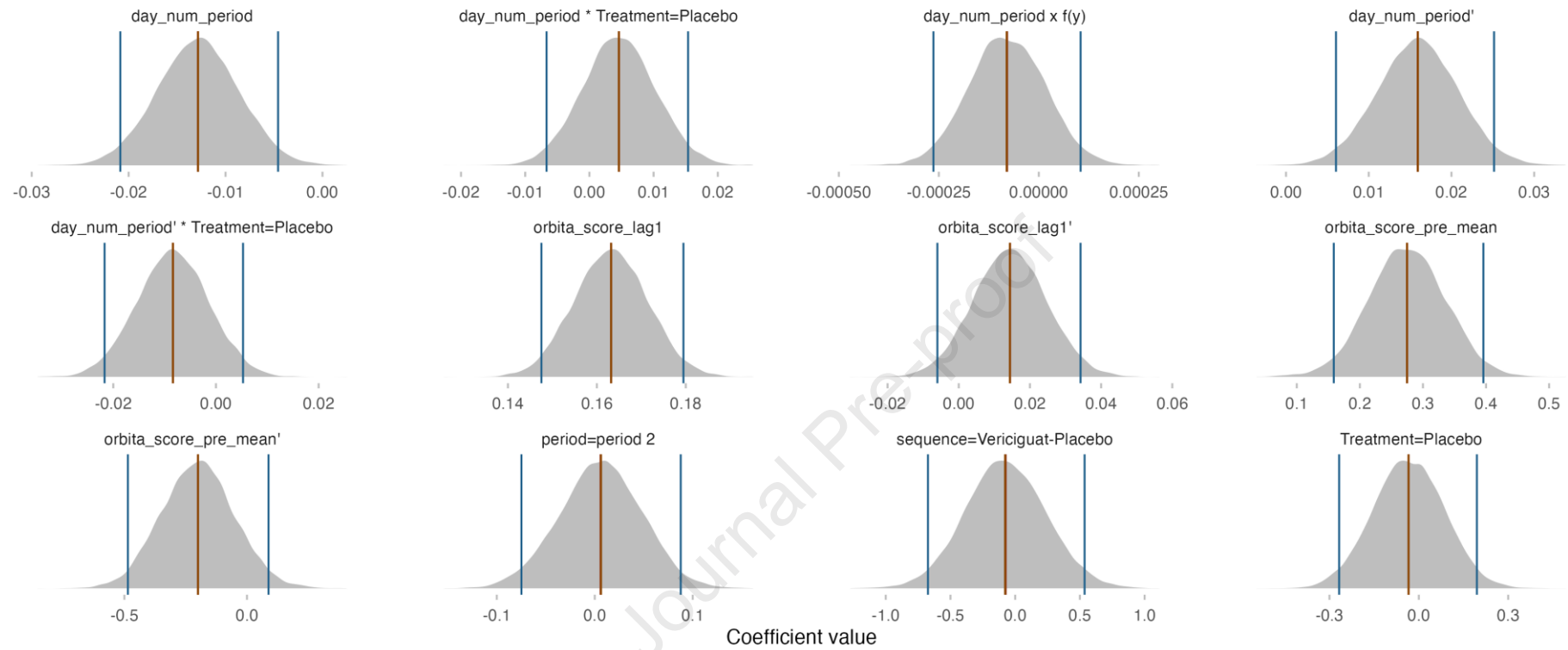
	Mixed Calibration/ Discrimination Indexes	Discrimination Indexes	Rank Discrim. Indexes
Obs $\hat{\alpha}$ $\in$ [7580	$\hat{B}\hat{\alpha}\in$ [0.117 [0.113, 0.123]	$\hat{g}\hat{\alpha}\in$ [3.655 [3.347, 3.99]	$\hat{C}\hat{\alpha}\in$ [0.844 [0.84, 0.847]
Draws $\hat{\alpha}\in$ [16000		$\hat{g}_p\hat{\alpha}\in$ [0.419 [0.401, 0.436]	$\hat{D}_{xy}\hat{\alpha}\in$ [0.687 [0.68, 0.694]
Chains $\hat{\alpha}\in$ [4		$\hat{E}\hat{V}\hat{\alpha}\in$ [0.557 [0.509, 0.605]	
Time $\hat{\alpha}\in$ [568.4s		$\hat{v}\hat{\alpha}\in$ [10.569 [8.858, 12.54]	
$\hat{p}\hat{\alpha}\in$ [11		$\hat{v}_p\hat{\alpha}\in$ [0.138 [0.125, 0.15]	
Cluster on $\hat{\alpha}\in$ [orbita_id			
Clusters $\hat{\alpha}\in$ [57			
$\sigma_1\hat{\alpha}\in$ [1.1304 [0.9228, 1.3712]			

	Mean I ²	Median I ²	S.E.	Lower	Upper	Pr(I ² >0)	Symmetry
y $\hat{\alpha}\in$ [1	$\hat{A}$ -0.4252	$\hat{A}$ -0.4282	$\hat{A}$ 0.4627	$\hat{A}$ -1.2903	$\hat{A}$ 0.5273	$\hat{A}$ 0.1735	$\hat{A}$ 1.03
y $\hat{\alpha}\in$ [2	$\hat{A}$ -1.1193	$\hat{A}$ -1.1213	$\hat{A}$ 0.4624	$\hat{A}$ -1.9948	$\hat{A}$ -0.1810	$\hat{A}$ 0.0089	$\hat{A}$ 1.03
y $\hat{\alpha}\in$ [3	$\hat{A}$ -1.4676	$\hat{A}$ -1.4708	$\hat{A}$ 0.4624	$\hat{A}$ -2.3405	$\hat{A}$ -0.5220	$\hat{A}$ 0.0021	$\hat{A}$ 1.03
y $\hat{\alpha}\in$ [4	$\hat{A}$ -1.5689	$\hat{A}$ -1.5726	$\hat{A}$ 0.4621	$\hat{A}$ -2.4571	$\hat{A}$ -0.6406	$\hat{A}$ 0.0009	$\hat{A}$ 1.03
y $\hat{\alpha}\in$ [5	$\hat{A}$ -2.1185	$\hat{A}$ -2.1206	$\hat{A}$ 0.4624	$\hat{A}$ -3.0572	$\hat{A}$ -1.2374	$\hat{A}$ 0.0000	$\hat{A}$ 1.03
y $\hat{\alpha}\in$ [6	$\hat{A}$ -2.7593	$\hat{A}$ -2.7620	$\hat{A}$ 0.4631	$\hat{A}$ -3.6410	$\hat{A}$ -1.8220	$\hat{A}$ 0.0000	$\hat{A}$ 1.04
y $\hat{\alpha}\in$ [7	$\hat{A}$ -2.9812	$\hat{A}$ -2.9852	$\hat{A}$ 0.4632	$\hat{A}$ -3.8775	$\hat{A}$ -2.0573	$\hat{A}$ 0.0000	$\hat{A}$ 1.03
y $\hat{\alpha}\in$ [8	$\hat{A}$ -3.2683	$\hat{A}$ -3.2707	$\hat{A}$ 0.4637	$\hat{A}$ -4.2002	$\hat{A}$ -2.3824	$\hat{A}$ 0.0000	$\hat{A}$ 1.03
y $\hat{\alpha}\in$ [9	$\hat{A}$ -3.9279	$\hat{A}$ -3.9312	$\hat{A}$ 0.4645	$\hat{A}$ -4.8700	$\hat{A}$ -3.0505	$\hat{A}$ 0.0000	$\hat{A}$ 1.03
y $\hat{\alpha}\in$ [10	$\hat{A}$ -4.3431	$\hat{A}$ -4.3456	$\hat{A}$ 0.4653	$\hat{A}$ -5.2778	$\hat{A}$ -3.4582	$\hat{A}$ 0.0000	$\hat{A}$ 1.02
y $\hat{\alpha}\in$ [11	$\hat{A}$ -4.5046	$\hat{A}$ -4.5066	$\hat{A}$ 0.4652	$\hat{A}$ -5.4509	$\hat{A}$ -3.6316	$\hat{A}$ 0.0000	$\hat{A}$ 1.02
y $\hat{\alpha}\in$ [12	$\hat{A}$ -4.9955	$\hat{A}$ -4.9980	$\hat{A}$ 0.4657	$\hat{A}$ -5.9367	$\hat{A}$ -4.1184	$\hat{A}$ 0.0000	$\hat{A}$ 1.02
y $\hat{\alpha}\in$ [13	$\hat{A}$ -5.3935	$\hat{A}$ -5.3950	$\hat{A}$ 0.4664	$\hat{A}$ -6.3270	$\hat{A}$ -4.5019	$\hat{A}$ 0.0000	$\hat{A}$ 1.02
y $\hat{\alpha}\in$ [14	$\hat{A}$ -5.5553	$\hat{A}$ -5.5584	$\hat{A}$ 0.4666	$\hat{A}$ -6.4604	$\hat{A}$ -4.6382	$\hat{A}$ 0.0000	$\hat{A}$ 1.02
y $\hat{\alpha}\in$ [15	$\hat{A}$ -5.9582	$\hat{A}$ -5.9599	$\hat{A}$ 0.4670	$\hat{A}$ -6.8478	$\hat{A}$ -5.0224	$\hat{A}$ 0.0000	$\hat{A}$ 1.02
y $\hat{\alpha}\in$ [16	$\hat{A}$ -6.1519	$\hat{A}$ -6.1537	$\hat{A}$ 0.4673	$\hat{A}$ -7.0599	$\hat{A}$ -5.2345	$\hat{A}$ 0.0000	$\hat{A}$ 1.02
y $\hat{\alpha}\in$ [17	$\hat{A}$ -6.2914	$\hat{A}$ -6.2940	$\hat{A}$ 0.4677	$\hat{A}$ -7.1787	$\hat{A}$ -5.3532	$\hat{A}$ 0.0000	$\hat{A}$ 1.02
y $\hat{\alpha}\in$ [18	$\hat{A}$ -6.7971	$\hat{A}$ -6.7982	$\hat{A}$ 0.4688	$\hat{A}$ -7.7097	$\hat{A}$ -5.8814	$\hat{A}$ 0.0000	$\hat{A}$ 1.02
y $\hat{\alpha}\in$ [19	$\hat{A}$ -6.9450	$\hat{A}$ -6.9444	$\hat{A}$ 0.4692	$\hat{A}$ -7.8738	$\hat{A}$ -6.0452	$\hat{A}$ 0.0000	$\hat{A}$ 1.02
y $\hat{\alpha}\in$ [20	$\hat{A}$ -7.0074	$\hat{A}$ -7.0070	$\hat{A}$ 0.4695	$\hat{A}$ -7.9356	$\hat{A}$ -6.1052	$\hat{A}$ 0.0000	$\hat{A}$ 1.02
y $\hat{\alpha}\in$ [21	$\hat{A}$ -7.4880	$\hat{A}$ -7.4893	$\hat{A}$ 0.4703	$\hat{A}$ -8.3921	$\hat{A}$ -6.5560	$\hat{A}$ 0.0000	$\hat{A}$ 1.03
y $\hat{\alpha}\in$ [22	$\hat{A}$ -7.6127	$\hat{A}$ -7.6150	$\hat{A}$ 0.4705	$\hat{A}$ -8.5098	$\hat{A}$ -6.6741	$\hat{A}$ 0.0000	$\hat{A}$ 1.03
y $\hat{\alpha}\in$ [23	$\hat{A}$ -7.7038	$\hat{A}$ -7.7050	$\hat{A}$ 0.4708	$\hat{A}$ -8.6155	$\hat{A}$ -6.7801	$\hat{A}$ 0.0000	$\hat{A}$ 1.03
y $\hat{\alpha}\in$ [24	$\hat{A}$ -8.1588	$\hat{A}$ -8.1589	$\hat{A}$ 0.4720	$\hat{A}$ -9.0911	$\hat{A}$ -7.2526	$\hat{A}$ 0.0000	$\hat{A}$ 1.03
y $\hat{\alpha}\in$ [25	$\hat{A}$ -8.5330	$\hat{A}$ -8.5345	$\hat{A}$ 0.4728	$\hat{A}$ -9.4759	$\hat{A}$ -7.6399	$\hat{A}$ 0.0000	$\hat{A}$ 1.03
y $\hat{\alpha}\in$ [26	$\hat{A}$ -8.5797	$\hat{A}$ -8.5807	$\hat{A}$ 0.4730	$\hat{A}$ -9.5007	$\hat{A}$ -7.6662	$\hat{A}$ 0.0000	$\hat{A}$ 1.02
y $\hat{\alpha}\in$ [27	$\hat{A}$ -8.7787	$\hat{A}$ -8.7790	$\hat{A}$ 0.4735	$\hat{A}$ -9.7023	$\hat{A}$ -7.8656	$\hat{A}$ 0.0000	$\hat{A}$ 1.02
y $\hat{\alpha}\in$ [28	$\hat{A}$ -9.0573	$\hat{A}$ -9.0586	$\hat{A}$ 0.4742	$\hat{A}$ -9.9786	$\hat{A}$ -8.1367	$\hat{A}$ 0.0000	$\hat{A}$ 1.01
y $\hat{\alpha}\in$ [29	$\hat{A}$ -9.3318	$\hat{A}$ -9.3323	$\hat{A}$ 0.4752	$\hat{A}$ -10.2670	$\hat{A}$ -8.4234	$\hat{A}$ 0.0000	$\hat{A}$ 1.01
y $\hat{\alpha}\in$ [30	$\hat{A}$ -10.6805	$\hat{A}$ -10.6803	$\hat{A}$ 0.4778	$\hat{A}$ -11.5790	$\hat{A}$ -9.7096	$\hat{A}$ 0.0000	$\hat{A}$ 1.02
orbita_score_lag1	$\hat{A}$ 0.1632	$\hat{A}$ 0.1632	$\hat{A}$ 0.0082	$\hat{A}$ 0.1475	$\hat{A}$ 0.1795	$\hat{A}$ 1.0000	$\hat{A}$ 1.02
orbita_score_lag1'	$\hat{A}$ 0.0144	$\hat{A}$ 0.0144	$\hat{A}$ 0.0103	$\hat{A}$ -0.0060	$\hat{A}$ 0.0342	$\hat{A}$ 0.9198	$\hat{A}$ 1.00
day_num_period	$\hat{A}$ -0.0128	$\hat{A}$ -0.0128	$\hat{A}$ 0.0042	$\hat{A}$ -0.0209	$\hat{A}$ -0.0046	$\hat{A}$ 0.0009	$\hat{A}$ 0.99
day_num_period'	$\hat{A}$ 0.0159	$\hat{A}$ 0.0159	$\hat{A}$ 0.0049	$\hat{A}$ 0.0060	$\hat{A}$ 0.0251	$\hat{A}$ 0.9994	$\hat{A}$ 0.99
Treatment=Placebo	$\hat{A}$ -0.0343	$\hat{A}$ -0.0345	$\hat{A}$ 0.1187	$\hat{A}$ -0.2676	$\hat{A}$ 0.1949	$\hat{A}$ 0.3902	$\hat{A}$ 0.97
orbita_score_pre_mean	$\hat{A}$ 0.2748	$\hat{A}$ 0.2747	$\hat{A}$ 0.0606	$\hat{A}$ 0.1587	$\hat{A}$ 0.3958	$\hat{A}$ 1.0000	$\hat{A}$ 0.98
orbita_score_pre_mean'	$\hat{A}$ -0.1996	$\hat{A}$ -0.1996	$\hat{A}$ 0.1478	$\hat{A}$ -0.4857	$\hat{A}$ 0.0878	$\hat{A}$ 0.0874	$\hat{A}$ 1.04
period=period 2	$\hat{A}$ 0.0059	$\hat{A}$ 0.0064	$\hat{A}$ 0.0417	$\hat{A}$ -0.0748	$\hat{A}$ 0.0878	$\hat{A}$ 0.5602	$\hat{A}$ 0.96
sequence=Vericiguat-Placebo	$\hat{A}$ -0.0734	$\hat{A}$ -0.0777	$\hat{A}$ 0.3097	$\hat{A}$ -0.6738	$\hat{A}$ 0.5364	$\hat{A}$ 0.4021	$\hat{A}$ 1.02
day_num_period A— Treatment=Placebo	$\hat{A}$ 0.0046	$\hat{A}$ 0.0046	$\hat{A}$ 0.0057	$\hat{A}$ -0.0067	$\hat{A}$ 0.0154	$\hat{A}$ 0.7944	$\hat{A}$ 1.00
day_num_period' A— Treatment=Placebo	$\hat{A}$ -0.0084	$\hat{A}$ -0.0084	$\hat{A}$ 0.0069	$\hat{A}$ -0.0217	$\hat{A}$ 0.0053	$\hat{A}$ 0.1133	$\hat{A}$ 0.98
day_num_period x t(y)	$\hat{A}$ -0.0001	$\hat{A}$ -0.0001	$\hat{A}$ 0.0001	$\hat{A}$ -0.0003	$\hat{A}$ 0.0001	$\hat{A}$ 0.2042	$\hat{A}$ 1.01

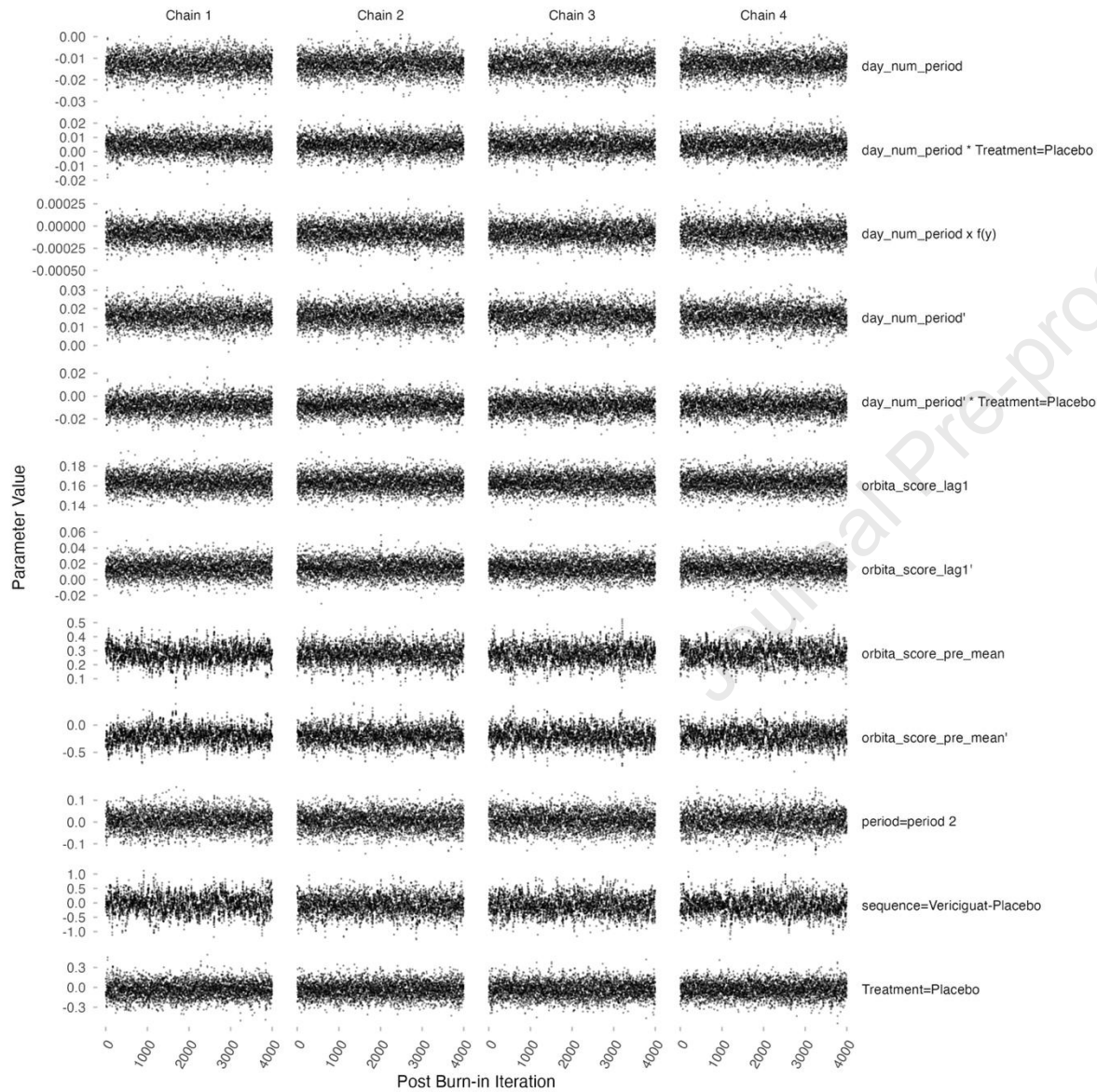
Contrasts Given Priors

```
[1] "list(c1 = list(Treatment = \"Vericiguat\"), c2 = list(Treatment = \"Placebo\"), \" \" contrast = expression(c1 - c2), sd = 0.842807127883599)"
```

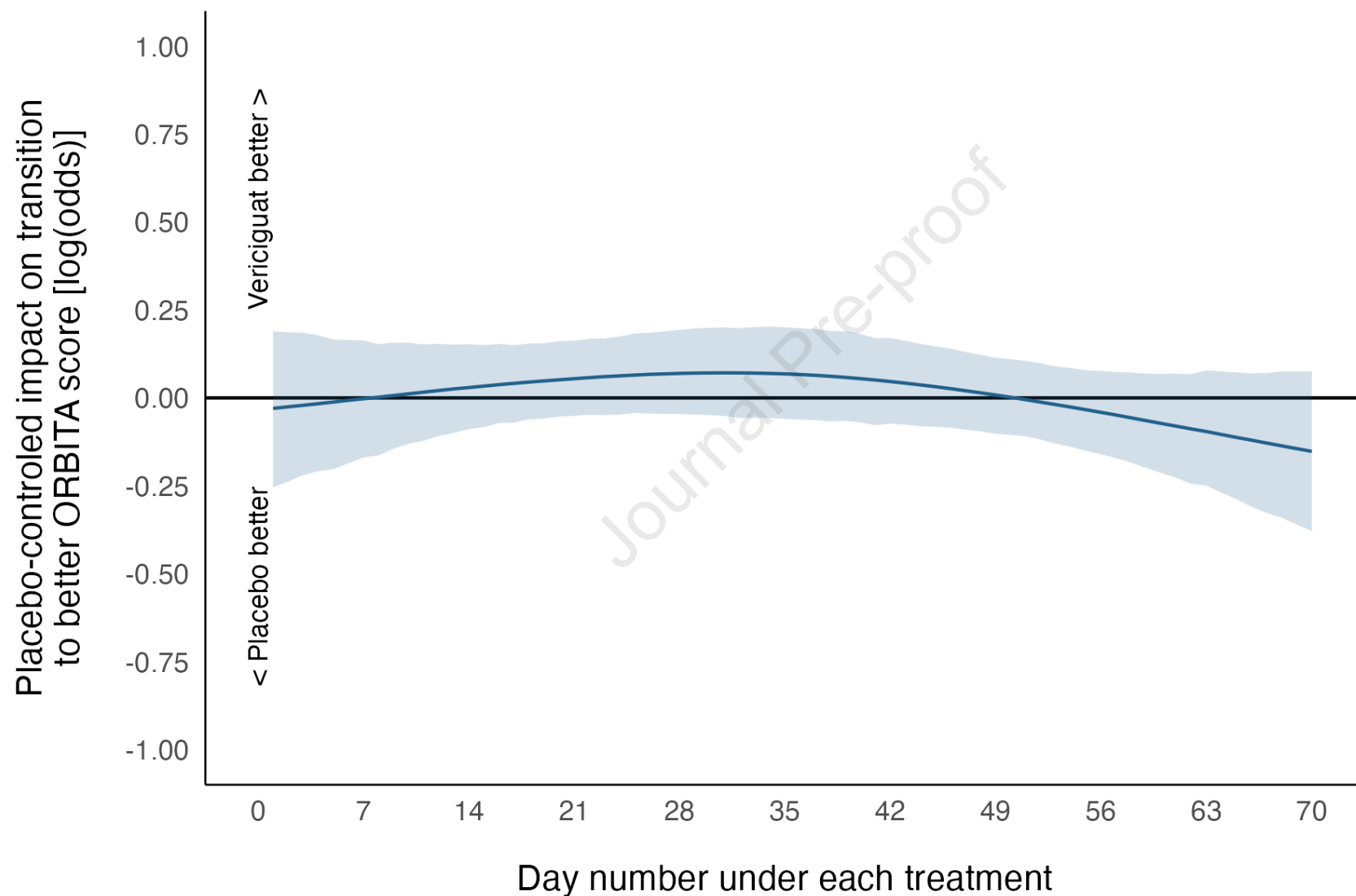
Supplementary Figure S12: Coefficient density plots for ORBITA score app model based on 57-patient intention-to-treat analysis



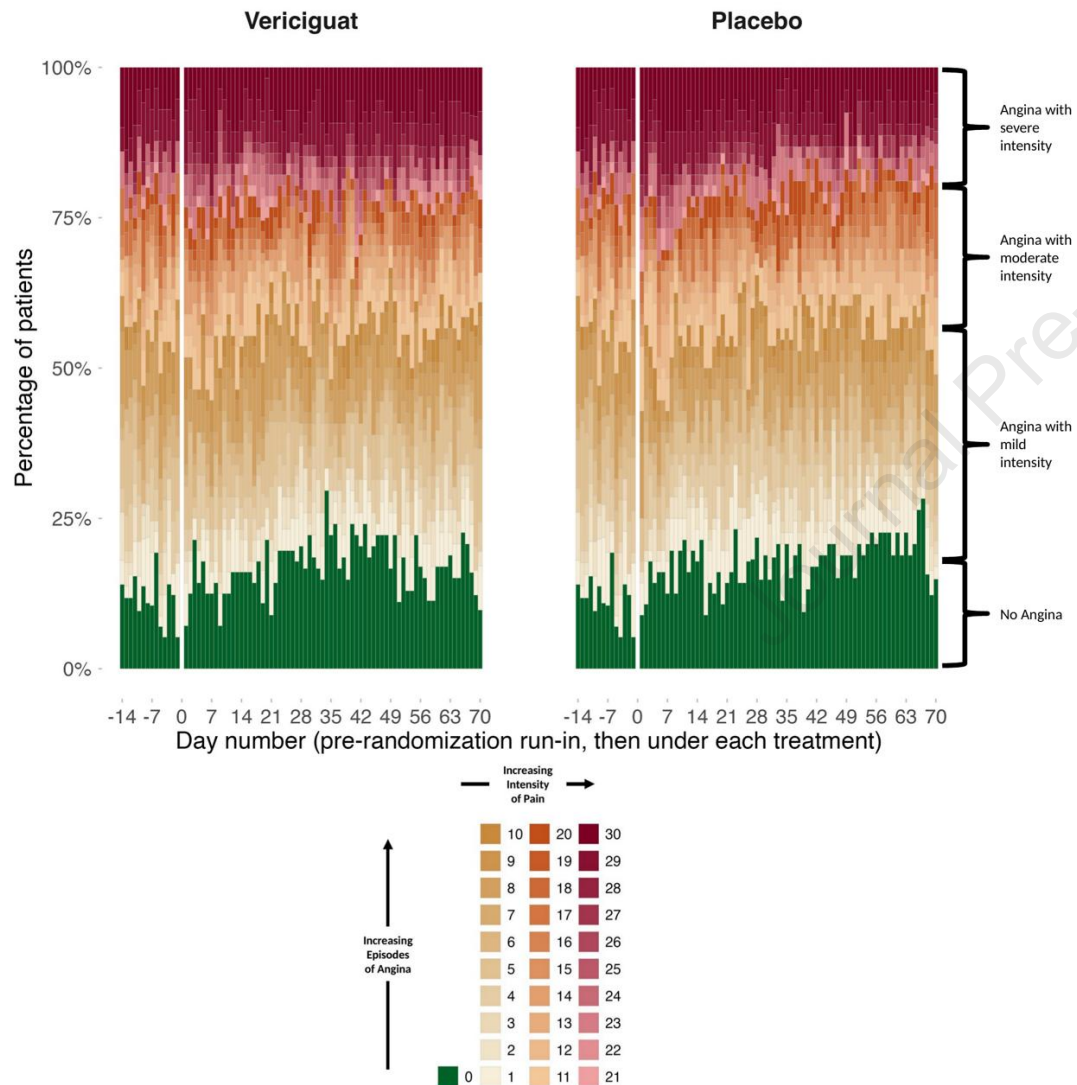
Supplementary Figure S13: Chain plot of MCMC draws for ORBITA score app model based on 57-patient intention-to-treat analysis



Supplementary Figure S14: Log-odds ratio for transitioning to better ORBITA score with vericiguat versus placebo, by day number under each treatment based on 57-patient intention-to-treat analysis



Supplementary Figure S15: Percentage of patients by ORBITA score on each day of treatment, for vericiguat and placebo separately, including the shared 14-day pre-randomization run-in (day -1 to -14)



Supplementary Figure S16: Regression model and coefficients for ORBITA-app symptom intensity model based on 57-patient intention-to-treat analysis

Model Summary

Bayesian Constrained Partial Proportional Odds Ordinal Logistic Model

Dirichlet Priors With Concentration Parameter 0.455 for Intercepts

```
blrm(formula = symptom_severity_num ~ rcs(symptom_severity_num_lag1,
c(1, 1.5, 2)) + rcs(day_num_period, 3) * Treatment + rcs(symptom_severity_num_pre_mean,
3) * period + sequence + cluster(orbita_id), ppo = -day_num_period,
cpo = function(y) y, data = model_d, pcontrast = pcon, backend = "cmdstan",
iter = 8000, chains = 4, refresh = 100, progress = file.path(out_dir,
"model.progress.txt"), loo = FALSE, ppairs = NULL, method = "sampling",
file = file.path(out_dir, "res1.blrm.rds"))
```

Frequencies of Responses

```
0 1 2 3
1315 1540 2896 1829
```

Frequencies of Missing Values Due to Each Variable

```
symptom_severity_num      symptom_severity_num_lag1      day_num_period      Treatment      symptom_severity_num_pre_mean      period      sequence      cluster(orbita_id)
367      361      0      0      0      0      0      0
```

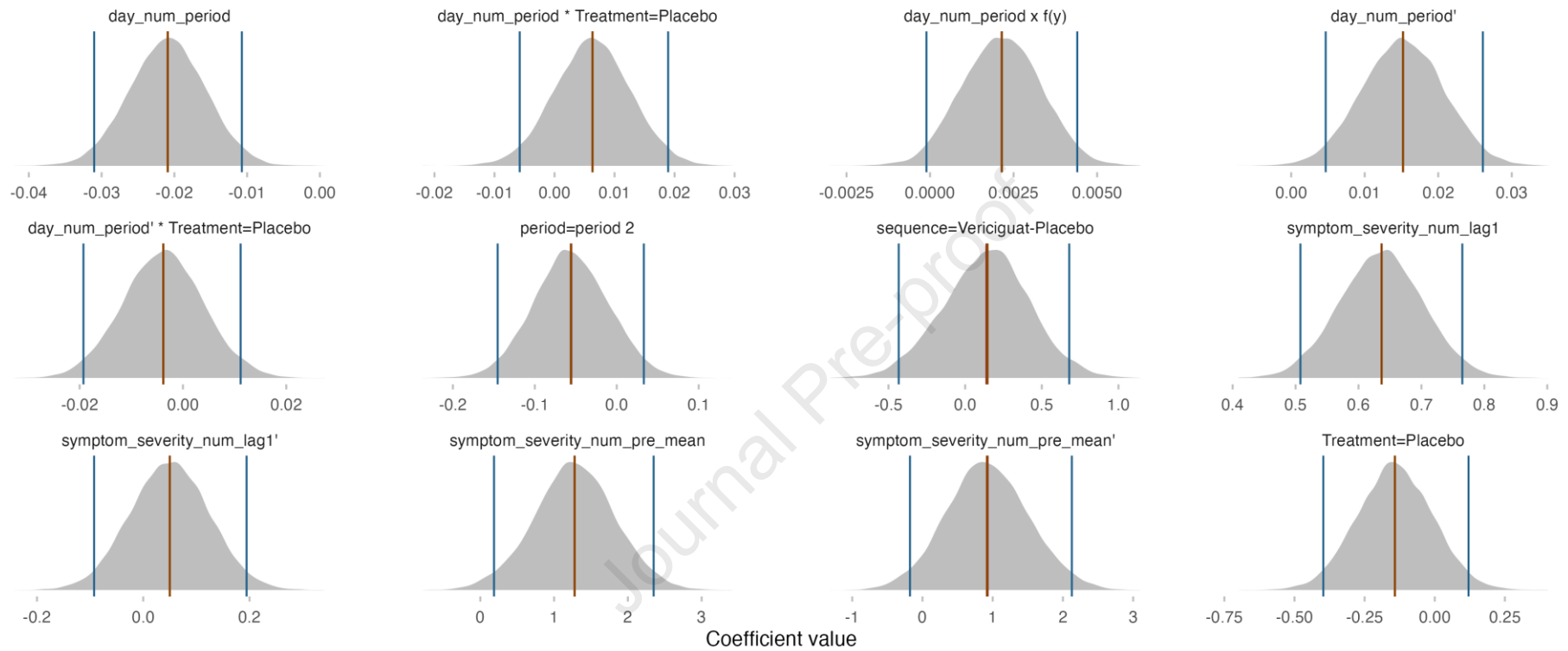
	Mixed Calibration/ Discrimination Indexes	Discrimination Indexes	Rank Discrim. Indexes
Obsâ€f7580	Bâ€f0.168 [0.163, 0.173]	gâ€f1.937 [1.683, 2.258]	Câ€f0.784 [0.777, 0.793]
Drawsâ€f16000		g _p â€f0.317 [0.288, 0.348]	D _{xy} â€f0.569 [0.555, 0.586]
Chainsâ€f4		EVâ€f0.338 [0.277, 0.397]	
Timeâ€f319.3s		vâ€f2.929 [2.161, 3.919]	
pâ€f11		vpâ€f0.079 [0.065, 0.095]	
Cluster onâ€f _{orbita_id}			
Clustersâ€f57			
σ _y â€f1.0221 [0.83, 1.2543]			

	Mean Î	Median Î	S.E.	Lower	Upper	Pr(Î>0)	Symmetry
yâ€f1	Â -1.0909	Â -1.0882	Â 0.8030	Â -2.6485	Â 0.4926	Â 0.0876	Â 1.01
yâ€f2	Â -2.8948	Â -2.8870	Â 0.8038	Â -4.4444	Â -1.2940	Â 0.0001	Â 1.01
yâ€f3	Â -5.6527	Â -5.6498	Â 0.8060	Â -7.2117	Â -4.0512	Â 0.0000	Â 1.01
symptom_severity_num_lag1	Â 0.6366	Â 0.6369	Â 0.0658	Â 0.5077	Â 0.7650	Â 1.0000	Â 1.01
symptom_severity_num_lag1'	Â 0.0499	Â 0.0501	Â 0.0737	Â -0.0925	Â 0.1945	Â 0.7501	Â 1.00
day_num_period	Â -0.0209	Â -0.0209	Â 0.0052	Â -0.0310	Â -0.0107	Â 0.0001	Â 1.00
day_num_period'	Â 0.0152	Â 0.0152	Â 0.0055	Â 0.0047	Â 0.0261	Â 0.9970	Â 0.99
Treatment=Placebo	Â -0.1419	Â -0.1424	Â 0.1333	Â -0.3973	Â 0.1205	Â 0.1447	Â 0.99
symptom_severity_num_pre_mean	Â 1.2782	Â 1.2782	Â 0.5450	Â 0.1855	Â 2.3494	Â 0.9898	Â 0.98
symptom_severity_num_pre_mean'	Â 0.9280	Â 0.9170	Â 0.5832	Â -0.1778	Â 2.1269	Â 0.9454	Â 1.03
period=period 2	Â -0.0555	Â -0.0561	Â 0.0456	Â -0.1453	Â 0.0330	Â 0.1163	Â 1.01
sequence=Vericigat=Placebo	Â 0.1386	Â 0.1462	Â 0.2808	Â -0.4333	Â 0.6790	Â 0.6976	Â 0.98
day_num_period Â— Treatment=Placebo	Â 0.0063	Â 0.0063	Â 0.0063	Â -0.0058	Â 0.0189	Â 0.8412	Â 1.00
day_num_period' Â— Treatment=Placebo	Â -0.0038	Â -0.0038	Â 0.0077	Â -0.0193	Â 0.0112	Â 0.3079	Â 1.02
day_num_period x f(y)	Â 0.0021	Â 0.0022	Â 0.0012	Â -0.0001	Â 0.0044	Â 0.9683	Â 1.01

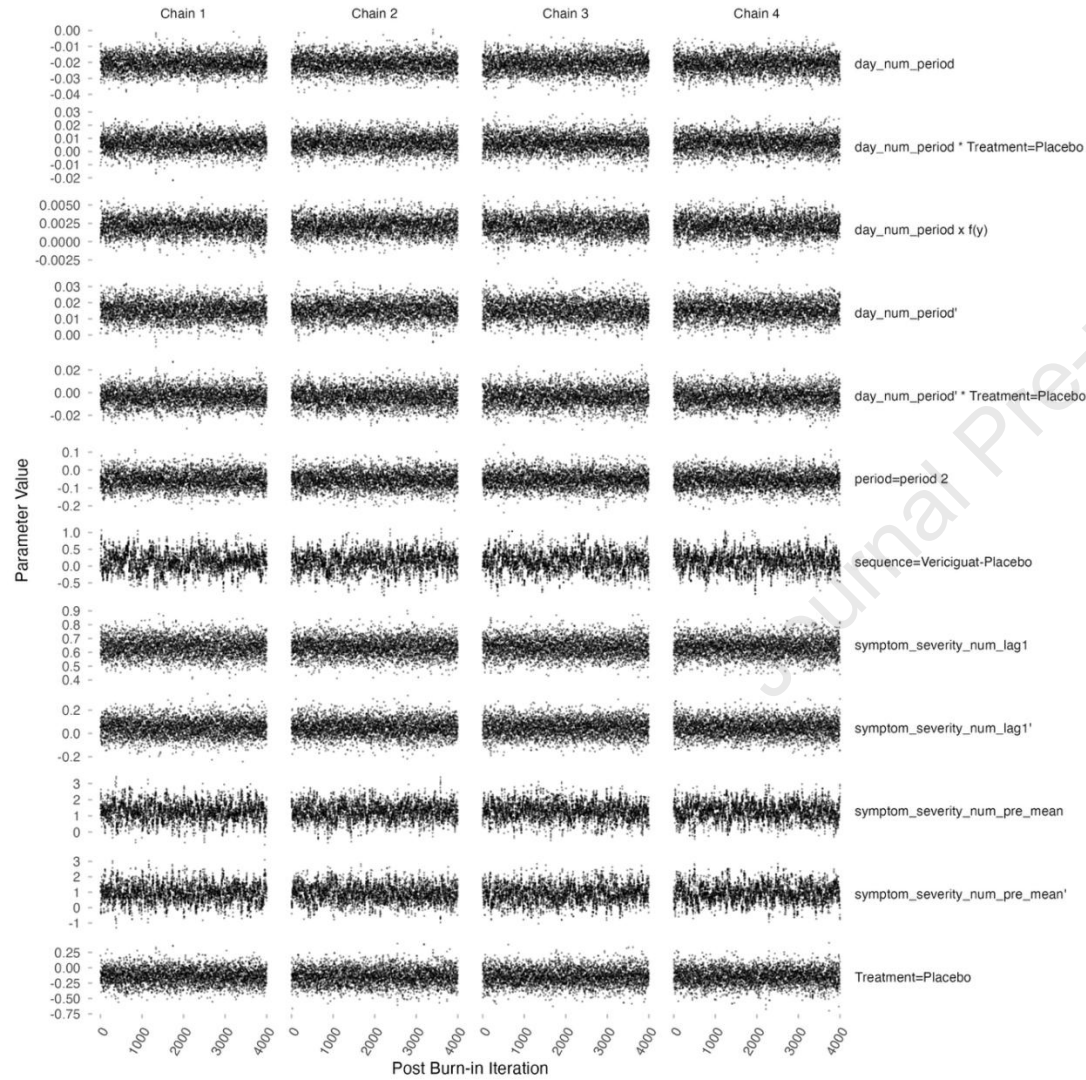
Contrasts Given Priors

```
[1] "list(c1 = list(Treatment = \"Vericigat\"), c2 = list(Treatment = \"Placebo\"), \" \" contrast = expression(c1 - c2), sd = 0.842807127883599)"
```

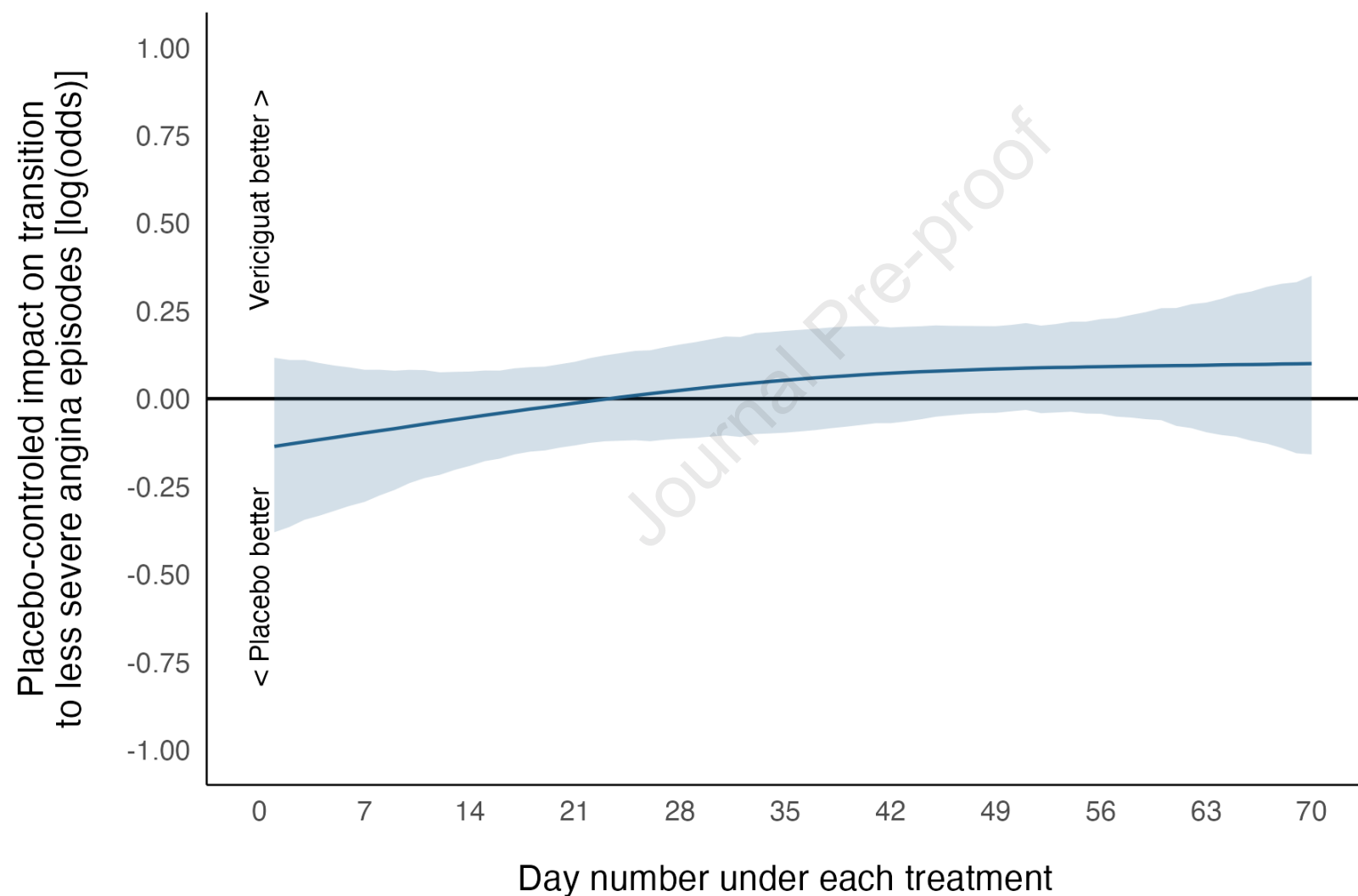
Supplementary Figure S17: Coefficient density plots for ORBITA-app symptom intensity model based on 57-patient intention-to-treat analysis



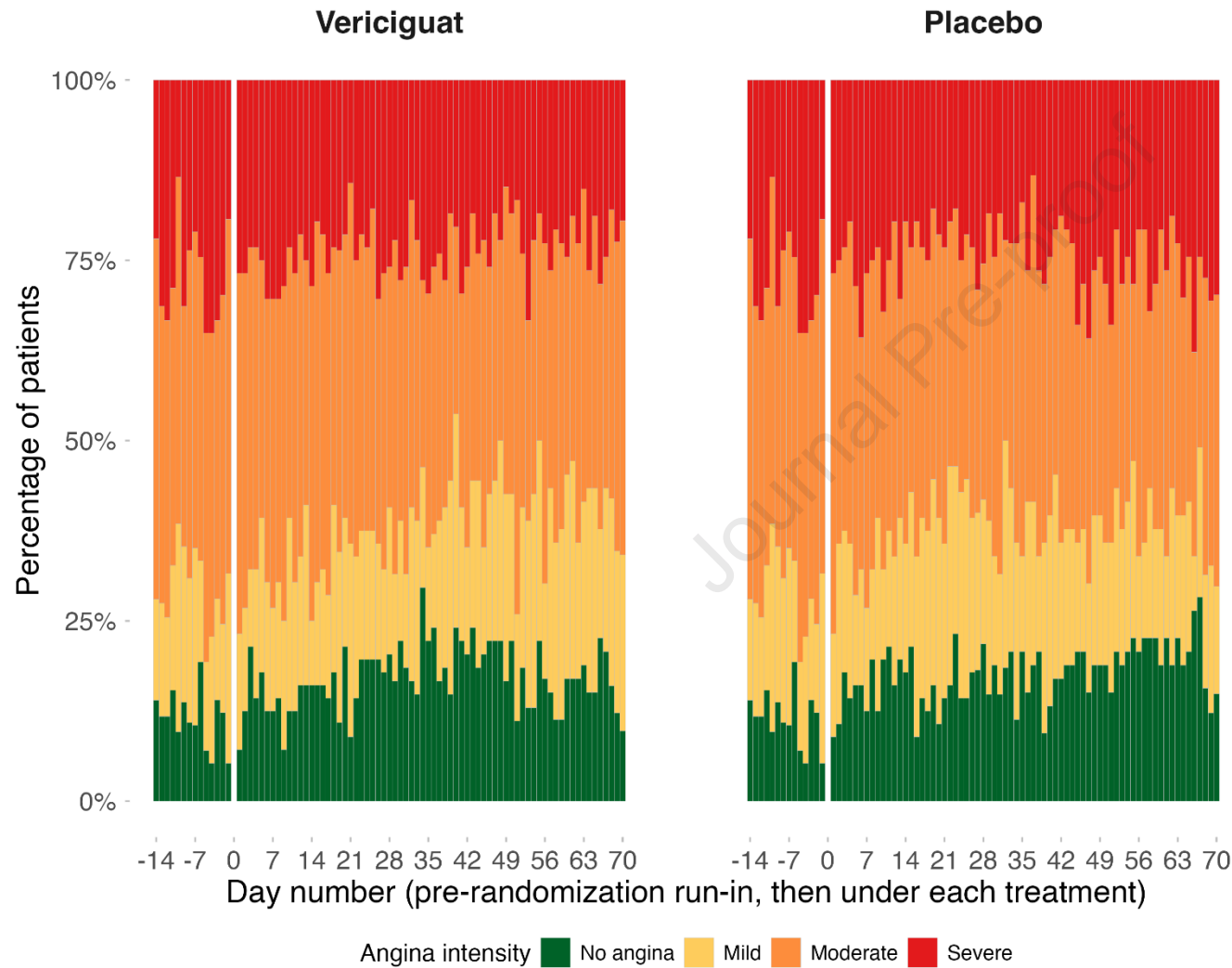
Supplementary Figure S18: Chain plot of MCMC draws for ORBITA-app symptom intensity model based on 57-patient intention-to-treat analysis



Supplementary Figure S19: Log-odds ratio for transitioning to better ORBITA-app symptom intensity with vericiguat versus placebo, by day number under each treatment based on 57-patient intention-to-treat analysis



Supplementary Figure S20: Percentage of patients by ORBITA angina intensity on each day of treatment, for vericiguat and placebo separately, including the shared 14-day pre-randomization run-in (day -1 to -14)



Supplementary Figure S21: Regression model and coefficients for ORBITA-app symptom frequency model stratified by vasospasm type based on 57-patient intention-to-treat analysis

Model Summary

Bayesian Constrained Partial Proportional Odds Ordinal Logistic Model

Dirichlet Priors With Concentration Parameter 0.215 for Intercepts

```
blrm(formula = symptom_frequency_num ~ rcs(symptom_frequency_num_lag1,
c(2, 5, 8)) + rcs(day_num_period, 3) * Treatment + ach_spasm_type *
Treatment + rcs(symptom_frequency_num_pre_mean, 3) + period +
sequence + cluster(orbita_id), ppo = ~day_num_period, cppo = function(y) y,
data = model_d, pconstrat = pcon, backend = "cmdstan", iter = 8000,
chains = 4, refresh = 100, progress = file.path(out_dir,
"model.progress.txt"), loo = FALSE, ppairs = NULL, method = "sampling",
file = file.path(out_dir, "res1.blrm.rds"))
```

Frequencies of Responses

0 1 2 3 4 5 6 7 8 9 10
1231 690 971 1000 718 435 386 273 338 178 1081

Frequencies of Missing Values Due to Each Variable

symptom_frequency_num 367
cluster(orbita_id) 0
symptom_frequency_num_lag1 361
day_num_period 0
Treatment 0
ach_spasm_type 0
symptom_frequency_num_pre_mean 0
period 0
sequence 0

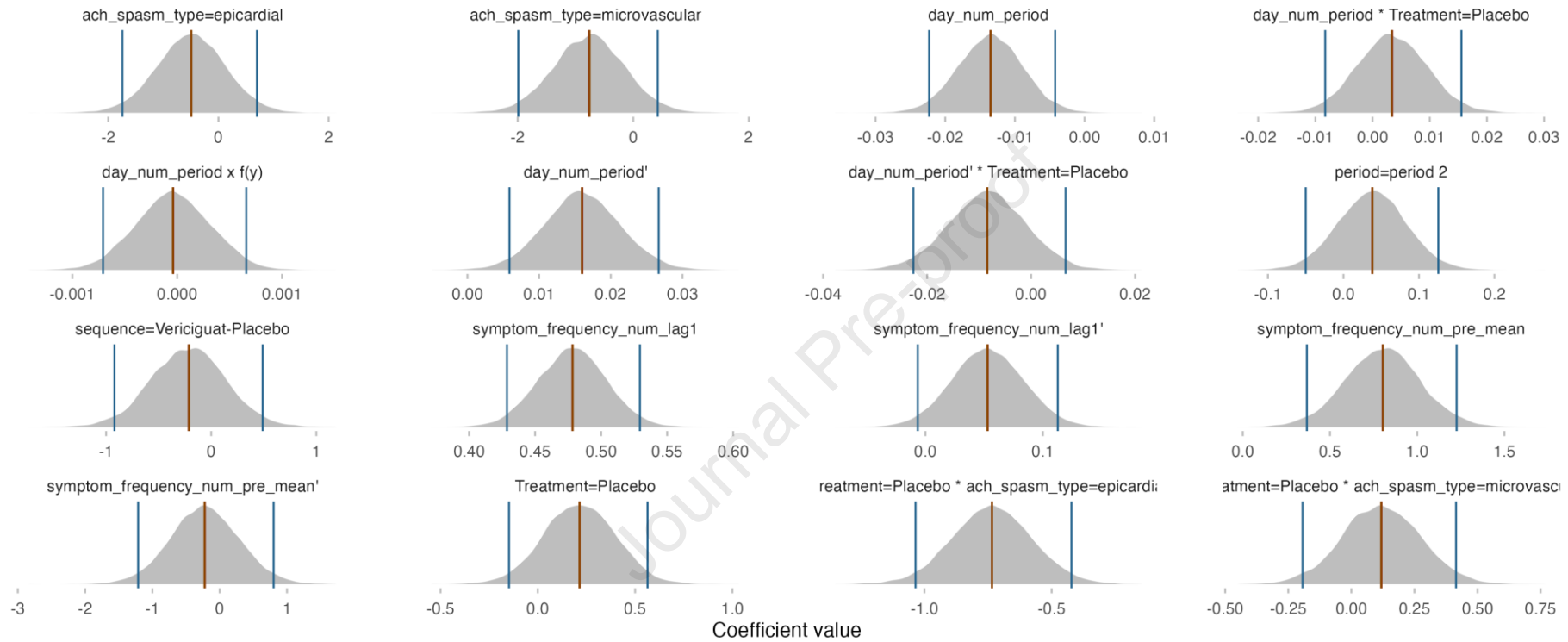
	Mixed Calibration/ Discrimination Indexes	Discrimination Indexes	Rank Discrim. Indexes
Obsâ€f7301	Bâ€f0.111 [0.105, 0.118]	gâ€f3.98 [3.575, 4.328]	Câ€f0.865 [0.858, 0.871]
Drawsâ€f16000		g _p â€f0.406 [0.377, 0.434]	D _{xy} â€f0.729 [0.717, 0.741]
Chainsâ€f4		EVâ€f0.551 [0.485, 0.608]	
Timeâ€f2741.1s		vâ€f12.515 [10.12, 14.755]	
pâ€f15		vpâ€f0.132 [0.111, 0.149]	
Cluster onâ€f _{orbita_id}			
Clustersâ€f55			
α _y â€f1.2862 [1.0359, 1.5843]			

	Mean I ²	Median I ²	S.E.	Lower	Upper	Pr(I ² >0)	Symmetry
yâ€f%W1	Â -0.0678	Â -0.0742	Â 0.8101	Â -1.6836	Â 1.5215	Â 0.4609	Â 1.01
yâ€f%W2	Â -1.1731	Â -1.1793	Â 0.8101	Â -2.8223	Â 0.3809	Â 0.0741	Â 1.01
yâ€f%W3	Â -2.6712	Â -2.6730	Â 0.8111	Â -4.3493	Â -1.1383	Â 0.0008	Â 1.00
yâ€f%W4	Â -4.1376	Â -4.1433	Â 0.8124	Â -5.7808	Â -2.5699	Â 0.0000	Â 1.01
yâ€f%W5	Â -5.2778	Â -5.2851	Â 0.8132	Â -6.9119	Â -3.7000	Â 0.0000	Â 1.01
yâ€f%W6	Â -6.1029	Â -6.1087	Â 0.8144	Â -7.7463	Â -4.5182	Â 0.0000	Â 1.02
yâ€f%W7	Â -6.9637	Â -6.9709	Â 0.8155	Â -8.5906	Â -5.3546	Â 0.0000	Â 1.01
yâ€f%W8	Â -7.6742	Â -7.6841	Â 0.8169	Â -9.2923	Â -6.0549	Â 0.0000	Â 1.01
yâ€f%W9	Â -8.6530	Â -8.6607	Â 0.8184	Â -10.2586	Â -7.0199	Â 0.0000	Â 1.01
yâ€f%W10	Â -9.2233	Â -9.2311	Â 0.8193	Â -10.8490	Â -7.6104	Â 0.0000	Â 1.01
symptom_frequency_num_lag1	Â 0.4783	Â 0.4784	Â 0.0258	Â 0.4287	Â 0.5294	Â 1.0000	Â 1.00
symptom_frequency_num_lag1'	Â 0.0531	Â 0.0528	Â 0.0305	Â -0.0065	Â 0.1127	Â 0.9588	Â 1.01
day_num_period	Â -0.0135	Â -0.0135	Â 0.0046	Â -0.0223	Â -0.0042	Â 0.0019	Â 1.01
day_num_period'	Â 0.0160	Â 0.0160	Â 0.0053	Â 0.0059	Â 0.0267	Â 0.9985	Â 1.00
Treatment=Placebo	Â 0.2145	Â 0.2143	Â 0.1834	Â -0.1478	Â 0.5639	Â 0.8812	Â 0.98
ach_spasm_type=epicardial	Â -0.4953	Â -0.4912	Â 0.6207	Â -1.7434	Â 0.6999	Â 0.2095	Â 0.98
ach_spasm_type=microvascular	Â -0.7643	Â -0.7559	Â 0.6181	Â -1.9903	Â 0.4246	Â 0.1039	Â 1.00
symptom_frequency_num_pre_mean	Â 0.8019	Â 0.8038	Â 0.2157	Â 0.3674	Â 1.2259	Â 0.9999	Â 0.99
symptom_frequency_num_pre_mean'	Â -0.2202	Â -0.2253	Â 0.5109	Â -1.2126	Â 0.8015	Â 0.3280	Â 1.02
period=period 2	Â 0.0380	Â 0.0385	Â 0.0449	Â -0.0500	Â 0.1256	Â 0.7998	Â 1.00
sequence=Vericiguat-Placebo	Â -0.2149	Â -0.2121	Â 0.3580	Â -0.9216	Â 0.4894	Â 0.2703	Â 0.98
day_num_period Â— Treatment=Placebo	Â 0.0034	Â 0.0034	Â 0.0061	Â -0.0083	Â 0.0155	Â 0.7136	Â 1.02
day_num_period' Â— Treatment=Placebo	Â -0.0085	Â -0.0084	Â 0.0075	Â -0.0227	Â 0.0066	Â 0.1272	Â 0.98
Treatment=Placebo Â— ach_spasm_type=epicardial	Â -0.7345	Â -0.7344	Â 0.1563	Â -1.0352	Â -0.4222	Â 0.0000	Â 1.00
Treatment=Placebo Â— ach_spasm_type=microvascular	Â 0.1193	Â 0.1180	Â 0.1538	Â -0.1933	Â 0.4148	Â 0.7826	Â 0.99
day_num_period x f(y)	Â 0.0000	Â 0.0000	Â 0.0004	Â -0.0007	Â 0.0007	Â 0.4540	Â 1.00

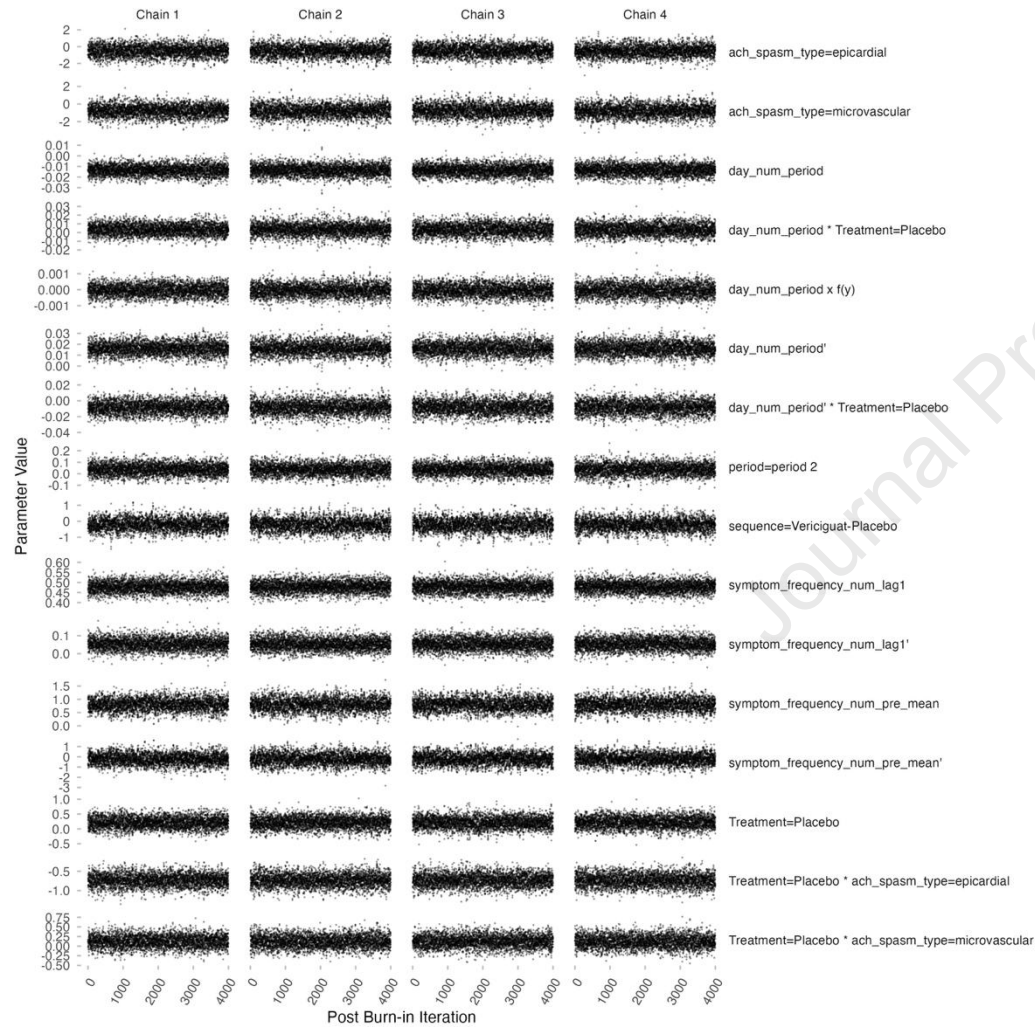
Contrasts Given Priors

```
[1] "list(c1 = list(Treatment = \"Vericiguat\"), c2 = list(Treatment = \"Placebo\"), \" \" contrast = expression(c1 - c2), sd = 0.842807127883599)\""
```

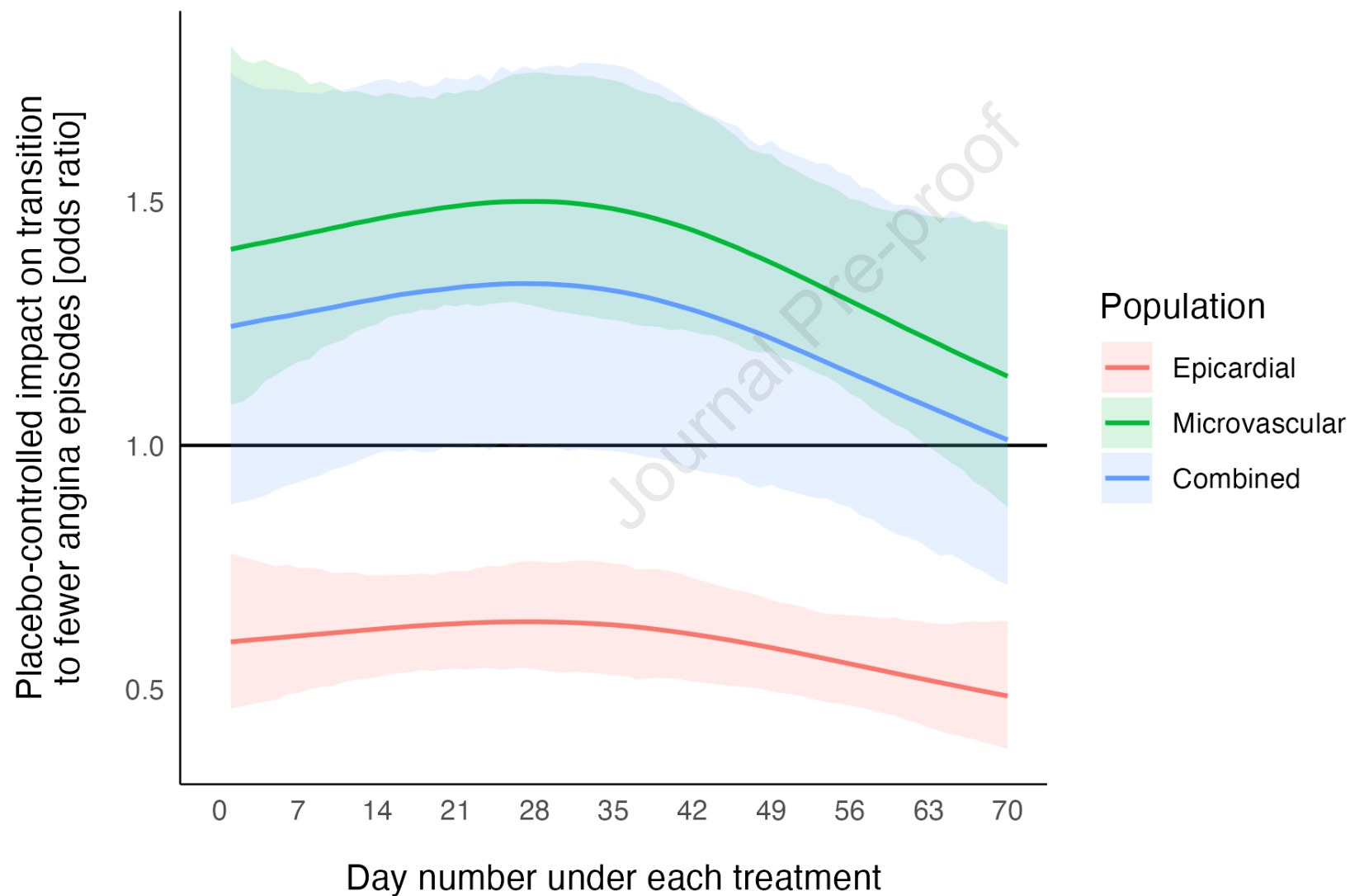
Supplementary Figure S22: Coefficient density plots for ORBITA-app symptom frequency model stratified by vasospasm type based on 57-patient intention-to-treat analysis



Supplementary Figure S23: Chain plot of MCMC draws for ORBITA-app symptom frequency model stratified by vasospasm type based on 57-patient intention-to-treat analysis



Supplementary Figure S24: Odds ratio for transitioning to better ORBITA-app symptom frequency with vericiguat versus placebo, by day number under each treatment and by vasospasm type, based on 57-patient intention-to-treat analysis



Supplementary Figure S25: Percentage of patients by ORBITA-app symptom frequency on each day of treatment, for vericiguat and placebo separately and stratified by vasospasm type, including the shared 14-day pre-randomization run-in (day -1 to -14)

